

Optimal Carbon Tax with Dynamic Skill Heterogeneity

Alexandre Pavlov*

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Abstract

How should a government price carbon when it also cares about redistribution? I extend the dynamic Mirrlees model to two goods, one polluting, over a life cycle of privately observed skill shocks. Under weak separability, the dirty good should be taxed at the social cost of carbon (SCC) while labor and savings distortions keep their standard forms. Only non-separability makes it nonlinear, raising it above the SCC when the dirty good is a weaker labor complement. In a US calibration with energy standing in for the dirty good however, uniform taxation captures 98% of the welfare gain of an optimal nonlinear carbon tax. When labor and capital distortions are frozen at the levels implied by the US tax system, this sufficiency result is overturned as the best uniform tax only captures 1.5% of the welfare gain of the best nonlinear carbon tax. Here, a flat tax at an SCC of \$190/tCO₂ would be welfare-reducing and the best flat tax would deflate to an equivalent of \$47/tCO₂. The numerical results suggest carbon tax revenue should then be recycled as a near-uniform lifetime dividend, equivalent to real-world fee-and-dividend designs.

Keywords: *Carbon tax, dynamic Mirrlees taxation, second-best environmental policy*

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*Université de Montréal: alexandre.pavlov@umontreal.ca

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1 Introduction

Carbon pricing is the economist’s favorite climate policy, and the voter’s least. France abandoned its carbon tax plans after the *gilets jaunes* protests and Canada repealed its consumer carbon tax in 2025 after it became an electoral wedge issue amid claims that it unfairly burdened households. This disconnect might in part be driven by the fact that most carbon policy is informed by integrated assessment models (Environment and Climate Change Canada (2016), Interagency Working Group on Social Cost of Greenhouse Gases (2021)), which are single-agent models that are not designed to address this concern.

Whether carbon taxes are regressive is a topic of intense study, with one aspect of the debate relating to the choice of measure. When households are ranked by annual income, taxes on gasoline and other energy appear to fall more heavily on the poor. However, when households are ranked by lifetime income, the burden appears much flatter because households smooth consumption and today’s low earners include the young and temporarily poor (Poterba 1989; Hassett et al. 2009). The welfare-relevant burden of a carbon tax is therefore a lifetime object rather than an annual one and it’s important to consider the timing of an optimal carbon tax.

However, most research focuses on the incidence of existing carbon taxes rather than asking how they should be designed in the first place. Barrage (2020) and Douenne et al. (2026) are the exceptions, but the first ties the tax system to an *ad hoc* revenue target rather than optimal redistribution and the second restricts attention to linear instruments even though income taxes are in practice nonlinear. How then should a government price carbon when it cares about redistribution, and does pollution change how income should be taxed?

The answer depends on how the income tax is modeled. The Ramsey tradition treats it as a schedule of linear instruments and usually finds that carbon should be priced below the Pigouvian level because the carbon tax compounds the distortions the rest of the tax system already imposes. The Mirrleesian tradition allows the government to levy nonlinear income taxes and usually finds that the carbon tax should be set at the Pigouvian level, unless labor supply is affected by consumption of the dirty good or by pollution.

I take the Mirrlees approach into a dynamic setting with two goods, one clean and one dirty, where agents draw privately observed skill shocks over their life cycle and their consumption of the dirty good feeds a stock externality. To my knowledge, this is the first dynamic-Mirrleesian derivation of the optimal consumer carbon tax. Compared to the static irrelevance theorems of Kaplow (2012) and Jacobs and Mooij (2015), dynamics introduce savings and history-dependence.

I find that the static irrelevance result holds in a dynamic environment. Under weak separability, the carbon wedge equals the social cost of carbon, the labor wedge keeps the Golosov et al. (2016) ABCD form, and the savings wedge keeps its inverse-Euler form. Redistributing through the optimal nonlinear income tax restores the split between corrective and redistributive taxation.

That is, the carbon price should not be adjusted to perform redistribution if nonlinear income taxes are available and optimal.

I then quantify this separation by solving the optimal mechanism, treating energy as a stand-in for the dirty good and calibrating the model to US data. I find that the optimal wedge on energy is the social cost of carbon scaled up by a history-dependent Corlett-Hague premium, starting at the Pigouvian level, rising with age with the labor wedge, and ending the working life about 9% above the Pigouvian level. That premium is however small in welfare terms. When the income tax is set optimally alongside it, a flat carbon tax at the social cost of carbon already captures 98% of the welfare gain of the full optimum.

This result however assumes that every margin is set optimally. I therefore also consider the case where the carbon wedge is set when labor and savings wedges are fixed to the levels implied by the US tax system. I find that the carbon wedge is pulled down when labor is overdistorted and up when it is underdistorted. The frozen savings margin instead rotates the carbon wedge across the life cycle: its contribution is positive at some ages and negative at others whatever the schedule, so it cannot move the price the same way at every age. Whether the price crosses the social cost of carbon depends on the size of the swing as well as the level of the labor wedge.

Solving this constrained optimum upends the previous analysis. A flat tax at the EPA's \$190/tCO₂ social cost of carbon now shrinks consumption by 0.115% relative to having no carbon price at all. In a jointly optimized tax system, even an unoptimized flat tax set at the social cost of carbon captures 98% of the welfare gain of a nonlinear tax. In this constrained setting, the best flat tax captures only 1.5% of it. Moreover, this tax is equivalent to \$47/tCO₂, about a quarter of the social cost of carbon. These results suggest that mitigation is constrained by a suboptimal income tax system rather than by suboptimal carbon pricing design.

The solution also answers the distributional worry this paper opens with, as it suggests that carbon tax revenue should be recycled as a near-uniform lifetime dividend, equivalent to a fee-and-dividend design. The planner returns about 97% of carbon tax revenue as a lifetime transfer that is nearly identical across histories. Against carbon payments that scale with lifetime spending, the net burden falls monotonically in lifetime earnings and the bottom two lifetime-earnings quintiles gain even from a welfare-reducing flat tax at the social cost of carbon.

Section 2 reviews the literature. Section 3 presents the modeling framework. Section 4 builds the two-good dynamic Mirrlees economy and derives the optimal distortions. Section 5 studies the constrained problems: the best carbon-only reform when the labor and savings wedges are frozen, the wedges a given carbon price induces, and the optimal flat instruments that bridge the two. Section 6 solves both cases numerically with a model calibrated to US data. Section 7 concludes.

2 Literature review

This paper sits between two literatures that reach opposite verdicts on how distortionary taxation should bend the carbon price and draws on a third that has the tools to adjudicate between them, yet has never been aimed at the environment. Should a government facing a distortionary tax system price carbon below marginal damages because the environmental tax compounds distortions elsewhere, above marginal damages because the environmental tax can be used to ease distortions elsewhere, or exactly at marginal damages?

The Ramsey tradition typically finds that the optimal carbon tax should lie below marginal damages, but that the structure of the optimal income tax is unchanged (Sandmo 1975; Bovenberg and Mooij 1994; Bovenberg and Smulders 1996). Barrage (2020) extends their static results with a dynamic climate-economy Ramsey model and finds that the optimal carbon tax should be 8-24% below the first-best Pigouvian level in the presence of distortionary income taxation. However, this analysis assumes a representative agent, focusing on efficiency and ignoring any distributional concerns.

Adding heterogeneous agents to this model overturns this result. Douenne et al. (2026) extend Barrage’s dynamic Ramsey model to heterogeneous agents, taking inspiration from the model developed in Werning (2007), and find that the optimal carbon tax is nearly Pigouvian. Douenne et al. (2025) show that this is robust to the inclusion of uninsurable income risk. Whether such a flat carbon tax is regressive and whether revenue recycling is progressive is still an open debate (Klenert et al. 2018; Goulder et al. 2019) and a parallel policy and ethics literature presses for differentiated carbon taxation (Chancel 2022; Dietsch 2024).

Without personalized carbon taxes, the exact restrictions on a uniform carbon tax reshape both its level and incidence. Using a heterogeneous agent microsimulation model calibrated to German data and focusing on revenue-neutral carbon tax reforms, Ploeg et al. (2025) find that the carbon tax should exceed marginal damages when agents have high inequality aversion. In a model calibrated to German data where households differ in income and energy efficiency, Hänsel et al. (2022) show that the first-best keeps the carbon tax uniform at the Pigouvian level and carries all redistribution through lump-sum transfers conditioned on the household’s income–energy type while leaving the income tax undistorted.

Separation can also fail for a reason orthogonal to instrument constraints. When climate damages fall unequally across households and compensation is incomplete, the social cost of carbon itself becomes distribution-dependent (Dennig et al. 2015; Kornek et al. 2021). I hold damages homogeneous to isolate the redistributive-instrument channel, so the separation I study is the Atkinson–Stiglitz one.

A static Mirrleesian literature establishes that separation, and overturns the Ramsey verdict in doing so: the below-damages result is not intrinsic to distortionary taxation but to holding the

tax system away from the optimum. Once the income tax is the optimal nonlinear schedule of Mirrlees (1971), the logic of Atkinson and Stiglitz (1976) applies and differential commodity taxes are redundant under weak separability. Cremer et al. (1998), Cremer et al. (2001), and Cremer et al. (2010) develop the environmental version of the problem where a nonlinear income tax alongside a strictly Pigouvian levy is efficient, so the public-funds correction of the Ramsey tradition disappears. Kaplow (2012) shows that any environmental reform can be made distribution-neutral by a type-by-type adjustment of the income tax, so a move to Pigouvian pricing is Pareto-improving whatever the schedule. Jacobs and Mooij (2015) show that, when goods and environmental quality are weakly separable from labor, the optimal pollution tax equals marginal damages exactly and the marginal cost of public funds is one.

When separability fails, the tax follows the logic established in Corlett and Hague (1953), which Mirrlees (1976) formalized in his model. A good is taxed above the Pigouvian level for a leisure complement, the case West and Williams (2007) substantiate for gasoline, and below it for a leisure substitute. Saez (2002) extends the benchmark to preferences that vary with ability. In a linear-tax setting, Jacobs and Ploeg (2019) show that the pollution tax stays Pigouvian when Engel curves are linear and slips below only when the poor spend disproportionately on the dirty good under nonlinear Engel curves. Every result in this literature is static, with no savings and no life-cycle dynamics. Therefore, it cannot speak to the age profile or the time path of a carbon tax.

The tools for this dynamic treatment come from the New Dynamic Public Finance literature. Golosov et al. (2003) derive a rationale for positive capital taxation in a dynamic Mirrlees economy, since it is harder to elicit future work from households that save privately. This overturns the classic zero-capital-tax benchmark common in the Ramsey tradition (Judd 1985; Chamley 1986). Moreover, the authors find that the Atkinson-Stiglitz uniform-commodity result carries over into the dynamic setting. Kocherlakota (2005) shows how to implement the capital wedge into a wealth-tax and Albanesi and Sleet (2006) derive a tractable recursive form of the problem. Kapička (2013) extends the first-order approach to persistent shocks, Pavan et al. (2014) give the general envelope characterization, Farhi and Werning (2013) characterize the labor wedge over the life cycle, and Golosov et al. (2016) extend the ABC form of the labor wedge derived in the static frameworks of Mirrlees (1971) and Saez (2001) into a dynamic ABCD form. The optimal wedges the dynamic Mirrlees model delivers vary with age, connecting to the age-dependent-tax literature (Weinzierl 2011; Bastani et al. 2013; Erosa and Gervais 2002), which prices age-varying labor rather than a commodity. None of this machinery has been aimed at commodity taxation under non-separable preferences or corrective taxation.

I extend the dynamic Mirrleesian apparatus from one consumption good to two, letting the second pollute, and characterize the optimal carbon tax alongside the labor and savings wedges over the life cycle. The heterogeneous-agent Ramsey results and this paper share the Jacobs and Mooij (2015) and Kaplow (2012) irrelevance result, which both approaches carry into a

climate economy by different routes. Douenne et al. (2026) carry it with an optimally-set linear transfer, showing analytically that the marginal cost of public funds averages to one across a balanced-growth path and equals one period by period when preferences are logarithmic. Douenne et al. (2025) find that this result still exists when agents face uninsurable labor risk. I consider uninsurable labor risk with a fully nonlinear income tax and obtain a carbon wedge equal to the social cost of carbon at every history under weak separability and any preference satisfying mild conditions, alongside a Mirrleesian labor wedge and a non-zero savings wedge following the inverse Euler condition. The two approaches converge in their policy conclusions on the condition that the tax system is jointly optimized. What matters is therefore not so much linearity of instruments, but whether other instruments are misset.

3 Environment

The economy is a two-good extension of the dynamic Mirrlees setting of Golosov et al. (2003), Farhi and Werning (2013), and Golosov et al. (2016). The two goods are a clean numéraire and a polluting good. To isolate how household income inequality shapes the optimal carbon tax, I abstract from international and intergenerational equity, and consider a single country as well as a single generation living for $T + 1$ periods.

Each household consumes a clean good c_t and a polluting good q_t , supplies labor l_t , and is exposed to the aggregate pollution stock S_t . Expected lifetime utility is

$$\mathbb{E}_0 \left[\sum_{t=0}^T \beta^t U(c_t, q_t, l_t, S_t) \right]$$

$\beta \in (0, 1)$ is the discount factor and utility is non-separable between its arguments. The clean good c is the numéraire, with prices normalized to 1. A unit of output transforms one-for-one into either good, so the producer price of the dirty good is also 1. The clean good is emission-free, whereas each unit of the dirty good emits a unit of carbon, so that date- t emissions equal aggregate dirty-good consumption Q_t . Emissions accumulate into a persistent pollution stock S_t , a function of current and past aggregate emissions.

$$S_t = S(Q_0, Q_1, \dots, Q_t)$$

where $Q_t = \int_0^\infty \mathbb{E}_0[q_t(\theta^t)|\theta] dF_0(\theta)$ is aggregate consumption of the dirty good at date t . I assume that S is differentiable and non-decreasing in each argument, but otherwise leave it unrestricted. The linear special case $S_t = \sum_{j=0}^t (1 - d_j) Q_{t-j}$ in which each past emission flow leaves an additive trace $(1 - d_j)$ in the current stock, is the carbon-cycle representation of

Golosov et al. (2014). The general form above nests it and allows nonlinear accumulation. A convenient leading example keeps a fraction δ of the stock from one period to the next.

$$S_t = \sum_{j=0}^t \delta^j Q_{t-j}$$

This is the recursion $S_t = \delta S_{t-1} + Q_t$ from an initial pollution stock. To keep the state finite, I assume emissions stop affecting the pollution stock after s periods.

$$S_t = S(Q_{t-s}, \dots, Q_t)$$

Aside from the polluting good, the model follows the standard dynamic Mirrlees setup. At $t = 0$, agents draw an initial skill (type) θ_0 from a distribution $F_0(\theta)$. Later skills follow a Markov process $F_t(\theta|\theta_{t-1})$. At period t , each agent knows their skill draws up to time t , which I denote by a skill history vector $\theta^t = (\theta_0, \dots, \theta_t)$. Θ^t is the set of all possible skill histories.

Agents retire at an exogenous date $T_R \leq T$, common to all and not a choice. Periods $t < T_R$ are working periods and periods $t \geq T_R$ are retirement. From T_R on the agent supplies no labor and draws no further skills, so period utility is the same U evaluated at $l = 0$. Nothing is left to screen: the incentive constraints below hold trivially, the costate vanishes (Appendix A.7), and the labor wedge has no content. The savings and commodity margins survive, so a retiree still splits consumption between the two goods and still faces a carbon price. This is the life-cycle structure of Farhi and Werning (2013), with one difference that matters later: there retirement is a pure consumption-smoothing problem, while here the retiree's bundle still contains the dirty good.

An agent of type θ_t who supplies l_t units of labor produces $y_t = \theta_t l_t$ units of output. Everyone observes total output, but skill θ_t and labor supply l_t are private. Consumption of both goods is also publicly observable¹. Let $c_t(\theta^t), q_t(\theta^t) : \Theta^t \rightarrow \mathbb{R}_+$ and $y_t(\theta^t) : \Theta^t \rightarrow \mathbb{R}_+$ be the agent's allocation of the clean good, the polluting good, and output, respectively. Let $\sigma^t(\theta^t) : \Theta^t \rightarrow \Theta^t$ describe how an agent with skill history θ^t reports it, and let Σ^t be the set of all possible reporting strategies. Capital transfers resources across periods through a linear technology with exogenous constant gross return $R > 0$. The social welfare function is

¹This is a plausible assumption for some goods such as electricity and air travel. For goods whose consumption cannot be observed, the planner is further constrained and the history-dependent commodity wedges of Sections 4 and 5.1 no longer apply. The anonymous price of Section 5.2 survives: an anonymous linear price requires observing total expenditure, not its composition. The assumption binds hardest on Section 5.1, where the commodity margin is the only instrument the planner still sets freely and Section 6 finds its history dependence to be where the welfare gains concentrate.

$$\int_0^\infty \alpha(\theta) \mathbb{E}_0 \left[\sum_{t=0}^T \beta^t U(c_t, q_t, l_t, S_t) \right] dF_0(\theta)$$

$\alpha(\theta)$ is a Pareto weight on the welfare of an agent with initial skill θ , normalized so that $\int_0^\infty \alpha(\theta) dF_0(\theta) = 1$. A utilitarian planner sets $\alpha(\theta) = 1$ for all θ . Let U_c and U_q denote the partial derivatives with respect to the clean and polluting goods, and U_l with respect to labor. Since $y = \theta l$, I can write $U(c, q, l, S)$ as $U(c, q, \frac{y}{\theta}, S)$. Let U_y and U_θ be the partial derivatives of utility with respect to y and θ .

Assumption 3.1 (Preferences).

U is twice continuously differentiable with $U_c, U_q > 0$, $U_l, U_S < 0$, $U_{cc}, U_{qq}, U_{ll}, U_{SS} \leq 0$, strictly quasi-concave in (c, q) with $2U_c U_q U_{cq} - U_c^2 U_{qq} - U_q^2 U_{cc} > 0$ (regular strict quasi-concavity: a strictly convex, non-degenerate indifference map), and satisfies Inada conditions in (c, q) . Single crossing holds at every S .

$$\frac{\partial}{\partial \theta} \frac{U_l(c, q, y/\theta)}{U_c(c, q, y/\theta, S)} \geq 0 \quad \text{for all } (c, q, y, \theta)$$

At the optimal allocation, $1 + \epsilon_t(\theta) - \gamma_{c,t}(\theta) > 0$, where $\epsilon_t, \gamma_{c,t}$ are defined in (1) below. The conditions involving labor are imposed on working periods, where the labor margin has content; from T_R on labor is zero, U is evaluated at $l = 0$, and the inverse Frisch elasticity ϵ_t of (1) is not defined.

Assumption 3.2 (Shock process).

For all $1 \leq t < T_R$, f_t is differentiable in both arguments with $f'_t = \partial f_t / \partial \theta$ and $f_{2,t} = \partial f_t / \partial \theta_{t-1}$. For all θ_{t-1} , the ratio

$$\frac{\theta_{t-1} \int_\theta^\infty f_{2,t}(x|\theta_{t-1}) dx}{\theta f_t(\theta|\theta_{t-1})}$$

is bounded, and $\lim_{\theta \rightarrow \infty} \frac{1 - F_t(\theta|\theta_{t-1})}{\theta f_t(\theta|\theta_{t-1})}$ is finite.

Any admissible allocation must be incentive compatible: the agent must weakly prefer truthful reporting to any other strategy. The pollution stock S_t is an aggregate, so no single agent's deviation moves it and it enters both sides of the constraint identically. For every initial type θ and every reporting strategy,

$$\begin{aligned} & \mathbb{E}_0 \left[\sum_{t=0}^T \beta^t U \left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t \right) \middle| \theta \right] \\ & \geq \mathbb{E}_0 \left[\sum_{t=0}^T \beta^t U \left(c_t(\sigma^t(\theta^t)), q_t(\sigma^t(\theta^t)), \frac{y_t(\sigma^t(\theta^t))}{\theta_t}, S_t \right) \middle| \theta \right] \end{aligned}$$

Over a finite horizon, global incentive compatibility is equivalent to the absence of a profitable one-shot deviation after every history (Fernandes and Phelan 2000; Kapička 2013). I follow the first-order approach of Kapička (2013), Farhi and Werning (2013), and Golosov et al. (2016), replacing these constraints with their local counterpart. Let $u_t(\theta^t)$ be the lifetime utility an agent obtains from reporting truthfully at every date. Truth-telling is locally optimal when the envelope condition holds.

$$\dot{u}_t(\theta^t) = U_\theta \left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t \right) + \beta w_{2,t}(\theta^t)$$

$w_{2,t}(\theta^t) = \int_0^\infty u_{t+1}(\theta^t, \theta_{t+1}) \partial_{\theta_t} f_{t+1}(\theta_{t+1} | \theta_t) d\theta_{t+1}$ is the marginal continuation value of type θ^t , and the dot denotes a derivative with respect to the current draw θ_t . The continuation term is the future information rent of the current draw: with i.i.d. shocks, $\partial_{\theta_t} f_{t+1} = 0$ and it vanishes, since today's skill then carries no information about tomorrow's. Under Assumptions 3.1 and 3.2, this local condition is necessary for incentive compatibility (Kapička 2013). The monotonicity requirements on the allocation belong to the sufficiency direction, not to necessity. The local conditions define a relaxed problem and are necessary but not sufficient for global incentive compatibility. Indeed, no general primitive restriction guarantees sufficiency in dynamic models with multiple goods (Golosov et al. 2016). I solve the relaxed problem throughout and verify global incentive compatibility numerically in Section 6.

The aggregate resource constraint must hold in every period,

$$\int_0^\infty \mathbb{E}_0[c_t(\theta^t) + q_t(\theta^t) | \theta] dF_0(\theta) + K_{t+1} \leq \int_0^\infty \mathbb{E}_0[y_t(\theta^t) | \theta] dF_0(\theta) + RK_t$$

where K_t is the aggregate capital stock, with K_0 given and $K_{T+1} = 0$. Dividing each period's constraint by R^t , summing over t , and using these boundary conditions to telescope the capital terms consolidates the sequence into a single intertemporal constraint,

$$\int_0^\infty \mathbb{E}_0 \left[\sum_{t=0}^T \frac{c_t(\theta^t) + q_t(\theta^t)}{R^t} \middle| \theta \right] dF_0(\theta) \leq \int_0^\infty \mathbb{E}_0 \left[\sum_{t=0}^T \frac{y_t(\theta^t)}{R^t} \middle| \theta \right] dF_0(\theta) + RK_0$$

The planner chooses allocations to maximize welfare subject to the local incentive constraint, the intertemporal resource constraint, and the pollutant law of motion.

$$\max_{\{c_t(\theta^t), q_t(\theta^t), y_t(\theta^t), S_t\}_{\theta^t \in \Theta^t, t=0, \dots, T}} \int_0^\infty \alpha(\theta) \left(\mathbb{E}_0 \left[\sum_{t=0}^T \beta^t U \left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t \right) \middle| \theta \right] \right) dF_0(\theta)$$

Three wedges, the implicit marginal taxes on the labor, savings, and commodity margins, describe the distortions the optimal allocation imposes.

$$\begin{aligned} 1 - \tau_t^y(\theta^t) &\equiv - \frac{U_l \left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t \right)}{\theta_t U_c \left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t \right)} \\ 1 - \tau_t^s(\theta^t) &\equiv \frac{1}{\beta R} \frac{U_c \left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t \right)}{\mathbb{E}_t \left[U_c \left(c_{t+1}(\theta^{t+1}), q_{t+1}(\theta^{t+1}), \frac{y_{t+1}(\theta^{t+1})}{\theta_{t+1}}, S_{t+1} \right) \right]} \\ 1 + \tau_t^q(\theta^t) &\equiv \frac{U_q \left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t \right)}{U_c \left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t \right)} \end{aligned}$$

I define six preference statistics used throughout. Two are the standard own-curvature elasticities, namely the inverse Frisch elasticity and the inverse intertemporal elasticity of substitution.

$$\epsilon_t \equiv \frac{U_{ll,t} l_t}{U_{l,t}} \quad \sigma_t \equiv - \frac{U_{cc,t} c_t}{U_{c,t}} \quad (1)$$

Four are cross-partials that measure how the marginal utilities of the two goods and of the pollutant move with labor as well as with each other. In order, they represent the labor complementarity with the clean good, the labor complementarity with the dirty good, the labor complementarity of environmental quality, and the cross-elasticity between the two goods.

$$\gamma_{c,t} \equiv \frac{U_{cl,t} l_t}{U_{c,t}} \quad \gamma_{q,t} \equiv \frac{U_{ql,t} l_t}{U_{q,t}} \quad \xi_t \equiv \frac{U_{Sl,t} l_t}{U_{S,t}} \quad \sigma_{cq,t} \equiv - \frac{U_{cq,t} q_t}{U_{c,t}} \quad (2)$$

Note that complementarity is measured with respect to labor rather than leisure here, so a positive γ signals a good whose marginal utility rises as the agent works more. To get the

leisure complementarity interpretation, it suffices to flip the sign. Most of the commodity-wedge results hinge on the gap $\gamma_{c,t} - \gamma_{q,t}$ between the clean and dirty goods' complementarities with labor. If the gap is positive, the clean good is the relative labor complement, meaning an agent who works more wants more of it. Likewise, the results pertaining to the social cost of carbon hinge on the gap $\xi_t - \gamma_{c,t}$, which measures how the marginal damage of pollution varies with labor supply. If the gap is positive, an agent who works more suffers more from pollution and a cleaner environment is a relative labor complement. Both complementarities are proportional to l_t , so both vanish in retirement: from T_R on $\gamma_{c,t} = \gamma_{q,t} = 0$ and the gap closes mechanically.

One combination of these statistics appears whenever the commodity margin is constrained:

$$\mathcal{I}_t \equiv \frac{\sigma_t}{c_t} - \frac{\sigma_{cq,t}}{q_t(1 + \tau_t^q)} \quad (3)$$

This is the income-effect gap evaluated at the prevailing consumer price of the dirty good. It is positive when the dirty good is normal since the marginal propensity to spend on it is proportional to $\mathcal{I}_t$ (Appendix C.3).

4 Optimal distortions

4.1 Planner's problem

First, following Kapička (2013)², we can write the planner's problem recursively. Appendix A.2 shows that the presence of externalities does not invalidate the approach. No individual report moves the aggregate pollution stock, so the agent's incentive problem is the one above with the pollution stock entering utility as a date-specific parameter. Aggregates are deterministic, so internalizing them appends a single new state, the emission window $\mathbf{S}_- = (Q_{t-s}, \dots, Q_{t-1})$ of lagged aggregate emissions feeding current and future pollution. The continuation value receives the updated window, with the current aggregate Q_t appended.

$$V_t(\hat{w}, \hat{w}_2, \theta_-, \mathbf{S}_-) = \min_{c, q, y, u, w, w_2, S} \int_0^\infty \left[c(\theta) + q(\theta) - y(\theta) + \frac{1}{R} V_{t+1}(w(\theta), w_2(\theta), \theta, \mathbf{S}) \right] f_t(\theta | \theta_-) d\theta$$

²Kapicka's consumption set is convex and compact (his Assumption 1), so consumption can be a vector rather than a scalar. This paper uses only the necessity direction, his Theorem 4, which is dimension-free and does not invoke his Assumption 5. That assumption, that U is affine in the consumption vector, carries the sufficiency direction, and he gives two routes to it: utility separable across the components and affine in the shock, $U(x, \theta) = U_1(x_1) + \theta U_2(x_2)$, which a non-separable $U(c, q, l, S)$ does not satisfy, or convexification by lotteries over consumption, which would replace the deterministic allocations solved here. Sufficiency is accordingly not assumed but verified numerically in Section 6.

$$\begin{aligned}
& \text{s.t.} \\
& \dot{u}(\theta) = U_\theta \left(c(\theta), q(\theta), \frac{y(\theta)}{\theta}, S \right) + \beta w_2(\theta) \\
& \hat{w} = \int_0^\infty u(\theta) \bar{\alpha}_t(\theta) f_t(\theta | \theta_-) d\theta \\
& \hat{w}_2 = \int_0^\infty u(\theta) f_{2,t}(\theta | \theta_-) d\theta \\
& u(\theta) = U \left(c(\theta), q(\theta), \frac{y(\theta)}{\theta}, S \right) + \beta w(\theta) \\
& S = S \left(\dots, \int q(\theta) f(\theta) d\theta \right)
\end{aligned}$$

The planner minimizes resource cost while offering an agent expected lifetime utility $\hat{w}$ with marginal continuation value $\hat{w}_2$, now internalizing what the assigned dirty good does to the pollution stock. $\mathbf{S}$ is the updated emission state, $\mathbf{S}_-$ with Q appended. $\bar{\alpha}_t(\theta)$ is the continuation Pareto weight attached to delivering utility to type θ . It equals the primitive Pareto weight in the initial period, $\bar{\alpha}_0(\theta) = \alpha(\theta)$, and $\bar{\alpha}_t(\theta) = 1$ for $t > 0$. As in Golosov et al. (2016), the lifetime Pareto weights enter only at date 0. From date 1 on, the weight collapses to one.

Two multipliers from this problem carry through the rest of the paper. $\lambda_{1,t}$ prices promise keeping, and the costate $\psi(\theta)$ prices the envelope condition, tied down at both ends of the type space by $\psi(0) = \psi(\infty) = 0$. Every wedge below is built on its normalized form $\psi U_{c,t}/(\theta f_t)$, the resources the planner gains by relaxing the local incentive constraint at a history. That object is zero wherever screening has no value, as in retirement (Appendix A.7).

4.2 Optimal distortions

Proposition 4.1 below describes the planner's optimality conditions in terms of labor supply, savings, and the split between the two goods.

Proposition 4.1 (Optimal wedges).

The labor wedge is

$$\frac{\tau_t^y}{1 - \tau_t^y} = A_t(\theta) B_t(\theta) C_t(\theta) + \beta R \frac{\tau_{t-1}^y}{1 - \tau_{t-1}^y} D_t(\theta)$$

$$\begin{aligned}
A_t(\theta) &= 1 + \epsilon_t(\theta) - \gamma_{c,t}(\theta) \\
B_t(\theta) &= \frac{1 - F_t(\theta)}{\theta f_t(\theta)} \\
C_t(\theta) &= \int_{\theta}^{\infty} \exp \left[\int_{\theta}^x \left(\sigma_t(\tilde{x}) \frac{\dot{c}_t(\tilde{x})}{c_t(\tilde{x})} + \sigma_{cq,t}(\tilde{x}) \frac{\dot{q}_t(\tilde{x})}{q_t(\tilde{x})} - \gamma_{c,t}(\tilde{x}) \frac{\dot{y}_t(\tilde{x})}{y_t(\tilde{x})} \right) d\tilde{x} \right] (1 - \lambda_{1,t} \bar{\alpha}_t(x) U_{c,t}(x)) \frac{f_t(x) dx}{1 - F_t(\theta)} \\
D_t(\theta) &= \frac{A_t(\theta) U_{c,t}(\theta)}{A_{t-1} U_{c,t-1}} \frac{\theta_{t-1} \int_{\theta}^{\infty} \exp \left[- \int_{\theta}^x \gamma_{c,t}(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] f_{2,t}(x) dx}{\theta f_t(\theta)}
\end{aligned}$$

Let $\widehat{U}_{c,t} \equiv U_{c,t}/(1 - \chi_t)$, with $\chi_t \equiv \frac{\gamma_{c,t}}{1 + \epsilon_t - \gamma_{c,t}} \frac{\tau_t^y}{1 - \tau_t^y}$ as the incentive credit. Savings satisfy the generalized inverse Euler equation

$$\frac{1}{\widehat{U}_{c,t}} = \frac{1}{\beta R} \mathbb{E}_t \left[\frac{1}{\widehat{U}_{c,t+1}} \right]$$

The carbon wedge departs from the social cost of carbon by

$$\frac{\tau_t^q - \text{SCC}_t}{1 + \tau_t^q} = (\gamma_{c,t} - \gamma_{q,t}) \left[B_t(\theta) C_t(\theta) + \beta R \frac{\tau_{t-1}^y}{1 - \tau_{t-1}^y} \frac{D_t(\theta)}{A_t(\theta)} \right] = \frac{\gamma_{c,t} - \gamma_{q,t}}{1 + \epsilon_t - \gamma_{c,t}} \frac{\tau_t^y}{1 - \tau_t^y} \quad (4)$$

with SCC_t characterized in Proposition 4.2.

Proof. See Appendices A.3, A.4, and A.5. □

The first formula is the standard ABCD labor wedge formula, as in Golosov et al. (2016) and Farhi and Werning (2013). None of the terms change, except for term C_t which represents the social value of the resources extracted from types above θ , where the exponential term captures the income effect induced on those types induced by the redistribution. This term now includes the cross-curvature between the two goods, such that the wedge rises all things equal when the goods are Edgeworth substitutes ($\sigma_{cq,t} > 0$) and falls when they are complements ($\sigma_{cq,t} < 0$). Intuitively, when the goods are substitutes, taking away the numéraire from these types renders the dirty good more pleasurable and allows the planner to extract more resources from them.

The second formula is the generalized inverse Euler condition, expressed as a reweighted inverse Euler condition similar to Hellwig (2021). When preferences are non-separable between labor and consumption, the cost of delivering utility is weighted by $1 - \chi_t$, where χ_t is the incentive credit. When the numéraire complements labor, a unit of consumption makes work less painful and shrinks the information rent that truth-telling requires, rebating χ_t from the delivery cost. Neither the second good nor the externality changes the structure of savings wedges.

The third formula describes how the planner allocates each good across types. It is the dynamic analogue to the optimal commodity wedge derived in Jacobs and Mooij (2015). The left-hand side is the commodity distortion net of the social cost of carbon, described in the next section, and the right-hand side is the labor distortion scaled by the relative labor complementarities of the two goods. When preferences are weakly separable, $\gamma_{c,t} = \gamma_{q,t}$ and the deviation vanishes, so the carbon wedge is exactly the social cost of carbon; with no externality this is the zero commodity wedge of Golosov et al. (2003). Otherwise, when one of the goods is relatively more complementary to labor than the other, the planner would like to distort the weaker complement to labor more heavily. Because the labor wedge is history dependent, so is the goods wedge, which then inherits both the shape and the time path of the labor wedge, bent by the ratio of relative labor complementarities and labor response elasticities. This is the dynamic version of the Corlett-Hague rule that states that the weaker complement of labor should be distorted more heavily since this acts as an indirect tax on leisure.

4.3 Social cost of carbon

The social cost of carbon in (4) is not the textbook Pigouvian price, but rather the discounted stream of marginal damages where each damage term is reweighted by how pollution interacts with labor distortions.

Proposition 4.2 (The social cost of carbon).

The shadow price of current emissions (13) is

$$\text{SCC}_t = - \sum_{j=0}^s \frac{1}{R^j} \mathbb{E}_t \int \left[1 + \left(\frac{\tau_{t+j}^y}{1 - \tau_{t+j}^y} \right) \frac{\xi_{t+j} - \gamma_{c,t+j}}{1 + \epsilon_{t+j} - \gamma_{c,t+j}} \right] \frac{U_{S,t+j}}{U_{c,t+j}} \Phi_j f_{t+j} d\theta$$

with $\Phi_j \equiv \partial S_{t+j} / \partial Q_t$ the effect of current aggregate emissions on the date- $(t+j)$ pollution stock ($\Phi_j = 0$ for $j > s$ by construction) and the expectation taken over future skill histories. The Pigouvian price of the same emission is the same stream with every bracket set to one,

$$\text{SCC}_t^P = - \sum_{j=0}^s \frac{1}{R^j} \mathbb{E}_t \int \frac{U_{S,t+j}}{U_{c,t+j}} \Phi_j f_{t+j} d\theta$$

and $\text{SCC}_t = \text{SCC}_t^P$ when $\xi_{t+j} = \gamma_{c,t+j}$ at every future history or when labor is undistorted.

Proof. See Appendix A.6. □

The bracket is a fiscal-adjustment factor, which is active whenever labor is distorted and pollution interacts with labor supply. Suppose that $\xi_{t+j}, \gamma_{c,t+j} > 0$. When $\xi_{t+j} > \gamma_{c,t+j}$,

pollution harm discourages labor supply more so than consuming more of the numéraire encourages it. Pollution then causes direct harm to utility and indirect harm through labor discouragement, both of which the planner would like to correct. Hence, the social cost of carbon rises above the Pigouvian level. When $\xi_{t+j} < \gamma_{c,t+j}$, pollution discourages labor less than consuming more of the numéraire encourages it, so the planner would like to underdistort the dirty good, shrinking the social cost of carbon. This correction operates on every term of the discounted stream of damages and is the dynamic extension of Jacobs and Mooij (2015), under different notation. The two statistics enter only through their difference, and that difference has a compact reading: $\xi_t - \gamma_{c,t}$ is the elasticity of the marginal rate of substitution U_S/U_c with respect to hours, so wherever labor is distorted the bracket exceeds one exactly when working more raises the household's valuation of a cleaner environment, measured in units of the numéraire.

4.4 Separation

Proposition 4.3 (Three-instrument separation).

Under weak separability, $\gamma_{c,t} = \gamma_{q,t}$, the corrective and redistributive margins separate.

$$\tau_t^q = \text{SCC}_t \quad \frac{\tau_t^y}{1 - \tau_t^y} = A_t B_t C_t + \beta R \frac{\tau_{t-1}^y}{1 - \tau_{t-1}^y} D_t \quad \frac{1}{\widehat{U}_{c,t}} = \frac{1}{\beta R} \mathbb{E}_t \left[\frac{1}{\widehat{U}_{c,t+1}} \right]$$

The carbon price carries no redistributive term: the income tax and the savings wedge do all the redistribution and insurance whereas the goods wedge is purely corrective.

Proof. Immediate from (4) at $\gamma_{c,t} = \gamma_{q,t}$. □

Corollary 4.1 (Dynamic Atkinson-Stiglitz).

With no externality, weak separability gives $\tau_t^q = 0$ and the goods wedge vanishes.

Proof. Proposition 4.3 at $\text{SCC}_t = 0$. □

Redistributing through an optimal nonlinear income tax separates externality correction from redistribution. The carbon price is set at the social cost of carbon, unlike a dynamic Ramsey planner with representative agents like that in Barrage (2020) who prices carbon below it. Douenne et al. (2026) recover an average near-Pigouvian price with linear instruments, whereas the separation here is exact at every history. However, the social cost of carbon may be below or above the Pigouvian level depending on whether a clean environment complements labor. Corollary 4.1 is the dynamic Atkinson-Stiglitz result of Golosov et al. (2003) and Proposition 4.3 generalizes the Mirrlees irrelevance result of Jacobs and Mooij (2015) to a dynamic setting.

5 Constrained optimum

The results so far let the government reach any incentive-compatible allocation, but real carbon taxes are set inside imperfect tax systems where the income tax may be more or less progressive than the dynamic Mirrlees optimum would prescribe. To model this restriction, I consider the planner problem in which labor and savings wedges are held at exogenous schedules. That is, of the incentive-feasible allocations, the planner may offer only those that price both margins at specific rates. This restriction only fixes marginal rates, so labor and saving still respond to a carbon reform since any change in the assigned bundles moves both through income effects.

The question of how to price carbon when the rest of the tax system is misset has some antecedents in the literature, but the exercise here differs in two ways. First, in a single-agent Ramsey model, Barrage (2020) computes optimal carbon taxes with the labor or the capital income tax held at its observed level, one margin at a time. Douenne et al. (2026) run the same one-margin freezes in a heterogeneous-agent Ramsey model. Because the model here features two goods, both income-side margins can be held at once and still leave the goods margin free. Second, the carbon margin is allowed to be nonlinear. The first difference makes the constraint stricter as it implicitly rules out using the carbon instrument to finance changes to the income tax system, such as the revenue-neutral swaps of British Columbia and Sweden, or to pay income-contingent transfers, such as France’s Chèque énergie. The second however is more permissive as instruments may be nonlinear such that whatever margin is free can do more to undo the inefficiency of the others. Since the planner chooses the best allocation the model allows, it implicitly recycles any carbon revenue optimally. The theory does not offer a revenue recycling rule, but the quantitative results of Section 6.6 suggest that the mechanism’s implicit recycling is close to a uniform lifetime dividend, consistent with the fee-and-dividend design of Canada’s now-repealed federal fuel charge and Switzerland’s CO₂ levy.

5.1 Best carbon-only reform

Suppose the labor wedge is held at a schedule $\bar{\tau}_t^y(\theta) < 1$, with Lagrange multiplier $\eta(\theta)f_t(\theta|\theta_-)$.

$$\Gamma(\theta) \equiv -\frac{U_l\left(c(\theta), q(\theta), \frac{y(\theta)}{\theta}, S\right)}{\theta U_c\left(c(\theta), q(\theta), \frac{y(\theta)}{\theta}, S\right)} - (1 - \bar{\tau}_t^y(\theta)) = 0 \quad (5)$$

Γ is the distance between the implemented labor wedge and the frozen schedule. Moreover, the savings wedge is held at $\bar{\tau}_t^s(\theta_t) < 1$, with Lagrange multiplier $\vartheta_{t+1}(\theta^t)f_t(\theta|\theta_-)$.

$$U_{c,t}(\theta^t) = (1 - \bar{\tau}_t^s(\theta_t)) \beta R \mathbb{E}_t[U_{c,t+1}(\theta^{t+1})] \quad (6)$$

Unlike the labor-wedge constraint, the savings-wedge constraint touches current and future consumption, so neither date's problem can enforce it alone. For a given marginal utility of current consumption, it requires tomorrow's marginal utility to average some value M . Today's consumption is also constrained by the inherited requirement $\hat{M}$ from yesterday's problem. This is needed because the planner can promise the same utility and marginal utility under two histories yet deliver them with different consumption paths. The planner's problem is that of Section 4.1 with wedge constraints (5) and (6), M in the continuation value, and state $\hat{M}$.

Lemma 5.1 (Recursive formulation with frozen wedges).

Neither constraint involves a reporting strategy or enters incentive constraints, so the first-order approach still applies. The savings-wedge constraint reaches the date- $(t+1)$ allocations only through $\hat{M}$, which enters as a new state with multiplier $-\vartheta_t$, such that $\vartheta_0 = 0$ and $\vartheta_{T+1} = 0$.

Proof. See Appendix B.1. □

Every first-order condition gains two terms, one for the requirement the date- t problem inherits and one for the requirement it passes forward (Appendix B.2). The two terms collect into a single shadow price of the intertemporal restriction:

$$\eta_t^s(\theta^t) \equiv \left[\vartheta_t - \frac{\vartheta_{t+1}(\theta^t)}{(1 - \bar{\tau}_t^s) \beta R^2} \right] U_{c,t}(\theta^t) \quad (7)$$

η_t prices the labor schedule and η_t^s the savings schedule. Both reach the carbon price through the net shadow price $\zeta_t \equiv \eta_t (1 - \bar{\tau}_t^y) - \eta_t^s$. η_t reaches it a second time, multiplied by the labor complementarity difference $\gamma_{c,t} - \gamma_{q,t}$ so that separability closes that channel.

Proposition 5.1 (Constrained-optimal carbon price).

Let $\mathcal{I}_t$ be the income-effect gap defined in equation (3), positive when the dirty good is normal, and let $\tau_t^{y,*}$ be the desired labor wedge. Then,

$$\frac{\tau_t^q - \text{SCC}_t}{1 + \tau_t^q} = \frac{\gamma_{c,t} - \gamma_{q,t}}{1 + \epsilon_t - \gamma_{c,t}} \frac{\tau_t^{y,*}}{1 - \tau_t^{y,*}} + (\gamma_{c,t} - \gamma_{q,t}) \frac{\eta_t}{y_t} - \zeta_t \mathcal{I}_t$$

The social cost of carbon keeps the usual form, with the adjustment term replaced by

$$1 + \left(\frac{\tau_{t+j}^{y,*}}{1 - \tau_{t+j}^{y,*}} \right) \frac{\xi_{t+j} - \gamma_{c,t+j}}{1 + \epsilon_{t+j} - \gamma_{c,t+j}} + (\xi_{t+j} - \gamma_{c,t+j}) \frac{\eta_{t+j}}{y_{t+j}} + \zeta_{t+j} \frac{\sigma_{t+j} + \xi_{c,t+j}}{c_{t+j}}$$

Proof. See Appendices B.9 and B.10, with Appendix A.7 for $\psi = 0$ at the retirement dates. □

Corollary 5.1 (Constrained-optimal carbon price with separable preferences).

Under weak separability or $\gamma_{c,t} = \gamma_{q,t}$,

$$\frac{\tau_t^q - \text{SCC}_t}{1 + \tau_t^q} = -\zeta_t \mathcal{I}_t$$

When the dirty good is normal, the carbon price is below (above) it where $\zeta_t > 0$ ($\zeta_t < 0$).

Proof. Proposition 5.1 at $\gamma_{c,t} = \gamma_{q,t}$. □

The first term in the carbon wedge expression is unchanged from the full optimum, except that the labor wedge is now the desired one. The second term is the same complementarity gap multiplied by the shadow price of the frozen labor schedule, per unit of earnings, η_t/y_t . The gap therefore prices two things where it priced one: what relaxing the incentive constraint is worth, which is the screening the planner wants, and what relaxing the frozen schedule is worth, which is the screening the schedule denies it. The third term subtracts the income-effect gap multiplied by ζ_t , the net shadow price of the two frozen wedges, and has no counterpart at the full optimum. Two objects therefore carry the freeze into the carbon price, the screening pair and ζ_t .

The second and third terms are the two halves of one derivative. Handing the agent a unit of either good moves the frozen hours margin twice: directly, because the good complements or substitutes for work, and again because it moves the marginal utility of the numéraire against which that margin is measured. The complementarity half is the second term. The marginal-utility half is the third, netted against the savings constraint because the frozen Euler equation binds on that same marginal utility. Weak separability closes the first half and leaves the second, which is Corollary 5.1.

Both halves answer the same problem, and they answer it in opposite directions. Take the labor freeze alone, at the calibrated signs: the dirty good normal, and the weaker complement of labor. Where the frozen schedule holds labor above the desired wedge, hours sit below where the planner wants them. Taxing the weaker complement harder is an indirect tax on leisure and pushes hours back up, which raises the carbon price. Shifting spending toward the good that displaces the marginal utility of the numéraire least raises that marginal utility and, with leisure normal, pushes hours up too, which lowers the carbon price. The net carries the sign of $\eta_t [(\gamma_{c,t} - \gamma_{q,t})/y_t - (1 - \bar{\tau}_t^y) \mathcal{I}_t]$.

The labor-wedge constraint prices a movement in $U_{c,t}$ at $\eta_t(1 - \bar{\tau}_t^y)$, because it displaces the labor supply margin, and the savings-wedge constraint prices it at $-\eta_t^s$. ζ_t is the net, and $\mathcal{I}_t$ is how far a dollar shifted between the goods moves $U_{c,t}$. At any history where the two prices are equal they cancel without either constraint being slack, and under Corollary 5.1 the carbon

price sits at the social cost of carbon there. The screening term of Proposition 5.1 survives that cancellation, and so does the modified damage weight wherever the multipliers are nonzero at future histories.

The damage weight carries the same two objects by a second route. The screening pair enters it weighted by $\xi_{t+j} - \gamma_{c,t+j}$ rather than by the complementarity gap. A unit of consumption moves the rate U_S/U_c at which damages are valued, in proportion by $(\sigma_{t+j} + \xi_{c,t+j})/c_{t+j}$, and under the freeze U_c is not the planner's to set, so that conversion carries the shadow price ζ_{t+j} too. It vanishes where consumption moves U_S and U_c in step. Setting $\eta_t = \eta_t^s = 0$ at every history returns both displays to Section 4.

Corollary 5.1 turns on ζ_t , which so far is only a pair of Lagrange multipliers and not something a tax authority would recognize. Neither display is a formula in primitives, no more than those of Section 4 are: τ_t^q appears on both sides and the damage weight carries the whole future allocation through ζ_{t+j} . Both characterize a fixed point. The next section gives the multipliers a reading, in statistics of the frozen margins.

5.1.1 Reading the multipliers

Both multipliers are pinned by conditions written in three statistics of the frozen labor margin.

$$\Gamma_{c,t} \equiv (1 - \bar{\tau}_t^y) \frac{\sigma_t}{c_t} - \frac{\gamma_{c,t}}{y_t} \quad \Lambda_t \equiv \frac{\Gamma_{c,t}}{1 - \bar{\tau}_t^y} \quad \Delta_t \equiv (1 - \bar{\tau}_t^y) \frac{\sigma_t}{c_t} + \frac{\epsilon_t - 2\gamma_{c,t}}{y_t} \quad (8)$$

Derived in Appendix B.3, these statistics measure how the labor wedge responds to a perturbation of the allocation. $\Gamma_{c,t} = \partial\Gamma/\partial c$ moves consumption alone at fixed labor supply and is positive when leisure is a normal good since extra consumption then raises the margin the agent requires to stay put. With the wedge frozen, labor supply falls to restore (5). Λ_t is $\Gamma_{c,t}$ rescaled into the units of the wedge ratio $\bar{\tau}_t^y/(1 - \bar{\tau}_t^y)$ so that the savings multiplier enters Lemma 5.2 in the same units as the labor wedge gap. Δ_t moves consumption and labor supply together along a utility-compensated path and carries the opposite sign to the second derivative of the agent's objective in labor supply, so $\Delta_t > 0$ is the second-order condition of her labor choice at the frozen wedge. I assume this throughout³.

Lemma 5.2 (The labor multiplier).

With $\tau_t^{y,*}$ the desired labor wedge of Proposition 5.1, what the planner's own labor condition returns at the constrained allocation, let $E_t \equiv \bar{\tau}_t^y/(1 - \bar{\tau}_t^y) - \tau_t^{y,*}/(1 - \tau_t^{y,*})$ be the labor gap,

³ $\Delta_t > 0$ is equivalent to a positive compensated earnings response, $s_{yy,t} = 1/((1 - \bar{\tau}_t^y) \Delta_t) > 0$. It fails only when the numéraire complements labor strongly enough that $2\gamma_{c,t}$ exceeds ϵ_t by more than $(1 - \bar{\tau}_t^y) \sigma_t y_t / c_t$, and the calibration of Section 6 clears it with room, at $\epsilon_t \approx 2$ against a $\gamma_{c,t}$ smaller by more than an order of magnitude. The progressive earnings-indexed schedule of Appendix B.13 weakens the requirement further, since Δ_t then gains p_t/y_t .

the distance at a history between the frozen wedge and the desired one. At every history with $t < T_R$,

$$\eta_t \Delta_t = E_t + \eta_t^s \Lambda_t$$

with Δ_t and Λ_t as in (8). Given η_t^s , the labor multiplier is pinned history by history. Since $\Delta_t > 0$, η_t is positive where the frozen schedule holds the labor wedge above the desired wedge corrected for the savings multiplier, negative where it holds it below, and zero where the two cross.

Proof. See Appendix B.4. □

The gap E_t is what the two multipliers have to account for, each scaled by how far the perturbation it controls moves the frozen labor margin. Both scales are properties of Γ : Δ_t for the labor multiplier, which moves consumption and labor together, and Λ_t for the savings multiplier, since the frozen Euler equation binds on consumption alone.

Zeroing one multiplier rather than both gives the single-freeze problems: at $\eta_t^s = 0$ the conditions describe a frozen labor schedule alone, and at $\eta_t = 0$ a frozen savings schedule alone, with the labor wedge solved rather than imposed.

Proposition 5.2 (Intertemporal condition).

Define the incentive credit and the savings-free delivery cost

$$\chi_t \equiv \gamma_{c,t} \frac{\psi U_{c,t}}{\theta f_t} - \eta_t \Gamma_{c,t} + \eta_t^s \frac{\sigma_t}{c_t} \quad \mathcal{G}_t \equiv 1 - \gamma_{c,t} \frac{\psi U_{c,t}}{\theta f_t} + \eta_t \Gamma_{c,t}$$

so that $1 - \chi_t = \mathcal{G}_t - \eta_t^s \sigma_t / c_t$, and define the desired savings wedge by $1 - \tau_t^{s,*} \equiv \mathcal{G}_t / \mathbb{E}_t[\mathcal{G}_{t+1}]$. At $\eta_t = \eta_t^s = 0$, χ_t collapses to the incentive credit of Proposition 4.1, which it generalizes. The generalized inverse Euler equation holds in the adjusted marginal utility $U_{c,t} / (1 - \chi_t)$,

$$\frac{1 - \chi_t}{U_{c,t}} = \frac{1}{\beta R} \mathbb{E}_t \left[\frac{1 - \chi_{t+1}}{U_{c,t+1}} \right]$$

and, read together with the frozen Euler equation, it pins the savings multiplier by a law of motion: for $t < T$, when the date- $(t+1)$ draw is degenerate,

$$\eta_t^s \frac{\sigma_t}{c_t} - (1 - \bar{\tau}_t^s) \eta_{t+1}^s \frac{\sigma_{t+1}}{c_{t+1}} = \mathcal{G}_{t+1} (\bar{\tau}_t^s - \tau_t^{s,*})$$

The level is not tied down here at all: it is tied down at the two ends of life by the endpoint conditions of Lemma 5.1, which (7) converts into conditions on η^s in Appendix B.6, where they are used. Away from a degenerate draw the date- $(t+1)$ terms enter as conditional expectations and the right-hand side carries an additional covariance term (Appendix B.5). Assume $\mathcal{G}_t > 0$ at every date; the desired wedge and the sign of the law of motion both need it, and nothing signs it in general.

Proof. See Appendix B.5. □

The intertemporal allocation is pinned by (6), so the inverse Euler equation no longer characterizes savings. It prices the restriction instead. Delivering utility through the numéraire costs $1/U_{c,t}$ in resources, discounted by $\gamma_{c,t} \frac{\psi U_{c,t}}{\theta f_t}$ when the numéraire complements labor. The labor schedule adds a second term: the same unit of consumption raises the margin $-U_l/(\theta U_c)$ by $\Gamma_{c,t}$, and with the wedge frozen, labor supply falls to restore (5). The planner values the induced fall at η_t , so where the schedule holds labor above the desired wedge and leisure is normal, consumption pushes hours further below where the planner wants them and delivery costs more. The savings schedule adds the third term by a different route: a unit of the numéraire moves $U_{c,t}$ itself, by σ_t/c_t in proportional terms, and $U_{c,t}$ is the object (6) constrains. Delivery is cheaper where $\eta_t^s > 0$ and dearer where $\eta_t^s < 0$, and Lemma 5.3 below forces both signs over the life cycle.

The law of motion differs from Lemma 5.2 in what the wedge gap determines. The gap is measured against a margin the mechanism itself delivers: setting $\eta_t^s = 0$ leaves $1 - \chi_t = \mathcal{G}_t$, so at a degenerate draw $\mathcal{G}_t/\mathcal{G}_{t+1}$ is the intertemporal margin the labor-freeze-only problem returns at this allocation. Like $\tau_t^{y,*}$, it is read off the constrained allocation and not off a counterfactual solve. A frozen labor wedge is a static restriction, so its gap sets η_t history by history. A frozen savings wedge restricts a ratio across dates, so its gap sets only the weighted *difference* of adjacent multipliers, while the level stays pinned at birth and death alone. Where the schedule sits at its desired counterpart at every date that difference vanishes, the recursion is homogeneous, and $\eta^s = 0$ solves it. Anywhere else the recursion is inhomogeneous between two fixed endpoints: a local gap can point the same way at every age, an accumulated one cannot.

Lemma 5.3 (Sign of the savings multiplier).

Fix a history. If $\eta_t^s \geq 0$ at every date along it, or $\eta_t^s \leq 0$ at every date along it, then $\vartheta_t = 0$ and $\eta_t^s = 0$ at every date along it. A schedule that leaves the multiplier alive anywhere therefore produces one that takes both signs over the life cycle. The argument uses only (7) with the two endpoint conditions of Lemma 5.1, and no first-order condition, so it holds whatever the labor schedule and whether or not the date- $(t + 1)$ draw is degenerate.

Proof. See Appendix B.6. □

The two results close on each other. By the law of motion at a degenerate draw, $\eta^s = 0$ forces $\bar{\tau}_t^s = \tau_t^{s,*}$ at every date, so the multiplier is alive at exactly the schedules that misprice the intertemporal margin. Away from that draw the covariance term can absorb a wedge gap instead, but wherever the multiplier is alive it changes sign, whatever the draw.

Both multipliers now have a reading, and ζ_t is their difference. Substituting one into the other separates them.

Lemma 5.4 (Decomposition of the net shadow price).

With E_t the labor gap of Lemma 5.2, for $t < T_R$,

$$\zeta_t = \frac{1 - \bar{\tau}_t^y}{\Delta_t} E_t - \frac{A_t - 1}{y_t \Delta_t} \eta_t^s \quad (9)$$

Since $\Delta_t > 0$ and $1 - \bar{\tau}_t^y > 0$, the coefficient on E_t is positive. Suppose $A_t > 1$ at every working age. The coefficient on η_t^s , taken with its sign, is then negative at every date of the life cycle and not only at those the display covers: $-(A_t - 1)/(y_t \Delta_t)$ during working life, and -1 in retirement, where $\zeta_t = -\eta_t^s$ (Corollary 5.3 below).

Proof. See Appendix B.8. □

Corollary 5.2 (The forced reversal and the free level).

Under the conditions of Lemma 5.4: (i) Fix a history. If η^s is nonzero at some date along it, the savings contribution to ζ_t is strictly positive at some dates along it and strictly negative at others: the multiplier takes both signs by Lemma 5.3, and a coefficient that never changes sign carries both through. (ii) No such restriction disciplines the labor contribution. Its sign profile is that of E_t , whatever profile the schedule delivers. (iii) Where the schedule overdistorts labor at every date, so that $E_t > 0$ throughout, the labor contribution is positive at every date, including at histories where η_t itself is negative. Whether raising the labor level shrinks the set of dates at which $\zeta_t < 0$ is not settled by the identity, which is read at one solved allocation and moves Δ_t , η_t^s and $\tau_t^{y,*}$ along with $\bar{\tau}_t^y$; Section 6.5.2 answers it numerically.

Proof. See Appendix B.8. □

The substitution is not a relabeling. In the raw form $\zeta_t = \eta_t (1 - \bar{\tau}_t^y) - \eta_t^s$ the savings-wedge constraint appears twice, once explicitly and once inside η_t by Lemma 5.2, so no part of the carbon-price deviation can be attributed to either margin. Equation (9) purges the implicit occurrence, and only in this form does each constraint enter through a single object. The freeze ladder of Section 6.4 attributes the two restrictions the same way, but it is a different operation and neither licenses the other: the identity splits ζ_t at the joint solution, both terms evaluated at the allocation the two constraints produce together, while the ladder differences separate solves and is order-dependent. That the two agree on what belongs to each margin is a check on both. What makes the identity collapse is that $\Gamma_{c,t}$ and Δ_t share the term $(1 - \bar{\tau}_t^y) \sigma_t / c_t$ exactly. The income effect on labor supply therefore cancels out of their difference and only labor-side curvature survives, as $A_t - 1 = \epsilon_t - \gamma_{c,t}$.

A schedule that holds the labor wedge too high throughout, paired with one that overdistorts saving, therefore gives a carbon price held below the social cost of carbon by an amount set by the labor gap and moved around that position by the savings multiplier, with the reversal

running from above to below as the cohort ages (Appendix B.7). That the multiplier reverses is draw-free; the order in which it does is not, since the ordering is read off the law of motion at a degenerate draw, which the random walk of Section 6 does not deliver over working life. The savings channel is a force rather than a guaranteed crossing: by (9) the deviation vanishes where the savings term *offsets* the labor term, not where η_t^s changes sign, so whether ζ_t itself reverses depends on the size of the savings swing against the labor level.⁴

The reversal is the commodity margin doing part of the savings margin's work, and the path of the carbon price is therefore an intertemporal price. A schedule that overdistsorts saving relative to its desired counterpart front-loads the consumption path the planner would otherwise assign, and the allocation answers by pricing the dirty good above the social cost of carbon early in life and below it late, leaning against the front-loading the frozen Euler equation enforces. That is why the savings contribution to the deviation of Corollary 5.1 runs from above to below as the cohort ages rather than holding one sign. Section 6.5 prices the repair.

Two things could break that reading. The first is the coefficient: a sign-changing multiplier scaled by a sign-changing coefficient need not contribute both signs, so $A_t > 1$ at working ages is what carries the savings reversal through.⁵

The second is the temptation to read the savings gap as the savings error. Only the labor channel is an "error" in the sense of a schedule sitting away from the desired wedge the constrained problem prices it against, and the distinction is not pedantic. At the calibration of Section 6 the savings *gap* is one-signed, since the frozen wedge exceeds $\tau_t^{s,*}$ at every age, while the savings *multiplier* is not, and reading the second as though it were the first replaces a reversal with a level shift and gets the late-life deviation backwards.

What the savings freeze does to the allocation has an exact aggregate measure. When the savings schedule does not vary with the type, taking unconditional expectations in (6) and iterating forward gives

⁴Equation (9) is a working-life statement, since ϵ_t is undefined once labor is zero and $\Gamma_{c,t}$, Λ_t and Δ_t all carry $1/y_t$. The retiree needs no decomposition, because $\eta_t = 0$ and $\zeta_t = -\eta_t^s$ by Corollary 5.3 below, where the labor term is gone and the savings multiplier is the whole deviation. The two regimes still meet: (40) shows the working-life coefficient tending to -1 as hours go to zero, so retirement's -1 is its limit rather than a separate case.

⁵Assumption 3.1 delivers only $A_t > 0$, and at the optimal rather than the constrained allocation. $A_t > 1$ is $\epsilon_t > \gamma_{c,t}$, the inverse Frisch elasticity above the numéraire's labor complementarity, which the calibration of Section 6 clears with room, $\alpha = 3$ giving $\epsilon_t \approx 2$ against a $\gamma_{c,t}$ smaller by more than an order of magnitude. The direction matters because retirement fixes the coefficient at -1 whatever the preferences, so only $A_t > 1$ signs the working-life coefficient the same way; $A_t < 1$ throughout would let the savings contribution keep one sign despite the multiplier's reversal. Under the earnings-indexed schedule of Appendix B.13, the requirement weakens to $A_t - 1 + p_t > 0$.

$$\mathbb{E}_0[U_{c,t}] = \prod_{j=0}^{t-1} \frac{1}{(1 - \bar{\tau}_j^s) \beta R} \mathbb{E}_0[U_{c,0}] \quad (10)$$

so the schedule pins the growth rate of the population mean of marginal utility at every date, whatever else the planner does, and leaves its level free. This is the allocation-side counterpart of the multiplier's level being tied down only at the two ends of life. At the calibration of Appendix D.1, $\beta R = 1$ and $\bar{\tau}^s \approx 0.009$: mean marginal utility then compounds at $\bar{\tau}^s/(1 - \bar{\tau}^s) \approx 0.9\%$ per period and rises by about seventy percent over the sixty-year life of the quantitative model. A frozen savings wedge does not so much distort the intertemporal margin date by date as bend the whole path of marginal utility. The carbon-price deviation of Corollary 5.1 prices that bending date by date (Appendix B.6 separates the accumulated level from the current increment).

5.1.2 Retirement

Retirement separates the two forces cleanly.

Corollary 5.3 (Retirement).

From the retirement date T_R of Section 3 on there is no labor to report, so $\psi = 0$ and (5) has no content, giving $\eta_t = 0$ and $\zeta_t = -\eta_t^s$. The retiree's carbon price satisfies

$$\frac{\tau_t^q - \text{SCC}_t}{1 + \tau_t^q} = \eta_t^s \mathcal{I}_t$$

whether or not preferences are weakly separable, since $\gamma_{c,t} = \gamma_{q,t} = 0$ mechanically at $l = 0$: the deviation is the savings multiplier alone. Corollary 5.6 below returns the retiree to a purely corrective price, but does so against an otherwise free income tax: what dies at T_R is the labor freeze, not the savings freeze.

Proof. See Appendix A.7 for $\psi = 0$, Appendix B.9 for the condition, and Appendix B.10 for SCC_t . Both are rederived at $l_t = 0$ rather than taken as a limit of Proposition 5.1. $\square$

The two freezes die differently. A miset labor schedule distorts the allocation of the numéraire across types within a date, so its damage dies with the labor margin. A miset savings schedule distorts the allocation of the numéraire across dates within a life, so it operates through retirement, where the desired savings wedge is zero and any nonzero frozen wedge keeps injecting error. Retirees' carbon price deviates from marginal damages for reasons that have nothing to do with screening their labor.

A history at which the two shadow prices cancel is observationally identical to one at the full optimum and has nothing to do with it. A carbon price sitting at the social cost of carbon

therefore does not certify that the income tax is set right. It certifies only that the two frozen margins are mispriced by offsetting amounts at that history, and the offset cannot hold at every age. In retirement there is no labor multiplier to do the offsetting, so $\zeta_t = -\eta_t^s$ by Corollary 5.3, and a savings multiplier vanishing at every retirement date would reduce the law of motion of Proposition 5.2, read at the retiree's $\tau_t^{s,*} = 0$, to $\bar{\tau}_t^s = 0$. Any nonzero capital income tax therefore breaks the cancellation somewhere.

5.1.3 What the freeze assumes and what it leaves free

In the quantitative exercise of Section 6, I take the income-side instruments to be the objects Appendix D.1 calibrates: differentiable labor income tax functions $T_t^y(y)$ and a linear capital income tax at rate τ^k on the interest component of saving so that households trade consumption across dates at the after-tax gross return $\bar{R} \equiv 1 + (R - 1)(1 - \tau^k)$. The carbon instrument is anything the government can add without touching these two objects. The instrument may be nonlinear and history-dependent, but may not condition on earnings, assets, or clean-good consumption. The following definition formalizes the class of instruments that are admissible under the freeze.

Definition 5.1 (Carbon-only reform).

A carbon-only reform is a sequence of dirty-good tariffs $\mathcal{T}_t(q^t)$, each measurable only in the household's purchase history $q^t = (q_0, \dots, q_t)$ and the date, but otherwise unrestricted. A tariff may be negative, nonlinear in the current purchase, and dependent on past purchases. This class contains the uniform tax, age-varying rates, block pricing, and any purchase-history-dependent schedule. It excludes instruments that condition on earnings, assets, or clean good consumption.

Lemma 5.5 (Carbon-only reforms with frozen income-side wedges).

Let a household facing $\{T_t^y\}$, τ^k , and any carbon-only reform choose both goods, hours, and saving at the return $\bar{R}$, with hours and saving interior at the dates where they are choices. Every optimum satisfies

$$-\frac{U_{l,t}}{\theta_t U_{c,t}} = 1 - T_t^{y'}(y_t) \quad \text{and} \quad U_{c,t} = (1 - \bar{\tau}^s) \beta R \mathbb{E}_t[U_{c,t+1}], \quad 1 - \bar{\tau}^s \equiv \frac{\bar{R}}{R}$$

The first applies at every working-life history and the second at every history. No tariff term enters either condition, which enters only the commodity margin where it can generate any wedge profile measurable in the purchase history and the date.

Proof. See Appendix B.11. □

A wedge is not a tax, and the lemma closes the gap between freezing the two. In one direction,

wedges are properties of allocations, invariant across the many tax systems that implement the same allocation, so constraints stated on wedges encode the system’s distortions without committing to the instruments behind them. The lemma supplies the other direction: within the class, no relabeling loosens the constraints, because nothing measurable in purchases and time enters the two frozen conditions. Nor is the constraint set larger than the class it stands for: the frozen-wedge constraints make the assigned bundle privately optimal on hours and saving at the statutory instruments, so the planner problem’s solution satisfies every first-order condition that membership of the class requires, and Appendix B.11 constructs the supporting tariff.⁶ The frozen-wedge constraints (5) and (6) are therefore equilibrium properties of every carbon-only reform, not assumptions laid beside the tax system.

Corollary 5.4 (No carbon-only reform reaches the full optimum).

Suppose the optimum of Section 4 carries a strictly positive savings wedge at some working-life history, as it does under skill risk (Goloso et al. 2003), and a zero savings wedge at retirement dates (Appendix A.7). Then no carbon-only reform implements it, at any constant τ^k , which is all the frozen instrument set allows: a nonzero rate fails against the zero retirement wedge, the zero rate fails against the positive working-life wedge, and no constant matches both. The labor margin fails independently wherever the optimal labor wedge is not measurable in earnings and age, the generic case under persistent shocks.

Proof. See Appendix B.11. □

The corollary answers a natural conjecture. Since many tax systems implement the same allocation, one might hope that a cleverly shaped carbon tax could reach the full optimum with the income taxes left where they stand, which would make the freeze vacuous. Within the class it cannot, and the obstruction is arithmetic rather than informational. The frozen Euler equation prices the intertemporal margin at one constant, while the optimum prices it at a path that is positive under risk and zero once risk resolves. The possibility returns only if the instrument conditions on earnings or assets: an energy rebate phased out with income, for instance, adds its phase-out rate to the effective labor wedge. Such an instrument is an income-tax reform under another name, excluded by the rigidity that motivates the freeze.

The two freezes do not have the same standing, and which results depend on which freeze is worth separating. The savings freeze is close to a description of markets. A household that trades freely at the after-tax return satisfies the Euler equation at that return whatever the planner intends, so the constraint would bind even if the planner wished it away; implementing the optimum’s intertemporal wedges instead requires the state-contingent wealth taxation of Kocherlakota (2005), an instrument no tax code carries, and the frozen Euler equation is the

⁶Its standing is that of the first-order approach maintained throughout the paper: a decentralized deviation that re-optimizes hours and saving alongside a misreport gains nothing to first order, and the global check is the one-shot audit of Section 6.

same feasibility fact that hidden borrowing and lending impose in Ábrahám and Pavoni (2008). That is why the restriction binds at $\tau^k = 0$ (Section 6.4): removing the capital income tax removes the mispricing, not the constraint.

The labor freeze has no market counterpart. A government could legislate a different income tax function, and holding it fixed is a statement about political rigidity. The asymmetry is worth stating precisely, because it does not run the same way for every result. What holds whatever the schedule, the forced reversal of Lemma 5.3 and the retiree's deviation of Corollary 5.3, rests on the freeze that markets enforce. What the political assumption carries is the labor gap, whose level and sign profile nothing disciplines (Corollary 5.2), and that gap is not a detail of the exercise: Section 6.5.2 re-runs the frozen world against labor anchors implied by Golosov et al. (2016), under which the labor channel reverses sign, nearly cancels the savings channel, and returns the best flat rate to the social cost of carbon.

One margin stays free in every rung of the exercise, and naming it keeps the exercise honest. The constraints fix marginal schedules; the planner still sets consumption levels history by history, and decentralized the levels are the tariff intercepts, the rebate side of a carbon package, conditioned on purchases rather than on earnings. The constrained optimum is therefore not the US system with a carbon tax bolted on: it is the best any tax system can do whose labor and savings distortions coincide with the US ones everywhere and whose new instrument prices the dirty good alone. The freedom is the realistic one, since actual carbon packages move rebates while leaving rate structures untouched. Rebates are optimal by construction, so when the flat tax at the social cost of carbon loses 0.115% of consumption in Section 6.5.2, none of the loss comes from a badly designed rebate; nor could transfers repair it, because they move the level of the marginal-utility path while the frozen Euler equation pins its growth (10). That is a statement about recycling *within* the class. Revenue that cuts marginal rates leaves the class, since it changes T_t^y , and Section 6.6.2 shows it reverses the sign of the same tax's welfare effect. Freezing the levels too, at the transfers the US system actually pays, is a positive question for a microsimulation rather than for a mechanism. Section 6.6 measures the free margin directly: the implied lifetime transfer each history receives, and the uniform dividend a flat carbon tax adds to it.

The class runs from the flat excise, which supports the flat-rate solves of Section 6, to the fully history-dependent wedge of Proposition 5.1, which no government would price off a lifetime purchase record. The welfare ladder of Section 6.5.2 should be read against that range.

5.2 A given carbon price

Real carbon prices are legislated rather than derived, anonymous rather than personalized, and set without reference to the income tax. The restriction that bites is anonymity: the instrument is one number where Section 5.1 priced a schedule in type. The wedge results below hold at any exogenous schedule, and anonymity is the member of that family flat in type, so the general

case comes first and the flat one where it earns its keep. This subsection prices the restriction against an income tax that is otherwise free, which is the mirror of the previous subsection rather than the realistic case. Section 5.3 sets the two exercises in one frame, and Section 6.5.2 imposes both restrictions at once. What the restriction costs turns on the cross-sectional dispersion of the history-specific wedge the planner forgoes.

The machinery mirrors the previous subsection. The planner holds the carbon wedge to an exogenous schedule $\bar{\tau}_t^q(\theta)$ and lets the income-side wedges respond. A common rate, the form every carbon tax in force approximates, is the member of that family flat in type; choosing that rate optimally rather than inheriting it is a second reading of the same problem. Both are read off the same multiplier. The configuration is Definition 5.1 with the roles exchanged: there the income-side instruments were data and the dirty-good tariff free, here the carbon price is data and the income-side margins free, and Lemma 5.6 below does for the given price what Lemma 5.5 did for the given income taxes, turning the instrument into a constraint on allocations.

The planner solves the problem of Section 4.1 subject to

$$\frac{U_q \left(c(\theta), q(\theta), \frac{y(\theta)}{\theta}, S \right)}{U_c \left(c(\theta), q(\theta), \frac{y(\theta)}{\theta}, S \right)} = 1 + \bar{\tau}_t^q(\theta) \equiv \bar{p}_t(\theta), \quad (11)$$

with multiplier $\eta^q(\theta)f_t(\theta|_{\theta_-})$. The constraint is static, so the multiplier is pinned by a static condition, and the regularity it needs is the second-order condition $\mathcal{D} < 0$ below, guaranteed by Assumption 3.1.

Three demand statistics organize the results, alongside the income-effect gap $\mathcal{I}_t$ and the normalized shadow price $\psi U_{c,t}/(\theta f_t)$ of the local incentive constraint. Let $\mathcal{D} \equiv U_{qq} - 2\bar{p}U_{cq} + \bar{p}^2U_{cc} < 0$ be the curvature of utility along the budget line. It is negative at whatever price the household faces, since the split sets $U_q = \bar{p}U_c$ and there $\mathcal{D} = -(2U_cU_qU_{cq} - U_c^2U_{qq} - U_q^2U_{cc})/U_c^2$, which Assumption 3.1 signs. Define

$$m_q \equiv \bar{p} \frac{\partial q^d}{\partial x} = \bar{p}^2 |s_{qq}| \mathcal{I}_t \quad s_{qq} \equiv \frac{\partial q^d}{\partial \bar{p}} \Big|_u = \frac{U_c}{\mathcal{D}} < 0$$

the marginal budget share of the dirty good and its compensated own-price response (Appendix C.3). Two more objects carry the price gap itself:

$$\varpi_t \equiv \frac{\bar{\tau}_t^q - \text{SCC}_t}{\bar{p}_t} \quad \text{and} \quad \varpi_t^* \equiv (\gamma_{c,t} - \gamma_{q,t}) \frac{\psi U_{c,t}}{\theta f_t}$$

ϖ_t is the screening component of the given price, the ad-valorem wedge net of the part that pays for damages. ϖ_t^* is the wedge the planner would set at that history if the instrument were

free, which is the commodity wedge of Proposition 4.1. The whole subsection turns on the gap between them.

When the schedule is flat in type, the restricted economy collapses to a one-good economy in expenditure.

Lemma 5.6 (Anonymity and the one-good reduction).

Suppose $\bar{p}_t(\theta) = \bar{p}_t$ for every type. An allocation satisfies (11) if and only if it assigns each type the private split of some pair (x, y) of expenditure and output at the common price. With indirect utility

$$\tilde{U}(x, l; \bar{p}) \equiv \max_{c, q \geq 0} U(c, q, l) \quad \text{s.t.} \quad c + \bar{p}q \leq x,$$

the split is unique since $\mathcal{D} < 0$, and incentive compatibility in the restricted economy is that of a one-good economy in (x, y) with period utility $\tilde{U}(x, y/\theta; \bar{p}_t)$. The envelope condition is unchanged,

$$\dot{u}(\theta) = -\frac{\tilde{U}_l l}{\theta} + \beta w_2(\theta) = -\frac{U_l l}{\theta} + \beta w_2(\theta)$$

Proof. See Appendix C.2. □

The reduction is what makes an anonymous carbon price tractable: the planner assigns expenditure and the agent splits it, so the two-good screening problem is a one-good screening problem in a reshaped utility. It fails for a schedule that varies with type, since then no single budget line rationalizes the assigned bundles. Section 6 solves the restricted mechanism through exactly this reduction.

Lemma 5.7 (The price of a given carbon wedge).

At every history,

$$\eta_t^q = \bar{p}_t |s_{qq,t}| (\varpi_t^* - \varpi_t)$$

The multiplier is the price gap converted into resources by the compensated substitution response: $\eta_t^q > 0$ where the schedule underprices the dirty good relative to the history's Corlett-Hague level, $\eta_t^q < 0$ where it overprices it, and $\eta_t^q = 0$ where the two cross.

Proof. See Appendix C.4. □

Histories at which substitution away from the dirty good is easy convert a given price gap into a large multiplier, which is the deadweight-loss weighting of the Corlett-Hague tradition. Every result below is this multiplier read at a different schedule.

Proposition 5.3 (Wedges under a given carbon price).

At every history the labor wedge and the savings adjustment are

$$\frac{\tau_t^y}{1 - \tau_t^y} = \frac{\tau_t^{y,*}}{1 - \tau_t^{y,*}} + \eta_t^q \bar{p}_t \left[\mathcal{I}_t - \frac{\gamma_{c,t} - \gamma_{q,t}}{(1 - \tau_t^y) y_t} \right] \quad \chi_t^q \equiv \gamma_{c,t} \frac{\psi U_{c,t}}{\theta f_t} - \eta_t^q \bar{p}_t \mathcal{I}_t$$

with the generalized inverse Euler equation holding in $U_{c,t}/(1 - \chi_t^q)$ and, to second order in the conditional fluctuation of $\ln U_{c,t+1}$ and to first order in χ^q (Appendix C.6),

$$\tau_t^s \approx \text{Var}_t(\ln U_{c,t+1}) + (\chi_t^q - \mathbb{E}_t \chi_{t+1}^q) + \text{Cov}_t(\chi_{t+1}^q, \ln U_{c,t+1})$$

The social cost of carbon keeps its discounted-damage form with the damage weight

$$1 + (\xi - \gamma_c) \frac{\psi U_c}{\theta f} + \eta^q \left[\bar{p} \frac{\sigma + \xi_c}{c} - \frac{\sigma_{cq} + \xi_q}{q} \right]$$

at each future history, where $\xi_{q,t} \equiv U_{S,q,t} q_t / U_{S,t}$ is the complementarity of the pollutant with the dirty good.

Proof. See Appendices C.5, C.6, and C.11. □

The labor wedge the proposition reports is the earnings-margin wedge alone, and it is not what the screening condition pins down. When an agent earns a marginal unit of output on a compensated path, the planner collects τ_t^y through the earnings margin and $\bar{\tau}_t^q dq^d/dy|_u$ through the commodity margin, where $dq^d/dy|_u \equiv (1 - \tau^y) \partial_x q^d + \partial_l q^d / \theta$ is the utility-compensated response of dirty-good demand to earnings; of the commodity part, $\text{SCC}_t dq^d/dy|_u$ only pays for the marginal damages the extra dirty consumption causes. Their sum net of that corrective piece

$$\Omega_t \equiv \frac{\tau_t^y}{1 - \tau_t^y} + \frac{\bar{\tau}_t^q - \text{SCC}_t}{1 - \tau_t^y} \frac{dq_t^d}{dy} \Big|_u = \tilde{A}_t \frac{\psi U_{c,t}}{\theta f_t} \quad (12)$$

is the comprehensive wedge the mechanism sets, the dynamic counterpart of the effective labor wedge of Jacobs and Boadway (2014), and $\tilde{A}_t$ is built on composite statistics that average the two goods' curvatures by their marginal expenditure shares (Appendix C.7). The comprehensive wedge inherits the full $ABCD$ structure of Proposition 4.1 in those composites (Appendix C.8), and Proposition 5.3 then dictates how Ω_t splits between the income-tax wedge and the commodity-revenue term.

The correction to the income-tax wedge has two parts of opposite sign. The expenditure part $(1 - \tau^y) \partial_x q^d$ is positive for a normal good. The labor part $\partial_l q^d / \theta$ is the statistic of Christiansen (1984), negative when the dirty good is a labor substitute, since working more

at fixed expenditure shifts spending away from it. When the labor-substitute part dominates, discouraging labor is fiscally cheaper than it looks, because the agent who works less spends more on the taxed good and part of the forgone output returns as commodity revenue.

Under weak separability the whole apparatus collapses to a single statistic.

Corollary 5.5 (The income tax collects the carbon the carbon price misses).

Under weak separability, $\gamma_{c,t} = \gamma_{q,t}$, we have $\varpi_t^* = 0$, so $\eta_t^q \bar{p}_t = -\bar{p}_t^2 |s_{qq,t}| \varpi_t$ and the corrections collapse to the marginal budget share times the uncollected carbon price:

$$\frac{\tau_t^y}{1 - \tau_t^y} = \frac{\tau_t^{y,*}}{1 - \tau_t^{y,*}} + m_{q,t} \frac{\text{SCC}_t - \bar{\tau}_t^q}{\bar{p}_t} \quad \chi_t^q = \gamma_{c,t} \frac{\psi U_{c,t}}{\theta f_t} - m_{q,t} \frac{\text{SCC}_t - \bar{\tau}_t^q}{\bar{p}_t}$$

A carbon price given at the social cost of carbon is harmless: $\bar{\tau}_t^q = \text{SCC}_t$ gives $\varpi_t = 0$, hence $\eta_t^q = 0$, and returns every wedge to its Section 4 form. A price given below it raises the labor wedge above the desired level; one given above it lowers the labor wedge.

Proof. See Appendix C.5. □

The mechanism is embedded carbon. A marginal compensated dollar of earnings is spent, a fraction $m_{q,t}$ of it on the dirty good, and each of those units carries uncollected marginal damage $(\text{SCC}_t - \bar{\tau}_t^q)/\bar{p}_t$. The income tax adds that charge, taxing the carbon content of marginal spending that the carbon price misses. The savings side applies the same charge intertemporally: delivering consumption at date $t + 1$ rather than t carries the uncollected damage of the extra dirty consumption it drags along, so the savings wedge rises where underpricing worsens over the life cycle and falls where pricing improves. A legislated price that stays flat while the social cost of carbon grows is the worsening case. Both corrections persist into retirement, since the price gap is a within-period object that never expires. This is the Mirrleesian counterpart of Hänsel et al. (2022), who find that a mandated uniform carbon tax reshapes the optimal income tax; here the reshaping has a closed form, and it is the marginal budget share times the price gap.

5.2.1 When the planner chooses the common rate

Suppose now that the price is anonymous but not inherited: the schedule is flat, $\bar{p}_t(\theta) = \bar{p}_t$, and the planner chooses its level at every date. This is the problem a government solves when it may legislate the carbon tax but not personalize it, and it is the configuration every carbon tax in force approximates.

Proposition 5.4 (The optimal anonymous price).

The optimal level sets the population average of the multiplier to zero at every date, $\mathbb{E}_0[\eta_t^q] = 0$. Equivalently, the screening component is the substitution-weighted average of the desired history-specific wedges,

$$\varpi_t = \frac{\mathbb{E}_0[|s_{qq,t}| \varpi_t^*]}{\mathbb{E}_0[|s_{qq,t}|]}$$

and the optimal price splits additively into a corrective and a screening part,

$$\tau_t^q = \text{SCC}_t + \bar{\tau}_t^{\text{screen}} \quad \frac{\bar{\tau}_t^{\text{screen}}}{1 + \tau_t^q} = \frac{\mathbb{E}_0\left[|s_{qq,t}| (\gamma_{c,t} - \gamma_{q,t}) \frac{\psi U_{c,t}}{\theta f_t}\right]}{\mathbb{E}_0[|s_{qq,t}|]}$$

To second order in the dispersion of ϖ_t^* , the date- t resource-equivalent loss from charging this common rate rather than the history-specific schedule of Proposition 4.1 is

$$\text{Loss}_t \approx \frac{1}{2} \frac{\bar{p}_t^3}{1 + \text{SCC}_t} \mathbb{E}_0[|s_{qq,t}|] \text{Var}_t^{|s|}(\varpi_t^*)$$

with $\text{Var}_t^{|s|}$ the $|s_{qq,t}|$ -weighted cross-sectional variance over date- t histories, minimized at the level above. The $1 + \text{SCC}_t$ in the denominator converts the ad-valorem wedge into its screening component, $d\bar{\tau}^q = \bar{p}^2 d\varpi / (1 + \text{SCC}_t)$, and drops out in the no-externality economy.

Proof. See Appendices C.9, C.11, and C.10. □

A single instrument averages the schedule the planner would set history by history, weighting each by how easily the consumer can substitute away from the dirty good there.⁷ The condition is implicit: the weights and the history-specific wedges are themselves evaluated at the restricted allocation, so the level is characterized as a fixed point rather than in closed form. The cost of not personalizing is the substitution-weighted cross-sectional variance of the wedge the planner forgoes. Because that wedge tracks the labor wedge by Proposition 4.1, its dispersion is first-order in the dispersion of τ^y , and what the tracking is worth is what anonymity gives up. Every object in the formula is a demand-side or tax-side statistic, so this is a sufficient-statistics answer to the cost-of-mispricing question. When the income tax is set optimally alongside the price, the dispersion the formula prices is that of the Corlett-Hague term alone, which is what the near-sufficiency of a flat carbon price in Section 6 looks like. Whether the same accounting survives a frozen income tax is not settled here. The loss formula is derived with the income-side wedges free, and against a freeze the schedule a flat rate forgoes is the whole deviation of Proposition 5.1, swept across ages by the labor gap and forced through both signs

⁷The construction parallels the many-person Ramsey rule of Diamond (1975) in collapsing a heterogeneous population into a single linear-tax condition: there the optimal excise rule, here $\mathbb{E}_0[\eta_t^q] = 0$. The weighting differs in kind. Diamond aggregates through the distributional characteristic, the covariance of demand with the net social marginal utility of income; here the weights are pure compensated substitution responses, with the redistributive content carried inside the history-specific wedges being averaged rather than in the weights.

by Lemma 5.3, so its dispersion has no reason to be small. That is a conjecture the frozen-world solves of Section 6.5.2 bear out and this section does not prove.

Corollary 5.6 (When the screening component vanishes).

The screening component of the optimal anonymous price is zero, and the price is exactly Pigouvian, whenever $\varpi_t^* = 0$ across the date- t cross-section. Two cases deliver this. Under weak separability, $\gamma_{c,t} = \gamma_{q,t}$ pointwise, so $\varpi_t^* = 0$ at every history and the restricted optimum coincides with the full optimum of Proposition 4.3, where the carbon price is the social cost of carbon. From the retirement date T_R on there is no labor to report, so $\psi = 0$ and again $\varpi_t^* = 0$: the screening component of the carbon price dies with the labor margin and retirees face pure Pigouvian pricing. Both cases are read against free income-side wedges; freeze the savings margin as well and the retiree's price carries the deviation of Corollary 5.3 on top. More generally the restriction is costless at date t if and only if ϖ_t^* is constant across the date- t cross-section, since only its dispersion enters the loss.

Proof. See Appendix A.7 for $\psi = 0$; the rest is immediate from Proposition 5.4. $\square$

The additive split is a statement about which margin the restriction binds. The corrective part of the optimal dirty-good price was uniform already at the full optimum, since a ton of emissions carries the same social cost whoever emits it and the emissions margin cannot screen. What anonymity constrains is only the Corlett-Hague screening part, forced from a history-dependent schedule down to a substitution-weighted scalar. A legislated uniform carbon price therefore sacrifices nothing on the corrective margin, and the whole cost of the restriction falls on screening, which the loss formula measures. The level of the social cost of carbon is nonetheless co-determined with the constrained allocation, its damage weight carrying the multiplier history by history through Proposition 5.3; this reprices carbon under the restriction but is not a loss on the corrective margin, which has no cross-sectional dispersion to forgo. This is the separation of Section 4.4 restated for the instrument governments actually use.

Between the two readings of this subsection lies the whole family. A price given at the social cost of carbon leaves every wedge at its Section 4 value. A price given anywhere else routes the uncollected damage into the income tax at the rate Corollary 5.5 sets. And a price chosen anonymously but optimally sits at the zero-average point, $\mathbb{E}_0[\eta_t^q] = 0$, where the errors cancel in the cross-section without vanishing history by history.

5.3 Mirrlees-Ramsey bridge

The section's exercises assemble into a bridge between the two traditions the literature review set against each other. Each margin now admits three readings. Left free, it carries its Mirrleesian wedge. Given flat with the level chosen optimally, it is a linear instrument of the Ramsey kind. Given anywhere else, the margin is frozen, and the multiplier's profile prices

the error. Only the middle reading is still missing on the income side: Proposition 5.4 supplied it for the carbon price, and the following is its counterpart for the labor and savings rates.

Proposition 5.5 (Optimal uniform schedules).

Let both frozen schedules of Section 5.1 be uniform, $\bar{\tau}_t^y(\theta) = \bar{\tau}_t^y$ and $\bar{\tau}_t^s(\theta) = \bar{\tau}_t^s$. The best uniform schedules satisfy

$$\mathbb{E}_0[\eta_t] = 0 \quad \text{and} \quad \mathbb{E}_0[\vartheta_{t+1} U_{c,t}] = 0$$

Equivalently, for the labor wedge,

$$\frac{\bar{\tau}_t^y}{1 - \bar{\tau}_t^y} = \frac{\mathbb{E}_0 \left[s_{yy,t} \left(\frac{\tau_t^{y,*}}{1 - \tau_t^{y,*}} - \eta_t^s \Lambda_t \right) \right]}{\mathbb{E}_0[s_{yy,t}]} \quad s_{yy,t} \equiv \frac{\partial y}{\partial (1 - \bar{\tau}_t^y)} \Big|_u = \frac{1}{(1 - \bar{\tau}_t^y) \Delta_t}$$

with $s_{yy,t}$ the compensated response of earnings to the net-of-wedge rate, holding the commodity split fixed (Appendix B.3), and $\mathbb{E}_0[\eta_t^s] = \mathbb{E}_0[\vartheta_t U_{c,t}]$ for the savings rate; that expectation is guaranteed to vanish only at $t = 0$.

Proof. See Appendix B.12. □

The uniform labor wedge is the compensated-earnings-response-weighted average of the desired Mirrleesian wedges, net of the savings multiplier at each history. It mirrors the optimal anonymous carbon price of Proposition 5.4, with s_{yy} in place of $|s_{qq}|$: a single rate averages the schedule the planner would set history by history, weighting each by how elastically its earnings respond. Even at the optimal uniform pair, ζ_t is generically nonzero history by history, so the carbon-price deviations of Corollary 5.1 do not vanish.

Optimality on all three margins therefore takes the same form, a zero-average condition,

$$\mathbb{E}_0[\eta_t^q] = 0 \quad \mathbb{E}_0[\eta_t] = 0 \quad \mathbb{E}_0[\vartheta_{t+1} U_{c,t}] = 0$$

The parallel is exact on the two static margins, where the object averaging to zero is the multiplier on the wedge constraint itself. The savings condition is the same idea one step removed: the rate controls only the requirement passed forward, so what averages to zero is the passed-forward multiplier weighted by current marginal utility, while the per-history price η_t^s averages to $\mathbb{E}_0[\vartheta_t U_{c,t}]$, zero at birth and drifting thereafter. Margin by margin, the optimal linear instrument is the zero-average point of the frozen-wedge family, and the full optimum of Section 4 is the point where every multiplier vanishes identically.

6 Quantitative analysis

How large is the Mirrleesian correction to the carbon price, how much welfare does it deliver over a flat carbon tax, and how much of the answer survives an income tax the planner did not choose? I answer all three by solving the model's optimal mechanism globally, calibrated to US data, rather than evaluating the wedge formulas at borrowed sufficient statistics.

The exercise runs in two worlds that differ in one respect. In the first the labor and savings wedges are set optimally alongside the carbon price, which is Section 4. In the second they are frozen at the schedules the US tax system implies, which is Section 5.1. Everything else is held fixed: the same preferences, the same calibration, the same solver family, and the same simulated skill histories. Nearly every result below is a comparison across those two worlds, and the contrast is sharper than the level of either; Section 6.6 adds a third comparison, inside the frozen world, across what the carbon revenue funds.

The solver extends the dual-costate recursion of Farhi and Werning (2013) and Golosov et al. (2016) to two consumption goods, a clean numéraire and a polluting energy good, and prices the externality as a constant marginal damage e entering the planner's resource constraint, so the social producer price of energy is $1+e$. Two assumptions take the social cost of carbon of Section 4.3 to that constant. First, the cohort is small relative to the world whose damages e measures: its emissions carry the full global social cost but do not move the world pollution stock, so nothing feeds back into its own utility and no pollution-stock state is carried. Second, damages are denominated in consumption units rather than utils, so a unit of energy costs $1+e$ out of the resource pool that finances consumption. This is exactly the restriction $\xi_t = \gamma_{c,t}$ of Proposition 4.2: the fiscal-adjustment bracket is one at every history, the social cost of carbon equals its Pigouvian counterpart, and the benchmark is $\tau^q = \text{SCC}^P = e$. The same specification makes the extra damage weight of Proposition 5.1 inert in the frozen world, since consumption-unit damages give $\xi_{c,t} = -\sigma_t$ and the term $\zeta_t(\sigma_t + \xi_{c,t})/c_t$ vanishes identically whatever the net shadow price does. That is what lets the frozen world be benchmarked against the same e .

Both are scope restrictions rather than results. The social cost of carbon that Proposition 4.2 derives is a within-cohort object, pricing the damage this cohort's emissions do to this cohort over this cohort's horizon. The number policy uses is not that object: the EPA's \$190 prices damages to people outside the cohort and beyond its life, which a single-cohort model cannot generate. Rather than derive a smaller and less relevant number, I import the policy one and hold it fixed. This splits the carbon price cleanly in two. Its level comes from the integrated-assessment literature and is exogenous here; its structure over the life cycle and across skill histories is endogenous, and is what this section measures. Every result below is a ratio to e or a deviation from it, so rescaling the social cost of carbon rescales the levels and leaves the structure intact. The pollution-stock dynamics of Section 4.3 are a separate exercise.

6.1 Model, solvers, and calibration

The life-cycle block is Farhi and Werning (2013)’s, replicated and validated before the two-good extension. Agents live sixty years, work for forty and retire for twenty, so $T = 59$ and $T_R = 40$, and period t is age $25+t$: labor-market entry is at 25, the last working year is 64, and retirement runs from 65 to 84. I report ages throughout; log-skills follow a geometric random walk with innovation variance $\hat{\sigma}^2 = 0.0095$; $\beta = 1/R = 0.95$, so $\beta R = 1$; skills are private information and the incentive constraints are handled by the first-order approach. The planner is utilitarian. Following Farhi and Werning (2013), newborns are ex ante identical, so the initial cross-section is the first innovation of the random walk: the initial skill dispersion is already the $\hat{\sigma}^2 = 0.0095$ above, there is no separate initial-skill variance to set, and the common initial promised value is pinned by the economy’s resource constraint. Each period the agent consumes a clean numéraire and an energy good (motor fuel, electricity, gas, fuel oil), both with producer price one. Period utility is

$$U(c, q, l) = \frac{1}{\rho} \log[(1 - b_q) c^\rho + b_q e^{-\kappa l^\alpha} (q - \bar{q})^\rho] - \frac{l^\alpha}{\alpha} \quad \rho \equiv 1 - \frac{1}{\varsigma}$$

a CES aggregate over the clean good and energy net of subsistence $\bar{q}$, inside the log shell that preserves Farhi and Werning (2013)’s balanced-growth homogeneity, with isoelastic labor disutility of curvature $\alpha = 3$ (Frisch elasticity one half). ς is the elasticity of substitution between the goods, b_q the energy weight, and $\bar{q} = 0$ in the homothetic benchmark; $\bar{q} > 0$ is the Stone-Geary necessity variant. The twist $e^{-\kappa l^\alpha}$ makes energy a relative leisure complement: its utility weight falls with labor effort, so $U_{ql} \neq 0$ with the sign, in the notation of Section 3, that makes the complementarity gap $\gamma_c - \gamma_q$ positive. This is the empirically relevant case (West and Williams (2007); Christiansen (1984)), and the one that pushes the optimal carbon tax above Pigouvian. The pollution stock does not appear in U because damages are denominated in consumption units: writing them as $U(c - D(S), q, l)$ gives $U_S/U_c = -D'$ and $U_{Sl}/U_S = U_{cl}/U_c$, so $\xi_t = \gamma_{c,t}$ holds identically, the fiscal-adjustment bracket of Proposition 4.2 is one at every history, and SCC_t coincides with its Pigouvian counterpart throughout this section; every derivative above is unchanged.

The two-good calibration is disciplined by four US moments (Table 1), two of which bind in the homothetic benchmark. The energy expenditure share is 7%, the round benchmark spanning the Consumer Expenditure Survey’s 7.6% in 2022 and 6.5% in 2023; motor fuel is half the composite and its price fell sharply between the two years. The own-price elasticity of energy is 0.4, which at a 7% share implies a substitution elasticity $\varsigma \approx 0.35$; I use $\varsigma = 0.4$, so the model’s own-price elasticity at the calibrated share is 0.44. This is a long-run value, the horizon a forty-year model asks for, and a conservative one: Labandeira et al. (2017) put the long-run elasticity of aggregate energy demand at about 0.6, against a short-run value roughly one third as large, and the welfare numbers scale with it. The income elasticity of energy is 0.45,

below one because energy is a necessity. This is a cross-sectional Engel gradient rather than a long-run elasticity, and it sits inside but well below the top of the range in the residential-energy meta-analyses, which report income elasticities from about 0.15 in the short run to near unity in the long run (Espey and Espey 2004; Zhu et al. 2018). Nothing in the paper’s headline turns on it: the homothetic benchmark sets it to one, and only the Stone-Geary variant imposes 0.45. The share moment pins the small CES energy weight b_q , and under Stone-Geary the income-elasticity moment implies a subsistence level $\bar{q} \approx 0.04$, about 57% of mean energy spending, so more than half of energy consumption is price-inelastic at the mean. The leisure complementarity $\varkappa$ is set so that, with the externality off, the full-optimum energy wedge at mid-career is about 10% ad valorem. West and Williams (2007) put the zero-externality gasoline tax at 33% of the pre-tax price and attribute 13 points of it to nonseparability, which is the part $\varkappa$ parameterizes; 10% is that figure tempered for the household-energy composite, whose labor cross-effects are weaker than motor fuel’s.⁸ The marginal damage is $e = 0.494$: the EPA’s central social cost of carbon of \$190 per ton of CO₂ (U.S. Environmental Protection Agency 2023), in 2020 dollars at a 2% near-term Ramsey discount rate for emissions in 2020, times the carbon intensity of household energy spending, 0.0026 tons per producer dollar.⁹ Because energy’s producer price is one, the Pigouvian wedge $\tau^q = e$ is a 49% ad valorem tax on energy.

The frozen schedules of the second world are the ones calibrated in Appendix D.1: the Heathcote et al. (2017) tax function at $p = 0.181$ for the labor margin, and a capital income tax at the US effective marginal rate $\tau^k = 0.18$ for the savings margin, which by (59) implies $\bar{\tau}^s = 0.009$ at $\beta R = 1$. Because the carbon price turns out to be far more sensitive to the savings schedule than to the labor schedule, Section 6.4 solves the model across a grid of τ^k running up to the 0.36 of the aggregate effective-rate measures; everything else in this section is at the benchmark.

Three solvers deliver the results, and each is anchored on the one before it. The *full optimum* assigns full consumption bundles, so the commodity wedge is unrestricted; this solver requires homothetic preferences. *Linear commodity taxation* layers an anonymous rate τ^q on the optimal nonlinear income and savings mechanism through the exact one-good reduction of Lemma 5.6, which is valid with subsistence and hence usable for the necessity calibration. The *frozen-world stack* adds the two frozen schedules, with the marginal-utility requirement $\hat{M}$ of Lemma 5.1 carried as a third state, and runs either with the commodity margin free or with it restricted

⁸They estimate a four-good demand system in gasoline, male leisure, female leisure, and other goods, so there is no home-energy counterpart to extrapolate from, and they caution against extrapolating to goods used differently from gasoline. The tempering is therefore an assumption, defensible only in that it errs toward understatement. Their sign is not unconditional either: dropping state fixed effects reverses gasoline from a leisure complement to a substitute, which would flip $\varkappa$ and with it the sign of the premium.

⁹The intensity aggregates 2023 emission factors and pre-tax prices for motor fuel, electricity, and natural gas; the unrounded spending-weighted average is nearer 0.0027, which would put e at about 0.51. I round down, so the benchmark is conservative, and e scales one-for-one with this coefficient. Pairing a social cost in 2020 dollars with an intensity whose denominator is a 2023 dollar understates e by a further margin, in the same direction.

Table 1: Calibration

parameter	value	moment or source
<i>life-cycle block (Farhi and Werning 2013)</i>		
discount factor $\beta = 1/R$	0.95	FW13 ($\beta R = 1$)
labor-disutility curvature α	3	Frisch elasticity 0.5
log-skill innovation variance $\hat{\sigma}^2$	0.0095	US male wages, via FW13
working / retirement periods	40/20	FW13
<i>two-good block</i>		
energy expenditure share	0.07	Consumer Expenditure Survey, 2022–23
own-price elasticity of energy	0.4	long run, energy-demand meta-analyses
income elasticity of energy	0.45	Engel gradient; unity in the homothetic benchmark
leisure complementarity $\varkappa$	0.25	mid-career zero-externality wedge $\approx 10\%$ (West and Williams 2007)
marginal damage e	0.494	\$190/tCO ₂ , 2020\$ (U.S. Environmental Protection Agency 2023) $\times 0.0026$ tCO ₂ /\$
<i>frozen schedules, Appendix D.1</i>		
labor progressivity p	0.181	Heathcote et al. (2017)
capital income tax τ^k	0.18	effective marginal rate, Burnham and Carloni (2022)
implied savings wedge $\bar{\tau}^s$	0.9% per year	(59) at $\beta R = 1$

to a flat rate. Two agreements discipline the stack. At $\gamma_c = \gamma_q$ the full optimum sets $\tau^q = e$ at every history and every age, the dynamic Atkinson-Stiglitz result, and coincides with a flat Pigouvian tax to solver precision. With the frozen schedules set at their desired counterparts the frozen-world solver returns the full optimum and its multipliers vanish, which is the $\eta_t \equiv \eta_t^s = 0$ limit of Section 5.1. All welfare comparisons use common random numbers, over 300,000 simulated histories for the full-optimum ladder and 60,000 for the frozen-world solvers, so within each comparison the same agents face every regime, and every regime pays for its own emissions in the planner’s budget at the social price $1 + e$ whatever the consumer price: no rung is flattered by unpriced damages. Flat rates are optimized by a bracketing search with the solver’s discretization and the simulated histories held fixed across rates, confirmed by a solve at the fitted optimum. Under the frozen income tax this gives $\tau^{q*} = 0.121$ at the benchmark capital tax and 0.087 at $\tau^k = 0.36$, each with a standard deviation of about 0.002 across simulation panels. Appendix D.2 documents the implementation and Appendix D.3 the

validation battery.

The incentive compatibility of the first-order-approach solution is verified ex post by a one-shot-deviation audit at simulated states, run on the full-optimum benchmark, on the Stone-Geary and placebo calibrations, and on the frozen-world stores. The frozen-world production grid is far coarser than the full optimum's, so its audit is read against an unconstrained solve at the identical grid, whose menu-interpolation noise bounds what the audit can resolve there: the frozen-wedge solutions show no deviation gains in excess of that control, and the retirement-junction rows, where the continuation value is closed-form and the audit is sharpest, are clean in every store. Appendix D.4 tabulates the audit.

6.2 The carbon wedge over the life cycle

Figure 1 is the paper's central quantitative fact, and the two curves in it are the same object computed under the same calibration.

When the income tax is set optimally, the carbon wedge barely leaves the social cost of carbon. At labor-market entry the labor wedge is near zero, there is no incentive motive yet, and the carbon wedge opens within a point of the Pigouvian level $e = 0.494$. It rises concavely to about 0.63 near retirement, a premium over the Pigouvian gross price that tops out around 9%, and at retirement it falls back to exactly 0.494: with no labor to report the costate vanishes, the screening term dies with it, and retirees face pure Pigouvian pricing (Appendix A.7). The whole path lies within a tenth of the social cost of carbon.

When the income tax is instead the one the US actually levies, the wedge starts in the same place and then rotates away from it. It opens at 51%, less than two points above the Pigouvian level, crosses it at 28, falls to 22% by 45 and -8% at 64, and reaches a 34% energy *subsidy* at 84. That is a rotation through 85 points of ad-valorem price, against the 14 points the full-optimum wedge covers over the same sixty years, and its working-life mean is 23%, about half the Pigouvian level. The freeze does not move the level at which the carbon price starts; it bends the whole path.

The premium in the first world is a scaled labor wedge (Table 2). Its first-order value is $\kappa l^\alpha \tau^y / (1 - \tau^y)$, and this identity tracks the solved premium to within 2.1% at every age. The identity sits just below the solved premium throughout, by about 1% through mid-career and by 2.1% at 64, where the labor wedge is flattest and the first-order approximation to it is worst. The non-Pigouvian part of the carbon wedge therefore inherits the labor wedge's level and its life-cycle rise, which is the proportionality result of Section 4.2 made quantitative. The income-side wedges are in turn essentially undisturbed by the second good: the labor wedge runs from 0.02 to 0.36 over the working life against 0.02 to 0.37 in the one-good economy, and the savings wedge falls from 0.56% to 0.02%, the Farhi and Werning (2013) pattern intact. The commodity margin adds a corrective and screening instrument without reshaping the other two,

Table 2: Cross-sectional mean wedges by age over the working life, income tax optimal. The premium is measured against the consumer price $1 + \tau^q$, the normalization of (4), so that it and the identity below it are the same object; against the Pigouvian gross price $1 + e$ it reaches 0.092 at 64.

age	25	35	45	55	64
labor wedge τ^y	0.021	0.180	0.280	0.341	0.363
carbon wedge τ^q	0.504	0.573	0.609	0.625	0.632
s.d. across histories	0.001	0.005	0.015	0.031	0.055
premium $(\tau^q - e)/(1 + \tau^q)$	0.0065	0.0505	0.0713	0.0806	0.0844
identity $\kappa l^\alpha \tau^y / (1 - \tau^y)$	0.0064	0.0499	0.0705	0.0796	0.0826

which is the separation showing up in the solved policies.

History dependence separates the two worlds as sharply as the path does. The full-optimum wedge's cross-sectional standard deviation fans out from 0.1 points early in life to 5.5 points near retirement, mirroring the spreading of the labor wedge as skill histories diverge. The frozen-world wedge's reaches 18 points in the early sixties, more than three times as wide, and the bands in Figure 1 separate from mid-career on. Against a given income tax, the price the mechanism would charge depends heavily on which history an agent has drawn.

6.3 Who bears the wedge

The optimal carbon wedge rises with skill in both worlds (Figure 2). Under an optimal income tax it is indistinguishable across quintiles early in life, when every history looks alike, and fans out from mid-career: by 64 it reaches about 70% for the top skill quintile against 57% for the bottom, a spread of about 14 points. Under the US income tax the same ranking holds and the spread is three times as wide, 44 points at 64, running from a 29% energy subsidy for the bottom quintile to a 15% tax on the top. High current productivity signals a history whose incentive constraints are expensive, and the commodity margin is the planner's lever on exactly those histories, so the screening value of the instrument concentrates on high-skill agents whether or not the income tax is optimal.

Two readings follow, and the second is the one that survives dropping the optimality assumption. The optimal carbon wedge is not regressive. The planner taxes the energy of high-skill households more, not less, so the redistributive worry about carbon pricing is misplaced at the level of the optimal type-contingent wedge. And this is not an artifact of assuming the rest of the tax system is right. Freezing the income tax at the observed US schedules leaves the progressivity intact and sharply strengthens it. The regressivity of real carbon taxes comes from elsewhere: from the flat-rate restriction, since an anonymous point-of-sale tax cannot condition on skill,

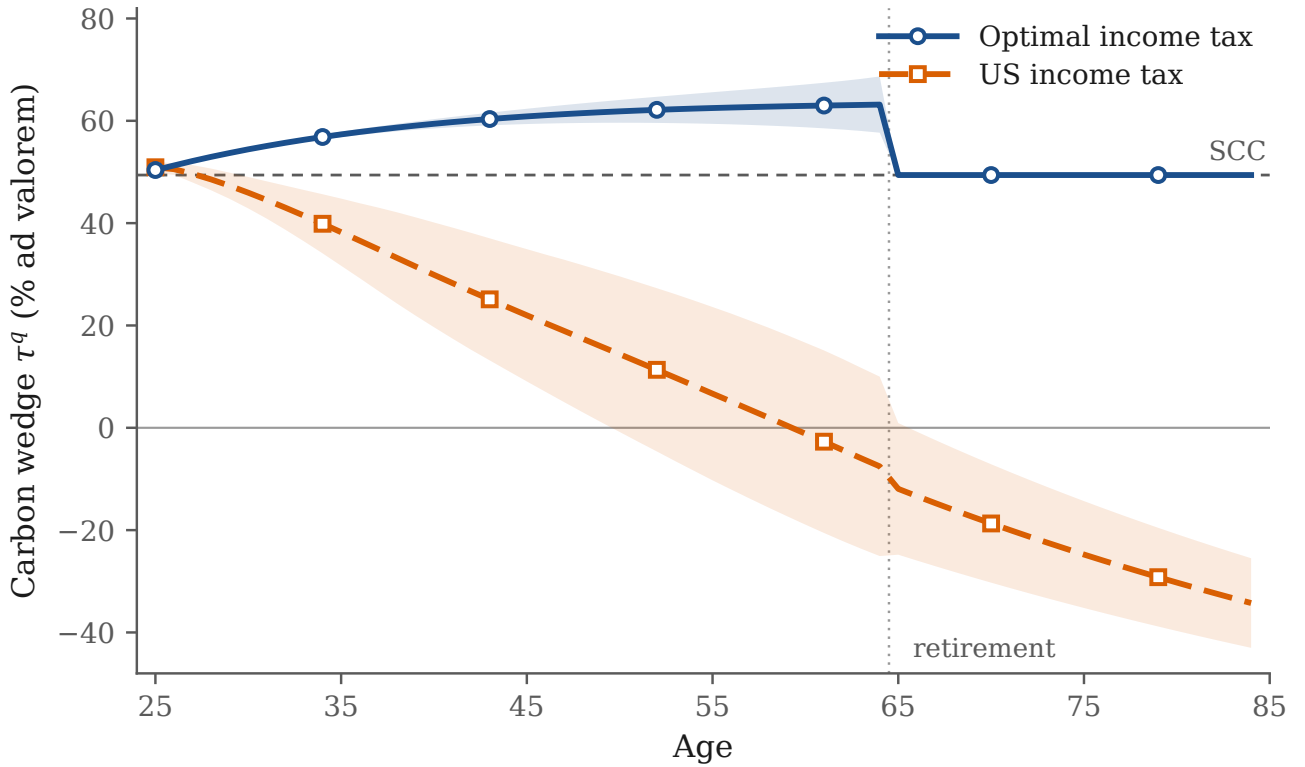


Figure 1: The optimal carbon wedge over the life cycle, benchmark calibration, when the income tax is set optimally alongside it and when it is frozen at the US schedules. Bands are ± 1 cross-sectional standard deviation across skill histories; the dashed horizontal line is the Pigouvian level $\tau^q = e = 0.494$. The retirement segment of the optimal-income-tax path is analytical rather than simulated (Appendix A.7).

and, in the necessity calibration, from energy's falling budget share.

Figure 2 is a schedule of marginal wedges by skill, not a map of the burden each household bears. That burden is set jointly by the income tax and transfers alongside the wedge, so the schedule's progressivity speaks to the corrective instrument itself, not to the full incidence of carbon pricing. The full incidence needs the rebate side, and Section 6.6 measures it once the welfare results are in hand.

6.4 Where the rotation comes from

Section 5.1 predicts that the two frozen margins reach the carbon price through one statistic, $\zeta_t = \eta_t(1 - \bar{\tau}_t^y) - \eta_t^s$, and that they enter it asymmetrically (Lemma 5.4 and Corollary 5.2). The labor gap is a within-date object whose sign profile follows the schedule's; here it holds each sign for roughly half the working life. The savings contribution can never hold one sign. Whatever

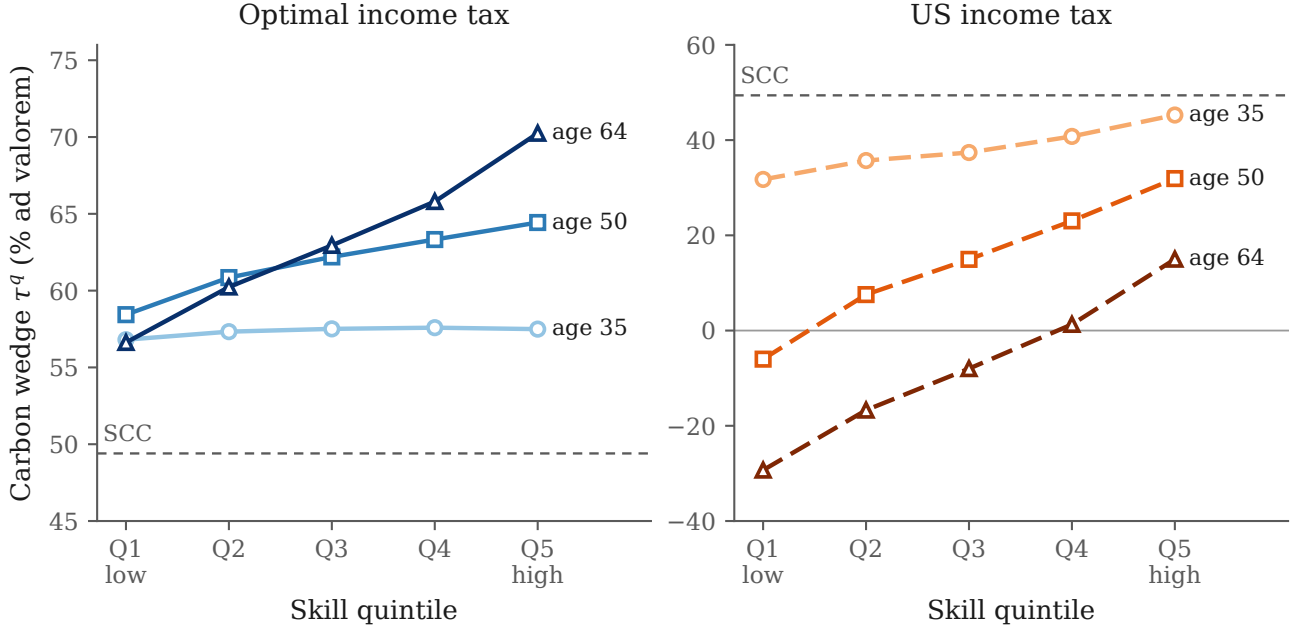


Figure 2: The optimal commodity wedge τ^q by skill quintile at ages 35, 50 and 64, when the income tax is set optimally (left) and when it is frozen at the US schedules (right). Shading runs light to dark in age. The dashed horizontal line is the Pigouvian level. Note the different vertical scales: the comparison the figure is drawn for is the slope in skill, whose magnitudes are given in the text.

the schedule, it raises ζ_t at some ages and lowers it at others (Lemma 5.3), so a savings freeze cannot move the whole profile one way. The two enter ζ_t additively, with coefficients that are one-signed under the condition $A_t > 1$ of Lemma 5.4, which the calibration below satisfies with room. Imposing the two restrictions in sequence separates their contributions. The first solve freezes the labor schedule and leaves the savings margin free, which isolates the labor channel. The second freezes both, which measures what the savings restriction adds. Figure 3 reports the sequence. Throughout, the *pull* of a restriction is the difference between consecutive solves, so the two pulls sum to the total by construction but their split depends on the order in which the restrictions are imposed. Only the total is order-invariant.

The labor gap is a mismatch in shape rather than in level. The frozen schedule is nearly flat in age, falling from 31% at 25 to 30% at 64, and its slope is a composition effect rather than a modeling choice, since the Heathcote et al. (2017) function conditions on earnings and earnings fall over the life cycle here. The Mirrleesian wedge meanwhile climbs from 2% to 36%. The two cross at 49. Labor is overdistorted for the first two decades of working life and underdistorted for the last, so by Proposition 5.1 the carbon wedge is pulled below its full-optimum counterpart early and above it late. It opens 27 points below that counterpart, sits 8 points above it

at 64, and the pull changes sign at 50. The two ages differ by a year because η_t is signed against the desired wedge of Lemma 5.2 evaluated at the constrained allocation, not against the full-optimum wedge that panel (a) draws. Nothing about this makes the carbon price low on average. With only the labor schedule frozen, the wedge runs from 23% at 25 to about 71% at 64, well above Pigouvian at the end of working life, and returns to the social cost of carbon at retirement.

The savings gap is one-signed and its pull is not. Panel (b) is drawn to keep the two apart. The frozen wedge is 0.9% per year against a full-optimum wedge that starts at 0.56% and decays to zero, so the US system overdistsorts the intertemporal margin at every age. A frozen savings wedge restricts a ratio of marginal utilities across dates rather than a level within a date, so what matters is the distortion accumulated by each age rather than the gap at that age. By (10) mean marginal utility is forced 71% above its date-25 value over the life cycle, where the optimal allocation leaves it essentially flat. The pull on the carbon wedge is *positive* at entry, 28 points, crosses zero at 34, and reaches -84 points at 84. That crossing is not the sign change of η_t^s discussed below, which sits five years later: a pull is a difference between two solves, and freezing the savings margin moves η_t as well.

The two pulls are equal and opposite at entry, 27 points down against 28 up, so the wedge opens at 51%. They cross at 39, and the savings pull's margin over the labor pull widens for the rest of life. The asymmetry at the other end of life is sharper: the labor pull dies at 65 and the savings pull does not (panel d), which is Corollary 5.3 in the solved policies.

Weighting each age by the emissions it carries converts the two pulls into prices. The labor freeze lowers the effective carbon price, the discounted-emissions-weighted mean of the wedge defined in Section 6.5.2, from \$218 to \$169 per ton. The savings freeze lowers it a further \$62, to \$107. Two fifths of that \$62 comes from the twenty retirement years alone, where the labor multiplier is dead and the savings multiplier runs unopposed, even though those years carry only 9% of discounted emissions.

Less than half of the savings channel is the capital-tax rate. The rest is the restriction itself, which binds even at a zero rate. Table 3 and the left panel of Figure 4 report the grid. At $\tau^k = 0$ the deviation is a level rather than a rotation: $\delta = -0.28$ at 25, climbing to zero only in the early sixties and staying within 0.003 of it through 64, with the working-life mean carbon price already down to 0.38 from a full-optimum mean near 0.60. A zero capital income tax is not a neutral benchmark on this margin: any constant $\bar{\tau}^s$ freezes the Euler equation, the full optimum satisfies the inverse Euler, and by Jensen's inequality the two cannot hold together under uncertainty, so the restriction binds at every τ^k including zero. Turning on the capital tax rotates the profile around that level, and the rotation grows monotonically in τ^k : at the benchmark $\tau^k = 0.18$ the deviation runs from $+0.01$ at 25 to -1.27 at 84, and at the $\tau^k = 0.36$ upper bound from $+0.29$ to -2.98 . The sign pattern is the one Appendix B.7 derives analytically at a degenerate draw, with the price above the social cost of carbon early

and below it late. Over the random walk of working life that argument delivers the reversal but not its date, so where the crossing falls is a solved property of this calibration rather than a consequence of it.

Two features of the rotation are worth separating from its size. The age at which $\hat{\eta}^s$ changes sign moves by at most a year across the capital-tax grid, 40–41 at the three lowest rates and 39–40 from the benchmark up, which is the grip of the endpoint conditions of Lemma 5.1 rather than of the size of the wedge gap. The crossing age of Table 3 is a different date and moves a great deal, from 28 at the benchmark to 36 at the upper bound, because by (9) the deviation vanishes where the savings term offsets the labor term rather than where η_t^s itself does. That gap between the two dates accounts for the top of the crossing column. At the three lowest rates the savings term barely reverses ζ_t . At $\tau^k = 0.025$ and 0.05 the deviation keeps one sign at every reported age, and at $\tau^k = 0$ it grazes zero in the early sixties and stays within 0.003 of it through 64, the table’s marginal crossing, before turning negative again in retirement, while η_t^s still changes sign between 40 and 41 in all three. The savings channel pushes toward a crossing without delivering a material one.

The deviation at labor-market entry is monotone in τ^k and crosses zero at $\tau^k \approx 0.17$, just below the benchmark (Figure 4, right panel). At the US effective marginal rate on capital income the labor and savings multipliers therefore very nearly cancel at entry: the young face a carbon price within a point of the social cost of carbon, and the whole frozen-world deviation is a rotation with age rather than a shift in level. This is a coincidence of the benchmark rather than a property of observed rates: such histories exist, but nothing says the US sits at one, and the entry deviation is -0.28 at a zero capital tax and $+0.29$ at the aggregate-measure upper bound, with Table 3 reporting the grid so the reading does not depend on where the benchmark falls.

Nothing in the section rests on that coincidence. Every qualitative feature above survives at every rate on the grid, and what the benchmark fixes is the size of the rotation and the age at which the price crosses the social cost of carbon, both of which rise with τ^k .

6.5 What carbon pricing is worth

The welfare accounting inverts between the two worlds, and the inversion is the section’s main result (Figure 5).

6.5.1 When the income tax is optimal

Table 4 reports consumption-equivalent gains over having no carbon price, under the optimal nonlinear income and savings mechanism throughout. Four findings, in order of size.

The value of a carbon price is overwhelmingly Pigouvian. At the EPA social cost of carbon,

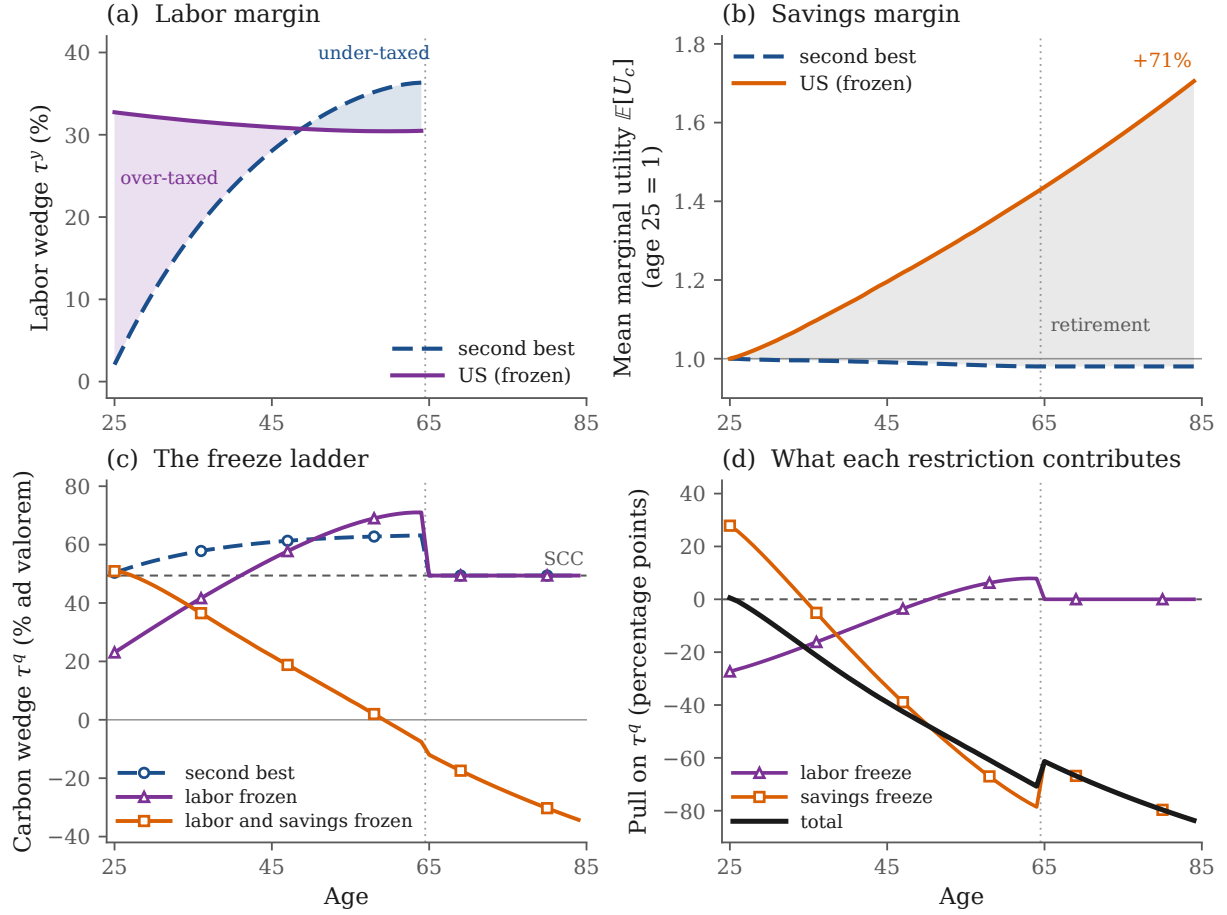


Figure 3: The two frozen schedules and what they do to the carbon wedge. The benchmark $\tau^k = 0.18$ throughout, and a *pull* is the difference between consecutive paths in (c). Blue is the full optimum, purple the labor restriction, orange the savings restriction, black their total; the full optimum is dashed and anything frozen is solid. (a) The frozen labor schedule at its age- t anchor against the full-optimum wedge at the same age; shading takes the color of whichever curve is on top, purple where the frozen schedule overdistorts labor and blue where it underdistorts. (b) The savings margin as the accumulated distortion: mean marginal utility under each regime, indexed to one at age 25, with the grey band the gap between them. Levels of U_c are not comparable across allocations; (10) restricts the growth rate, which is what the panel plots. The shading is neutral because a frozen savings wedge constrains a ratio across dates, so its multiplier changes sign (Lemma 5.3) and this gap's sign does not map into the sign of the pull. Both retirement segments are analytical: the full optimum holds $\mathbb{E}[U_c]$ constant once risk resolves, and the frozen path compounds at $1/(1 - \bar{\tau}^s)$. (c) The carbon wedge with nothing frozen, with the labor schedule frozen, and with both margins frozen; the retirement segments of the first two paths are analytical (Appendix A.7). (d) The two pulls, which sum to the total at every age.

Table 3: The carbon price against a given income tax, across the capital-tax grid. The labor schedule is frozen at the Heathcote et al. (2017) function in every row of the grid, and the savings margin is frozen in every row of the grid including the first: $\tau^k = 0$ sets $\bar{\tau}^s = 0$, which still restricts the Euler equation. The memo row is the full optimum, the reference the frozen rows move from. The labor restriction’s pull alone is isolated by the second rung of Figure 3, not by this table.

τ^k	$\bar{\tau}^s$ (%/yr)	δ at 25	crossing age	δ at 64	δ at 84	working-life mean τ^q
0	0	−0.28	62	0.00	−0.07	0.38
0.025	0.13	−0.25	—	−0.08	−0.22	0.34
0.05	0.25	−0.22	—	−0.18	−0.38	0.30
0.18	0.90	+0.01	28	−0.62	−1.27	0.23
0.36	1.80	+0.29	36	−1.31	−2.98	0.22
<i>memo</i> : second best	free	+0.01	—	+0.08	0	0.59

Note: $\text{SCC} = e = 0.494$, and $\delta_t \equiv (\tau_t^q - \text{SCC})/(1 + \tau_t^q)$ is the deviation of (4), evaluated at the date- t cross-sectional mean carbon price over simulated histories. Since δ is convex in τ^q this is not the mean of δ across histories; it is reported this way so that it inverts exactly into the τ^q levels of Section 6.2. “Crossing” is the age at which the deviation changes sign, and a dash means it keeps one sign over the whole life cycle. In the memo row the same deviation statistic is the premium of Table 2, computed from the same solve; its retirement value is exactly zero (Appendix A.7), and *free* marks the margins the full optimum leaves unfrozen. $\tau^k = 0.18$ is the benchmark of Table 1 and $\tau^k = 0.36$ the upper bound of Appendix D.1. Every frozen row is solved at the same grid resolution. The $\tau^k = 0$ crossing is marginal: the deviation stays within 0.003 of zero from age 60 on.

having any carbon price is worth 0.31% of consumption, and a flat rate at the social cost of carbon alone captures 98% of the full-optimum value while cutting emissions by 13.5%. For scale, Farhi and Werning (2013) put the entire insurance value of their optimal income-tax apparatus over laissez-faire at 1.56% of consumption: the carbon instrument is worth about a fifth of that at this social cost of carbon.

The optimal flat rate exceeds the social cost of carbon by about 12%, $\tau^{q} = 0.555$ against $e = 0.494$, or \$214 per ton against \$190, the flat reflection of the Corlett-Hague premium. But the welfare gain from that uplift over pricing at the social cost of carbon is 0.002%: the objective is very flat in the rate, so “price at the social cost of carbon” is a near-optimal rule of thumb even with nonseparable preferences.*

Sophistication buys almost nothing beyond the level. Making the rate age-dependent adds about 0.001 percentage points over the optimal flat rate, and full history dependence about 0.001 more, both at the numerical resolution. The premium varies over life and across histories by a few points of an equilibrium budget share near 6% with a substitution elasticity of 0.4, so the deadweight loss from mistracking it is second-order squared.

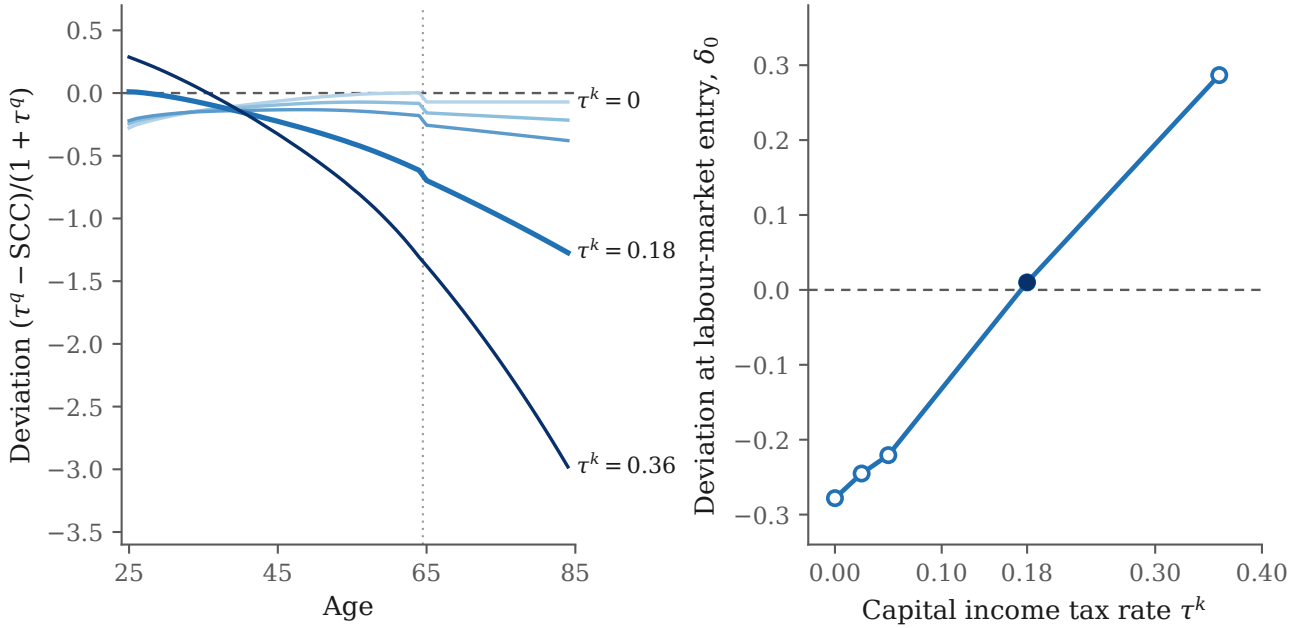


Figure 4: Left: the deviation $\delta_t = (\tau_t^q - \text{SCC}) / (1 + \tau_t^q)$ over the life cycle across the capital-tax grid, with the labor schedule frozen at the US function throughout. Shading runs light to dark in τ^k over the grid of Table 3; the benchmark $\tau^k = 0.18$ is drawn heavier. The dashed horizontal line is Pigouvian pricing and the dotted vertical line is the retirement date. Right: the same deviation at labor-market entry, plotted against τ^k , with the benchmark rate marked by a filled circle.

The pure Corlett-Hague motive is tiny, and necessity halves the stakes. With the externality off, the entire commodity-wedge apparatus of the full optimum is worth 0.004% of consumption, and its optimal flat rate is about 4%, materially positive as in West and Williams (2007) but negligible in welfare and barely identified, since the whole column sits at the numerical resolution. Under the Stone-Geary calibration, in which energy's budget share falls with income, the value of the flat carbon tax falls to 0.15% and its abatement to 6%, because the subsistence part of energy does not respond to price, while the optimal flat rate is essentially unchanged. This is the calibration in which a flat carbon tax is regressive, yet the mechanism absorbs the incidence through the income side, which is why the flat instrument loses so little welfare despite its incidence. As a check on the whole apparatus, switching the leisure complementarity off ($\gamma_c = \gamma_q$) makes the full optimum and the flat Pigouvian tax coincide in welfare at the numerical resolution and exactly in emissions: dynamic Atkinson-Stiglitz, and the exact knife-edge of Proposition 4.3, reproduced through two structurally different solvers.

Table 4: Welfare value of carbon pricing under progressively simpler rules, income tax optimal (consumption-equivalent gain over $\tau^q = 0$, % of consumption)

	benchmark ($e = 0.494$)	Stone-Geary (necessity)	$\gamma_c = \gamma_q$ (placebo)	no externality ($e = 0$)
second best (nonlinear τ^q)	0.313	—	0.252	0.0041
age-dependent linear $\tau^q(t)$	0.312	—	—	0.0032
flat τ^{q*} (optimized)	0.311	0.152	—	0.0019
optimal flat rate τ^{q*}	0.56	0.56	—	0.038
flat Pigouvian $\tau^q = e$	0.309	0.150	0.253	—
emissions vs $\tau^q = 0$: second best	−15.0%	—	−13.5%	−1.8%
flat Pigouvian	−13.5%	−5.7%	−13.5%	—

Note: welfare entries are in % of consumption; the optimal-flat-rate row is an ad-valorem rate, and the final two rows report emissions changes relative to $\tau^q = 0$. The rungs are separate solves whose simulated budgets agree only to about 0.003 points of consumption, the ladder’s numerical resolution: differences below it measure that alignment as much as any policy. Sampling error is far smaller. With common random numbers the seed-to-seed standard deviation is about 0.0003 points on each entry and below 0.0001 on any difference between rungs, so sub-resolution differences are stable across simulation panels; they are just not resolved. The no-externality column sits at the resolution throughout. The optimized flat rate abates slightly more than the Pigouvian one where the two differ: 6.3% under Stone-Geary and 1.4% in the no-externality column, where no Pigouvian rate exists. Dashes mark three cases. *Not computed*: the full optimum requires homothetic preferences, so it and the age-dependent rung are blank under the non-homothetic Stone-Geary calibration. *Degenerate*: under the $\gamma_c = \gamma_q$ placebo the full optimum already coincides with the flat Pigouvian tax, so the intermediate rungs add nothing. *Not applicable*: with no externality ($e = 0$) there is no Pigouvian benchmark.

6.5.2 When the income tax is the one the US levies

Every one of those conclusions reverses. Table 6 runs the same rate grid with the two income-side wedges frozen, and Figure 5 plots the two rate curves side by side. Both panels come from the same solver family at the same grid with the same simulated histories, so the only difference between them is the freeze.

A flat carbon tax at the social cost of carbon is welfare-negative. It costs 0.115% of consumption relative to having no carbon price at all, and 0.155% at the $\tau^k = 0.36$ upper bound. This is not an artifact of ignoring the damages the tax avoids: the planner’s budget prices energy at the social price $1 + e$ in every regime, including the one with no carbon tax, so the comparison is damage-inclusive by construction. With the income tax stuck at US levels, legislating the social cost of carbon as a flat rate overshoots the constrained optimum badly enough that the economy would rather have no carbon price. The average is an efficiency statement, not a distributional one: with the rebate side at its optimum, the bottom two lifetime-earnings quintiles gain even at this rate (Section 6.6).

The optimal flat rate is a quarter of the social cost of carbon, and it is worth almost nothing. It is 0.12, against 0.56 when the income tax is optimal, and it buys 0.02% of consumption over charging nothing. Welfare gains run out entirely at 0.254, near \$98 per ton, interpolated between solves on the same geometry at 0.24 and 0.32. The social cost of carbon is therefore four times the best flat rate and twice the highest flat rate that does no harm at all. A single rate must average a profile that runs from above the social cost of carbon at entry to a deep subsidy in retirement, weighted by substitution responses, and the twenty retirement years drag that average far down. The below-damages prescription of the Ramsey tradition reappears here, and more starkly than in the linear-instrument literature, because a frozen income tax cannot absorb any of the burden.

Ambition falls by much less than the flat rate suggests. Within the same frozen world the nonlinear wedge abates 8.0% of discounted emissions against the 4.3% the best flat rate delivers, and its effective carbon price, the average rate a ton of emissions actually faces,¹⁰ is 0.28 rather than 0.12: \$107 per ton against \$47. Emissions concentrate in mid-career, where the wedge is still well above zero, and only 9% of discounted emissions fall in the retirement years the subsidy reaches. What a miset income tax mostly destroys is the flat instrument, not the case for pricing carbon.

The near-sufficiency of flat carbon taxation is overturned. When the income tax is optimal, the entire nonlinear fine structure of the wedge is worth 0.002 percentage points over the best flat rate. When it is frozen at the US schedules, the nonlinear wedge is worth 1.38 percentage points, more than five hundred times as much and four times the entire full-optimum value of carbon pricing. Under a miset income tax the commodity margin's history dependence performs income-tax repair, and no anonymous rate can imitate it.

That last number is best read as a statement about commodity taxation rather than about carbon. Energy is the only margin the constrained planner still controls freely, so it absorbs the whole repair job; with more consumption goods the same logic would spread across every commodity wedge and each would move less.

The left panel of Figure 6 puts both subsections on one scale, in dollars per ton. The full-optimum mechanism charges \$218 on an emissions-weighted average. Freezing the labor schedule takes off \$49 and freezing the savings margin a further \$62, leaving the \$107 that the best carbon-only reform collects. Restricting the remaining instrument to a flat rate takes off \$60 more and lands at the \$47 of Table 6. That last restriction costs under \$5 when the income tax is set optimally alongside the price, since the best flat rate is then within a few dollars of

¹⁰The discounted-emissions-weighted mean of the age-varying wedge, $\sum_t \beta^t E_t \tau_t^q / \sum_t \beta^t E_t$, with E_t the cohort's date- t emissions under the policy being evaluated and $\beta = 1/R$. Equivalently, the flat rate that would collect the same amount per ton over the same emissions path, which is what makes a life-cycle profile comparable to a single rate. It exceeds the unweighted mean of τ_t^q because emissions concentrate at ages where the wedge is high, and each regime carries its own emission weights.

the \$218 the nonlinear mechanism collects. The flat-rate restriction is nearly free against an optimal income tax and expensive against the one in force. This is the near-sufficiency result and its reversal priced per ton rather than measured as a share of a welfare gain, and the two metrics do not compress the contrast equally: \$5 against \$60 is a factor of twelve where the welfare shares differ by three orders of magnitude.

Every dollar figure in that ladder conditions on where the US schedule sits against the desired one, and the right panel re-runs the entire ladder in the configuration that reverses the labor gap: the GR16-implied anchors of Section 6.7, under which the US schedule underdistorts labor over most of working life. Table 5 reports the two worlds side by side. The savings pull barely moves, $-\$60$ against $-\$62$: the frozen savings wedge compounds over the same retirement horizon whatever the labor anchors. The labor pull flips from $-\$49$ to $+\$54$ and nearly cancels it, so the best carbon-only reform prices carbon at \$211 per ton, within \$7 of the full optimum. Re-optimizing the flat rate in that world moves it from a quarter of the social cost of carbon to the social cost of carbon itself: $\tau^{q*} = 0.49$, \$187 per ton against \$190, and a flat tax at the social cost of carbon costs 0.0001 points of consumption against the best flat rate rather than destroying 0.115%. Ambition follows the level: the best flat rate abates 14.0% of discounted emissions there, against 4.3% under the US freeze. The nonlinear wedge is worth more in that world, not less, +1.74% of consumption against +1.40%, so the flat instrument's recovery is about its level landing on the social cost of carbon, not about history dependence losing its value: a frozen income tax leaves the same order of repair on the table in both worlds, and what the calibration decides is whether a flat rate can collect any of it. The practical content is symmetric and stark. Whether the best flat carbon tax is \$47 or \$190 per ton turns not on damages, which are identical in the two columns, but on which side of the desired schedule the income tax errs.

6.6 Where the revenue goes

Every solve above disposes of the carbon revenue silently: the planner chooses consumption levels, and the levels are the rebate. This subsection makes the rebate explicit, twice. Inside the class of Definition 5.1 the mechanism itself picks the design, and the pick turns out to be a uniform dividend. Outside the class sits the design the freeze rules out, revenue-neutral marginal-rate cuts, and pricing it overturns the ordering of the margins: what the revenue funds matters more than the rate that raises it.

6.6.1 The rebate the mechanism chooses

The full incidence of carbon pricing is computable in the frozen world, because the rebate side of the constrained optimum is solved rather than assumed. Section 5.1 leaves the levels free, and decentralized the levels are the tariff intercepts: the transfer each history receives is an outcome of the mechanism. Valuing each simulated history at the statutory system, spending

Table 5: The frozen world under two labor-schedule calibrations: US anchors against the GR16-implied anchors of Section 6.7

	US schedules	GR16-implied anchors
<i>Effective carbon price (\$/tCO₂, common emission weights)</i>		
full optimum (nonlinear)	\$218	\$218
labor freeze	−49	+54
savings freeze	−62	−60
best carbon-only reform	\$107	\$211
<i>The flat-rate restriction, within each frozen world</i>		
best flat rate τ^{q*}	0.12 (\$47)	0.49 (\$187)
its welfare gain (%)	+0.021	+0.250
flat Pigouvian, $e = 0.494$ (%)	−0.115	+0.250
nonlinear wedge (%)	+1.401	+1.744

Note: the upper block is the effective carbon price of Figure 6, the discounted-emissions-weighted mean wedge on the benchmark emission path, held common across regimes and columns; freeze rows are steps between levels. The GR16 column transplants the Golosov et al. (2016) wedge path onto the model’s desired schedule as described in Section 6.7, leaving preferences, damages, the capital tax, the solver family, and the simulated histories unchanged. Labor-freeze rungs are quoted at the labor-only stage’s canonical resolution; re-solving the GR16 stage at the joint solves’ finer productivity grid raises its level by about \$10, the resolution skew of the exhibit, and no conclusion turns on it. The lower block re-optimizes the flat rate within each column’s own frozen world; welfare entries are consumption-equivalent gains over that column’s $\tau^q = 0$, in % of consumption, on paired simulated histories; dollar rates convert at \$190 per ton = 0.494. Every GR16 store passes the verdict battery of the US solves, and its one-shot-deviation audit sits below the same-grid unconstrained control’s noise floor (Appendix, IC audit).

at consumer prices net of retained earnings, discounted at the after-tax return, gives the implied lifetime transfer $B(\theta^T)$.¹¹ Under the US schedules with no carbon tax, these transfers carry the system’s redistribution: B averages 19% of lifetime spending in the bottom lifetime-earnings quintile, 2% in the top, and its correlation with lifetime earnings across histories is -0.99 .

Adding the carbon tax and re-solving, the increment is a uniform dividend. At the best flat rate the mean transfer rises by 97% of the carbon revenue, and the increment varies across histories by 2.4% of its mean: sorted by lifetime earnings, the bottom quintile receives 0.97 of the mean revenue and the top 0.98 (Table 7). Nothing is withheld. The tax also erodes the labor-income-tax base, by about 7% of the carbon revenue, so the dividend exceeds the government’s net revenue gain, and the difference is carried by the resources abatement frees, which the planner’s budget prices at the social cost. In the all-flat family of Section 5.3 the

¹¹Retained earnings integrate the marginal schedule (43) in normalized earnings, so B is exact against the schedule the solver freezes and carries the detrending caveat of Section 5.1. Only the present value is identified: the household smooths at $\tilde{R}$, so rebate paths with equal present value implement the same allocation.

Table 6: Welfare of a flat carbon tax when the income-side wedges are frozen at the US schedules (consumption-equivalent gain over $\tau^q = 0$ of the same regime, % of consumption). The “income tax optimal” column is the same rate grid solved with the income and savings wedges free, at the same grid resolution and on the same simulated histories, and is reported for comparison.

flat rate τ^q	income tax optimal	income tax given
0 (no carbon price)	0	0
0.10	+0.119	+0.021
0.20	+0.204	+0.014
0.35	+0.280	−0.035
Pigouvian, $e = 0.494$	+0.311	−0.115
0.65	+0.310	−0.226
best flat rate τ^{q*}	0.56	0.12
its welfare gain	+0.315	+0.021
nonlinear τ^q	see Table 4	+1.401

Note: entries are consumption-equivalent gains in % of consumption; the best-flat-rate row is an ad-valorem rate. The right-hand column is the benchmark $\tau^k = 0.18$; at the $\tau^k = 0.36$ upper bound the same rows read +0.011 at the best flat rate 0.087, −0.155 at the Pigouvian rate, and +1.649 for the nonlinear wedge, so nothing in the reading turns on the capital-tax measure. The nonlinear entry is left to Table 4 in the left-hand column because the full optimum is solved on a finer grid than this rate sweep and the two levels are not directly comparable; the within-column comparison there gives 0.313 against 0.311 for the best flat rate.

uniformity is exact, the solved transfer constant across histories up to discretization, and the 2.4% here is an upper bound on true dispersion, since the retirement-junction discretization contributes at the same order. Within the class of Definition 5.1, the optimal rebate design is fee-and-dividend: charge the flat rate, return the revenue as an equal per-history payment, timing free.

Payments are not uniform, so the net incidence is progressive at every flat rate. Lifetime carbon payments scale with lifetime spending, 1.35 times the mean revenue in the top quintile against 0.74 at the bottom, while the dividend is flat, so the net position falls monotonically in lifetime earnings. The welfare incidence follows in the last columns of Table 7: the best flat rate hands the bottom quintile +0.27% of consumption and the top −0.19%. The pattern survives the rate that should not be levied at all. The Pigouvian flat tax destroys 0.115% of consumption on average (Section 6.5.2), yet the bottom two quintiles still gain, 0.77% and 0.17%, because the dividend they receive exceeds what they pay. Regressivity is not a property of carbon pricing in this economy; it is a property of designs that do not return the revenue. One margin is silent here. Lifetime energy intensity sits between 7.1 and 7.2% of spending in every quintile under homothetic preferences, so payments track total spending rather than composition, and the Stone-Geary calibration, in which the energy share falls with income, is where the payments

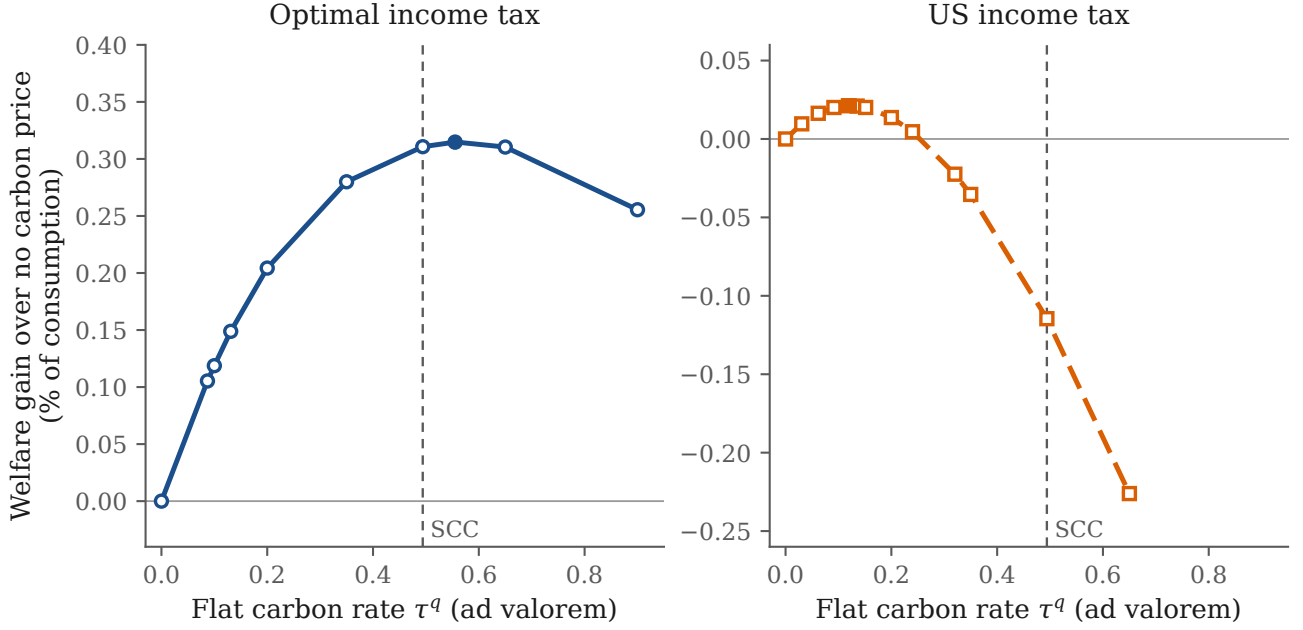


Figure 5: Consumption-equivalent welfare gain from a flat carbon tax, against the rate, when the income tax is set optimally alongside it (left) and when it is frozen at the US schedules (right). Both panels use the same solver family and the same simulated histories; only the freeze differs. The filled marker is the welfare-maximizing rate and the dashed vertical line is the social cost of carbon. Note the different vertical scales.

column would steepen.

Rebate shape is worth almost nothing beyond its level. The class forbids conditioning the rebate on earnings, and the solved profile shows the planner does not miss it: forcing the increment to one common dividend by fiat would cost at most 2×10^{-5} percent of consumption, four orders of magnitude below what the marginal-rate restrictions cost in Section 6.5.2. Transfers move the level of the marginal-utility path while the frozen Euler equation pins its growth (10), so recycling redistributes and insures across histories but repairs no margin. Within the carbon-only class, the design question is the rate, not the rebate.

6.6.2 Recycling into rate cuts

The dividend is the best rebate the class allows, but the class is the binding constraint: its freezes leave the rebate side as the only fiscal margin a reform can move. The revenue-neutral swaps of British Columbia and Sweden sit outside it by construction, spending carbon revenue on marginal rates, and this subsection steps outside deliberately to price the exclusion. The swap instrument is one scalar. The frozen labor schedule's net-of-tax rate is scaled by s at

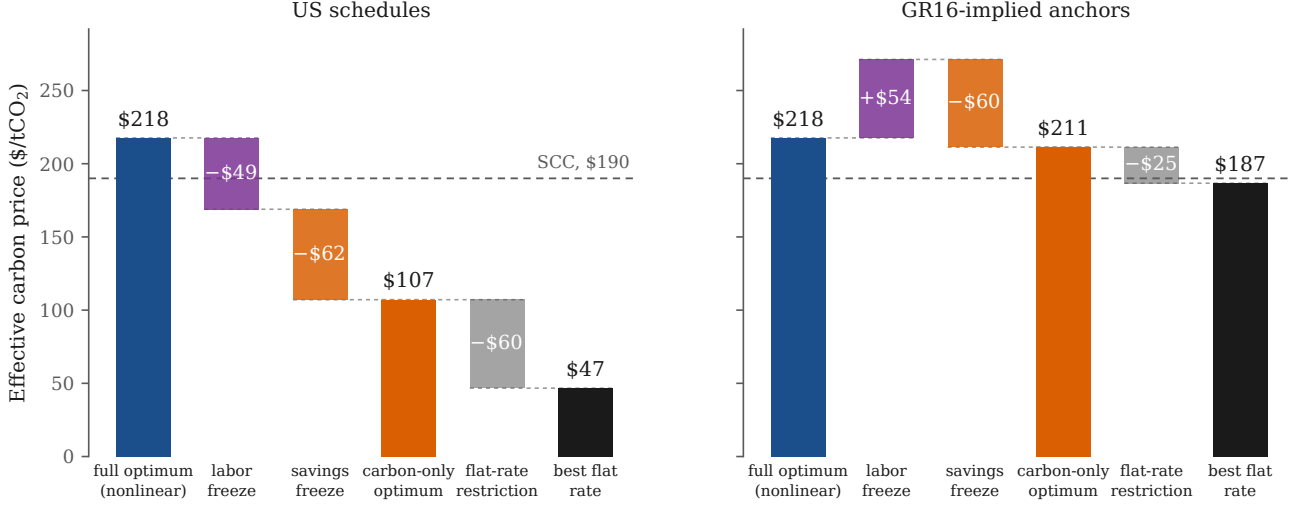


Figure 6: The effective carbon price, one restriction at a time, under the US schedules (left) and the GR16-implied anchors (right). The level bars are discounted-emissions-weighted means of the solved wedge, $\sum_t \beta^t E_t \tau_t^q / \sum_t \beta^t E_t$; the step bars are differences between them; the last bar in each panel is that world’s own re-optimized flat rate (Tables 6 and 5). The weights E_t are the benchmark emission path, held common across regimes and panels so that every level is a reweighting of one object. Using each regime’s own path instead, as in Section 6.5.2, moves the second-best level by under a dollar. Each step nets pulls that change sign over the life cycle, so a step is not a statement about any single age; Figure 3(d) shows the age profiles behind the left panel’s first two, and the right panel’s labor step runs upward because the GR16-implied schedule underdistorts labor over most of working life. The final step is not an averaging operation either: a flat rate solves a different problem rather than summarizing the nonlinear wedge, and in the left panel the flat rate matching the nonlinear wedge’s abatement is \$94 rather than \$47.

every age and every earnings level, $1 - \bar{\tau}^y \mapsto s(1 - \bar{\tau}^y)$, a uniform marginal-rate cut that leaves progressivity untouched, and the flat carbon tax finances it: s is set so that the mean implied lifetime transfer $B(\theta^T)$ stays at its no-carbon-tax level, so the revenue the dividend design returns as an equal per-history payment is spent on rates instead. Everything else is held as in the flat-rate solves: the savings wedge stays frozen at $\tau^k = 0.18$, on the same solver family and the same simulated histories.¹²

¹²Solve and comparison discipline: 22 swap-family stores, each passing the verdict battery of the flat-rate solves; welfare on paired common-random-number histories against the shared $\tau^q = 0$ baseline; $s = 1$ placebo solves within the swap family reproduce the canonical dividend stores to ± 0.001 points of consumption, so nothing here is a solver-geometry artifact. Welfare and the neutrality residual are measured locally linear in $\log s$; the $\tau^q = 0.131$ and 0.494 rows of Table 8 are solved at neutrality directly, with a residual transfer below 4×10^{-4} of carbon revenue, and the remaining rows interpolate between solved stores along those measured lines.

Table 7: The optimal recycling profile: implied lifetime transfers by lifetime-earnings quintile when the income-side wedges are frozen at the US schedules ($\tau^k = 0.18$)

quintile	B_0	best flat rate, \$47/t			Pigouvian, \$190/t		
		rebate	paid	ΔCE	rebate	paid	ΔCE
1 (lowest)	19.4	0.97	0.74	+0.27	0.96	0.74	+0.77
2	13.6	0.97	0.86	+0.10	0.96	0.87	+0.17
3	10.1	0.97	0.96	+0.01	0.96	0.96	-0.16
4	6.8	0.98	1.08	-0.08	0.97	1.08	-0.47
5 (highest)	2.1	0.98	1.35	-0.19	0.97	1.34	-0.87

Note: B_0 is the implied lifetime transfer of the solved optimum with no carbon tax, in % of the quintile’s lifetime spending, both in present value at the after-tax return $\tilde{R}$. “Rebate” is the reform’s transfer increment and “paid” the lifetime carbon payments, each relative to mean carbon revenue per history, so a rebate entry of 1.00 is exactly the fee-and-dividend payment. ΔCE is the quintile’s mean consumption-equivalent welfare change from the reform, in % of consumption, on paired simulation histories ($N = 60,000$, common random numbers). The mean rebate is 97% of revenue at both rates; the difference tracks the labor-tax base and the damage account. Quintiles are ranked by pre-reform lifetime gross earnings. Rates: the best flat rate 0.12, about \$47 per ton, and the Pigouvian 0.494, \$190 per ton.

Two facts organize the results. A one-percent rise in the net-of-tax level, about 0.7 points off the marginal rate, costs the budget only 0.79 of its static revenue, because hours respond; and it is worth +0.13% of consumption, a value that is almost independent of the carbon rate financing it. The swap therefore dominates the dividend at every rate, and by more than carbon pricing itself is worth (Table 8, Figure 7). At $\tau^q = 0.131$, the family of the dividend design’s best rate, the premium is +0.163% of consumption, eight times the +0.020% the dividend design delivers there in total. A flat tax at the social cost of carbon, which destroys 0.115% of consumption when its revenue is returned as a dividend, gains +0.43% when the same revenue cuts rates: the welfare-negativity of Section 6.5.2’s flat Pigouvian tax is a property of the rebate design, not of the freeze. And the best swap-financed rate is not a quarter of the social cost of carbon but above it, near 0.68 — about \$262 per ton, interpolated between solves on the same geometry — worth +0.47%, with gains still at +0.45% at $\tau^q = 0.8$, three times the rate at which the dividend design’s gains run out. The above-damages prescription that Ploeg et al. (2025) obtain when carbon revenue may reform a parametric income tax reappears here from a different source: theirs requires high inequality aversion, while this one comes from the frozen schedule’s misplaced marginal rates and survives at the utilitarian weights used throughout.

The source of the premium is the labor gap of Section 6.4, and naming it keeps the result honest. The frozen schedule overdistorts labor for the first two decades of working life, so the population-average labor multiplier prices a rate cut positively through age 42 and negatively after; a uniform cut buys repair at the young ages faster than it loses at the old ones, and carbon revenue merely finances the purchase. The swap’s value is income-tax repair under a

carbon label, not carbon value, and the recycling ladder brackets that repair: the dividend repairs no margin and delivers +0.02%, the one-scalar swap delivers +0.43% at the social cost of carbon, and the personalized wedge of Section 6.5.2, which performs the repair without the statutory constraint, reaches +1.40%. Within the carbon-only class the design question was the rate, not the rebate; across classes that ordering reverses, and the recycling design is worth an order of magnitude more than either the rate or the rebate’s shape.

Incidence orders the designs the other way. At the social cost of carbon the neutral swap raises welfare in every baseline lifetime-earnings decile, from +0.28% of consumption at the bottom to +0.55% at the top: the gross benefit of a rate cut scales with earnings, 0.42 to 1.14 times mean carbon revenue across deciles, while carbon payments and the planner’s residual intercepts, which lean toward the bottom, claw back only part of the tilt. The dividend design at the same rate hands the bottom quintile +0.77% and the top −0.87% (Table 7). This is the equity-efficiency trade-off of the recycling literature (Goulder et al. 2019; Klenert et al. 2018; Fried et al. 2024), reproduced inside a Mirrleesian frozen world, with one twist: the efficient design here is not merely less progressive, it is decile-Pareto-improving relative to no carbon tax, so the choice between designs is about how gains tilt rather than about who loses. Abatement is the swap’s other cost. The rate cut expands labor and spending, and discounted emissions fall 13.1% against the dividend’s 14.3% at the social cost of carbon, an eight-percent give-back.

Two cautions bound the reading. The swap’s value follows the labor gap, which Section 6.7 shows is not robust to the life-cycle calibration, and re-running the swap under the GR16-implied anchors prices that dependence: the premium over the dividend collapses from +0.55 to about +0.04 points of consumption at the social cost of carbon, and the improving direction remains a small rate cut rather than reversing, because discounting weights the young ages, where even that schedule overdistorts. The neutral swap there is progressive, +0.47% of consumption in the bottom lifetime-earnings decile against +0.16% at the top, the mirror of the tilt above. When the frozen schedule is nearly right on a discounted average, the recycling design stops mattering: how much the design is worth, like the carbon price itself, is a statement about how far the tax code sits from the desired schedule. And the swap is a different exercise, not a revision: within the carbon-only class, the \$47 best flat rate, the fee-and-dividend reading, and every number of Sections 6.6–6.5.2 stand. This subsection measures what the class boundary costs, which is the question the swaps of British Columbia and Sweden pose.

6.7 Reading and limitations

The practical hierarchy for carbon policy depends on a condition that is easy to leave implicit. If the income tax is set optimally alongside the carbon price, the hierarchy is clean: get the carbon price to the social cost of carbon, worth 0.31% of consumption; nudge it up by the premium for leisure complementarity, worth a further 0.002%; and spend the sophistication budget on the income tax, where Farhi and Werning (2013)’s own accounting shows the returns to age- and

Table 8: The revenue-neutral tax swap against fee-and-dividend: the same flat carbon tax, recycled as a uniform marginal-rate cut (s scales the frozen schedule’s net-of-tax level) versus as the uniform dividend of Table 7 ($\tau^k = 0.18$; consumption-equivalent gains over $\tau^q = 0$ of the same regime, % of consumption)

flat rate τ^q	$\approx \$/\text{tCO}_2$	dividend	swap s^*	rate cut (pp)	swap	swap – dividend
0.1	38	+0.021	1.0094	0.6	+0.144	+0.123
0.131	50	+0.020	1.0122	0.8	+0.182	+0.163
0.2	77	+0.014	1.0182	1.3	+0.254	+0.240
0.35	135	−0.035	1.0304	2.1	+0.367	+0.402
Pigouvian, $e = 0.494$	190	−0.115	1.0412	2.8	+0.432	+0.547
0.65	250	−0.226	1.0521	3.6	+0.465	+0.691
0.8	308	−0.357	1.0616	4.2	+0.453	+0.810

Note: “dividend” is the canonical flat-rate family of Table 6 ($s = 1$); “swap” holds the mean implied lifetime transfer at its no-carbon-tax level, so carbon revenue finances the rate cut. s^* is the revenue-neutral scale and “rate cut” its marginal-rate equivalent at the mean frozen rate, $(s^* - 1)(1 - \bar{\tau}^y)$, in percentage points. Dollar figures convert at $\$190$ per ton $= 0.494$. The $\tau^q = 0.131$ and 0.494 rows are solved at neutrality directly (residual transfer $\leq 4 \times 10^{-4}$ of carbon revenue); the other rows interpolate linearly in $\log s$ between solved stores, all on the same grid and simulated histories ($N = 60,000$, common random numbers). At $\tau^q = 0.8$ no canonical dividend store exists and the swap family’s $s = 1$ solve stands in; the two coincide to ± 0.001 points wherever both exist.

history-dependence are two orders of magnitude larger. The redistributive value embedded in the carbon tax is small, which is the separation principle in welfare terms, and it agrees with Douenne et al. (2026)’s near-Pigouvian conclusion, reached here from the Mirrleesian rather than the Ramsey side.

If the income tax is the one actually in force, that hierarchy does not hold, and the direction of the failure is informative. The below-damages prescription is not a property of distortionary taxation as such: with the income tax optimal, the carbon price is exactly the social cost of carbon at every history under weak separability. It is a property of income-tax misspecification. Set the income tax right and the correction vanishes; set it wrong and the correction is large, larger than the Ramsey literature’s 8–24%, and a flat tax at the social cost of carbon becomes worse than no carbon tax. The two results are one claim: the case for pricing carbon at the social cost of carbon is a case for fixing the income tax, and the two reforms are not separable in practice even though the wedges separate in theory.

This does not rehabilitate weakening the carbon price on its own. A government that lowers the carbon price while leaving the income tax misset moves along the right-hand panel of Figure 5 and recovers at most 0.02% of consumption; one that fixes the income tax moves to the left-hand panel and gains 0.31% from the carbon price alone, on top of the income tax’s own insurance value. The ordering of the two reforms matters, and it runs from the income tax to

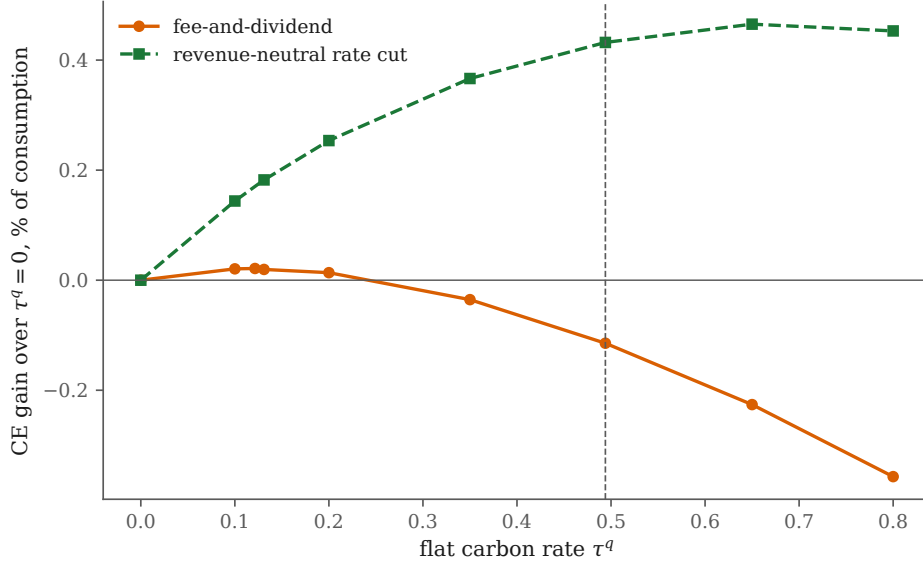


Figure 7: Welfare of a flat carbon tax when the income-side wedges are frozen at the US schedules, under the two recycling designs: revenue returned as a uniform dividend (the carbon-only class of Definition 5.1) and revenue spent on a revenue-neutral uniform marginal-rate cut. Consumption-equivalent gain over $\tau^q = 0$, in % of consumption; the dashed vertical line is the social cost of carbon. Both curves come from the same solver family on the same simulated histories; Table 8 reports the points.

the carbon price rather than the other way around. Section 6.6.2 makes the same point inside one bill: letting the carbon tax finance even a one-scalar repair of the marginal-rate schedule is worth twenty times the best carbon-only flat reform.

Six limitations bound the numbers. The split of Figure 6 between the two margins is the least secure of them, in two respects. It is order-dependent in the way Section 6.4 describes. And it depends on the life-cycle block. Farhi and Werning (2013) give newborns identical ex ante skills and a driftless random walk thereafter, so the Mirrleesian labor wedge starts near zero and mean earnings fall rather than rise with age. Both features widen the labor gap at young ages, which discounting and the mid-career concentration of emissions weight most heavily. Golosov et al. (2016), calibrating initial dispersion to the data, report a labor wedge at average earnings running from about 25% early in life to about 65% late, against the 2% to 36% here. Those are the two endpoints they report; the path between them is interpolated linearly. Against a schedule of that shape the US function would underdistort labor over most of the working life. The GR16-implied anchors solve the frozen world in exactly that configuration: the frozen schedule is shifted so that the frozen-minus-desired gap reproduces, age by age, the gap between the US function and their wedge path (overdistortion of 8 points at 25, a crossing at 32, underdistortion of 35 points by 64), leaving everything else fixed. Table 5 carries the

result: the labor rung flips from $-\$49$ to $+\$54$, the savings rung stays at $-\$60$, and the best flat rate moves from a quarter of the social cost of carbon to the social cost of carbon itself. The $\$49$ and the $\$47$ downstream of it are therefore calibration-contingent down to their signs. What survives the calibration is the direction rule of Proposition 5.1, overdistortion pulling the carbon price down and underdistortion pushing it up, and the savings pull, which is anchor-invariant. The externality is a constant marginal damage, so the pollution-stock dynamics of Section 4.3, which move the *level* of the tax rather than the wedge structure measured here, are a separate exercise. The full optimum requires homothetic preferences, so the necessity calibration is read off the exact linear-tax reduction rather than the full nonlinear mechanism. The log consumption index is required by the balanced-growth homogeneity the solver relies on. The energy good carries a single carbon intensity, so within-energy heterogeneity is outside the model. And the frozen-world exercise freezes the savings wedge through retirement as well as through the working life, which is the right reading of a capital income tax that does not stop at 65 but is nonetheless a modeling choice doing real work: it is what keeps the deviation growing after T_R and what pulls the optimal flat rate down.

None of these bears on the qualitative results, though the results rest on different footings. The premium's proportionality to the labor wedge is structural, the identity $\varkappa l^\alpha \tau^y / (1 - \tau^y)$, as is the near-sufficiency of a flat Pigouvian price under an optimal income tax, which follows from the objective's flatness in the rate. The rotation itself is structural too: Lemma 5.3 forces the savings multiplier to reverse along every history whatever the draw, and Appendix B.7 signs the direction of that reversal at a degenerate draw; the solved profiles reproduce both. What is calibration-contingent is the premium's size, the rotation's magnitude, and the age at which the crossing falls, which the degenerate-draw argument does not date under a random walk. They rest on energy being a leisure complement ($\varkappa > 0$), the empirical input from West and Williams (2007), and on the capital income tax rate. The premium scales roughly linearly in $\varkappa$, so $\varkappa = 0.45$ about doubles the mid-career premium and $\varkappa = 0$ collapses it to the Atkinson-Stiglitz benchmark; the welfare stakes in the full-optimum world stay small across this range. That upper bound is generous. Applying West and Williams (2007)'s nonseparability component to gasoline alone, with no tempering for the composite, would put $\varkappa$ nearer 0.33.

7 Conclusion

A pollution externality placed inside a dynamic Mirrlees economy leaves the corrective and redistributive tasks separate. When preferences are weakly separable, the carbon wedge is at the social cost of carbon and the labor and savings wedges keep their standard forms, so the below-Pigouvian result that the Ramsey tradition attributes to fiscal distortions does not survive an optimal nonlinear income tax. In a setting in which the planner is free to allocate labor and consumption with private information on skill as its only constraint, only nonseparability in preferences justifies setting the carbon price away from the social cost of

carbon. The dirty good is overdistorted when it is a labor substitute and underdistorted when it is a labor complement, and the carbon price then becomes nonlinear and history dependent. The social cost of carbon rises above the Pigouvian price when working more raises the household's valuation of a cleaner environment, and falls below it when working more lowers that valuation.

But solving a model calibrated to US data, with energy as a stand-in for the dirty good, shows that the welfare cost of a flat instrument rather than an optimal nonlinear one is small. A flat instrument set at the social cost of carbon already captures 98% of the welfare gain from the optimal nonlinear instrument. This assumes distortions are optimized jointly. When labor and savings distortions are fixed at the levels the US tax system implies, the best flat rate falls to a quarter of the social cost of carbon, from \$190/tCO₂ to \$47/tCO₂, and captures only 1.5% of the welfare gain of a fully optimal nonlinear tax. Worse, a flat tax at the social cost of carbon reduces welfare, as does any flat tax above \$98/tCO₂. The rebate side answers a second question. The constrained optimum returns about 97% of carbon revenue as a lifetime dividend that is nearly uniform across histories, and against payments that scale with lifetime spending the net incidence is progressive at every flat rate, with the bottom two lifetime-earnings quintiles gaining even at the welfare-reducing Pigouvian rate. Within the class of instruments that leave the income tax untouched, the design question is the rate, not the rebate.

Both exercises answer the paper's opening question through different lenses, and which one applies is a judgment about the social welfare function as much as about the tax code. If the US income tax is already optimal in that sense, a flat carbon tax slightly above the social cost of carbon approximates the optimum well. If it is not, and nonlinear instruments are unavailable, the best flat carbon tax is \$47/tCO₂, a quarter of the social cost of carbon. I do not read that number as an explanation of observed carbon prices. The model prices a US household energy composite over a single cohort, no jurisdiction's carbon price has that base, and the level is not robust to the life-cycle calibration, down to its sign relative to damages. Under the GR16-implied anchors of Section 6.7, under which the US schedule underdistorts labor over most of working life, the best flat rate is the social cost of carbon itself (Table 5). If nonlinear instruments were available, the wedge would price carbon at an effective \$107/tCO₂, still below the social cost of carbon but over twice the best flat rate. Instrument linearity therefore dampens climate ambition when the income tax cannot be optimized. Either move the carbon tax closer to a nonlinear form, or move the income tax closer to optimality so that a more ambitious flat carbon tax suffices.

Three questions stay open. The paper characterizes wedges, and implementation is settled only for the flat instrument, whose rebate side Section 6.6 measures; the nonlinear carbon wedge would require tariffs conditioned on purchase histories, which no government runs. Where income distortions are optimal that gap costs little, but where they are badly misset the nonlinear repair is where the welfare sits and implementation becomes the binding constraint.

The level of the social cost of carbon is exogenous here because a single cohort in a single country cannot endogenize it; overlapping generations would let carbon dynamics feed back into the cohort that emits. And outside the environmental context, a commodity wedge that varies with age raises the question of age-dependent commodity taxation. Age-based pricing of goods and services is already common practice, and whether it is optimal is unsettled.

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A Optimal wedge derivations

A.1 The problem and first-order conditions

Because $U(c, q, l, S)$ can be written as $U(c, q, y/\theta, S)$, I can write U_θ as $-U_l l/\theta$. The Hamiltonian of the planner's problem of Section 4.1 is as follows, with $f_{2,t} \equiv \partial f_t / \partial \theta_{t-1}$ the persistence derivative of the transition density (Assumption 3.2), written f_2 where the date is unambiguous:

$$\begin{aligned} \mathcal{H} = & \left[c + q - \theta l + \frac{1}{R} V_{t+1}(w, w_2, \theta, \mathbf{S}) \right] f \\ & + \psi \left[-U_l(c, q, y/\theta, S) \frac{l}{\theta} + \beta w_2 \right] \\ & - \lambda_1 u(\theta) \bar{\alpha}_t f_t \\ & + \lambda_2 u(\theta) f_2 \\ & + \phi [u - U(c, q, y/\theta, S) - \beta w] \\ & - \mu [S - S(\dots, Q)] \end{aligned}$$

This yields the following first-order conditions, in which $S_q \equiv \partial S / \partial Q$ is the marginal effect of current aggregate emissions on the current pollution stock.

$$\begin{aligned} \frac{\partial \mathcal{H}}{\partial c} : & f - \psi U_c \frac{l}{\theta} = \phi U_c \\ \frac{\partial \mathcal{H}}{\partial q} : & f - \psi U_q \frac{l}{\theta} + \left[\mu S_q + \frac{1}{R} \frac{\partial V_{t+1}}{\partial Q} \right] f = \phi U_q \\ \frac{\partial \mathcal{H}}{\partial l} : & -\theta f - \phi U_l = \frac{1}{\theta} \psi [U_{ll} l + U_l] \\ \frac{\partial \mathcal{H}}{\partial u} : & \dot{\psi} = \lambda_1 \bar{\alpha}_t f - \lambda_2 f_2 - \phi \\ \frac{\partial \mathcal{H}}{\partial w} : & \frac{1}{R} \frac{\partial V_{t+1}}{\partial w} f = \phi \beta \\ \frac{\partial \mathcal{H}}{\partial w_2} : & \frac{1}{R} \frac{\partial V_{t+1}}{\partial w_2} f = -\psi \beta \\ \frac{\partial \mathcal{H}}{\partial S} : & \int_0^\infty \left(-\psi U_{Sl} \frac{l}{\theta} - \phi U_S \right) d\theta = \mu \end{aligned}$$

Here $S_q \equiv \partial S / \partial Q$ is the marginal effect of current aggregate emissions on the current pollution stock and the density f in the first-order condition for q comes from differentiating the aggregate

$Q = \int q(\theta)f(\theta) d\theta$ which enters both the current accumulation constraint and, as the newest entry of the emission state $\mathbf{S}$, the continuation value through $\partial V_{t+1}/\partial Q \equiv \partial V_{t+1}/\partial Q_t$. The scalar current stock S is not itself an argument of $V_{t+1}(\cdot, \mathbf{S})$; the continuation value sees current emissions only through $\mathbf{S}$, via $\partial V_{t+1}/\partial Q$, which is why no V term appears in the first-order condition for S . Define the full shadow price of current emissions,

$$\nu_t \equiv \mu_t S_q + \frac{1}{R} \frac{\partial V_{t+1}}{\partial Q}. \quad (13)$$

The three wedges can be derived from here. The labor wedge is derived in Section A.3, the savings wedge in Section A.4, and the commodity wedge in Section A.5.

A.2 Validity of the recursive formulation

Section 4.1 takes its recursive form from Kapička (2013), whose consumption set is an arbitrary convex and compact set, so his reduction covers the consumption vector (c, q) . The externality adds two ingredients his environment lacks: the pollution stock enters preferences and the planner internalizes its accumulation from aggregate emissions. This subsection shows that neither disturbs the validity of the recursive approach.

An allocation is a collection of measurable functions $c_t(\theta^t)$, $q_t(\theta^t)$, and $y_t(\theta^t)$ for $t = 0, \dots, T$. Aggregate emissions at an allocation are

$$Q_t = \int_0^\infty \mathbb{E}_0[q_t(\theta^t)|\theta] dF_0(\theta) \quad t = 0, \dots, T$$

and the implied pollution stocks are $S_t = S(Q_{t-s}, \dots, Q_t)$, with pre-period emissions given. As in the main text, $\Phi_j \equiv \partial S_{t+j}/\partial Q_t$, with $\Phi_j = 0$ for $j > s$ by the truncation of the carbon cycle. Multipliers dated beyond T are zero.

Lemma A.1 (Incentive constraints with an aggregate pollution stock).

Fix a deterministic emission sequence $(Q_0, \dots, Q_T)$ and let $S_t = S(Q_{t-s}, \dots, Q_t)$. Define the dated utility

$$\tilde{U}_t(c, q, l) \equiv U(c, q, l, S_t)$$

Given this sequence, (i) an allocation is incentive compatible in the economy with pollution if and only if it is incentive compatible in the no-externality economy, the $\nu_t = 0$ case of Section 4.1 in which the dirty good is dirty in name only, with period utility $\tilde{U}_t$, and (ii) the local incentive constraint is

$$\dot{u}_t(\theta^t) = U_\theta \left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t \right) + \beta w_{2,t}(\theta^t)$$

with no term in the pollution stock.

Proof. Step 1: deviations leave the pollution stock unchanged. Incentive compatibility compares one agent's payoffs under truthful reporting and under a reporting strategy σ , holding the allocation and all other reports fixed. Aggregate emissions integrate the allocation over the population and a single agent has measure zero in that integral, so Q_t and with it S_t take the same values under truth-telling and under the deviation. Every payoff entering the incentive constraint is therefore evaluated at the same pollution-stock sequence.

Step 2: the incentive constraints are those of a dated no-externality economy. By Step 1, the incentive constraint of Section 3 reads for every initial type θ and every reporting strategy σ ,

$$\begin{aligned} & \mathbb{E}_0 \left[\sum_{t=0}^T \beta^t U \left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t \right) \middle| \theta \right] \\ & \geq \mathbb{E}_0 \left[\sum_{t=0}^T \beta^t U \left(c_t(\sigma^t(\theta^t)), q_t(\sigma^t(\theta^t)), \frac{y_t(\sigma^t(\theta^t))}{\theta_t}, S_t \right) \middle| \theta \right] \end{aligned}$$

with the same fixed S_t on both sides. Substituting the definition of $\tilde{U}_t$ on both sides,

$$\begin{aligned} & \mathbb{E}_0 \left[\sum_{t=0}^T \beta^t \tilde{U}_t \left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t} \right) \middle| \theta \right] \\ & \geq \mathbb{E}_0 \left[\sum_{t=0}^T \beta^t \tilde{U}_t \left(c_t(\sigma^t(\theta^t)), q_t(\sigma^t(\theta^t)), \frac{y_t(\sigma^t(\theta^t))}{\theta_t} \right) \middle| \theta \right] \end{aligned}$$

The two displays are the same statement, so an allocation satisfies one if and only if it satisfies the other. The second is the incentive constraint of the no-externality economy with period utility $\tilde{U}_t$, which proves (i).

Step 3: the local constraint. Part (i) places the reporting problem in the environment of Kapička (2013) with period utility $\tilde{U}_t$. His requirements bear on the period utility date by date and $\tilde{U}_t$ inherits them from Assumption 3.1 because S_t enters U as a fixed parameter: $\tilde{U}_t$

is twice continuously differentiable, increasing in both goods, jointly concave in (c, q, l) , and differentiable in the type. The dating of $\tilde{U}_t$ adds nothing beyond the date dependence Section 4.1 already carries in f_t and $\bar{\alpha}_t$ and the horizon is finite, so the one-shot-deviation reduction of Fernandes and Phelan (2000) and the envelope theorem of Kapička (2013) apply date by date:

$$\dot{u}_t(\theta^t) = \frac{\partial}{\partial \theta_t} \tilde{U}_t \left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t} \right) + \beta w_{2,t}(\theta^t)$$

The derivative passes through the pollution stock, which does not vary with the agent's type:

$$\frac{\partial}{\partial \theta_t} \tilde{U}_t \left(c_t, q_t, \frac{y_t}{\theta_t} \right) = U_\theta \left(c_t, q_t, \frac{y_t}{\theta_t}, S_t \right) + U_S \left(c_t, q_t, \frac{y_t}{\theta_t}, S_t \right) \frac{\partial S_t}{\partial \theta_t} = U_\theta \left(c_t, q_t, \frac{y_t}{\theta_t}, S_t \right)$$

since $\partial S_t / \partial \theta_t = 0$. This proves (ii). $\square$

The lemma is the formal version of the observation in Section 3 that the pollution stock enters both sides of the incentive constraint identically. To the agent, the pollution stock is a date-specific preference parameter, not a state she controls. The planner is different because it moves the pollution stock. The next proposition shows what internalizing it costs.

Proposition A.1 (Recursive formulation with an externality).

Let the Kapička (2013) reduction be valid for the two-good economy without an externality. The relaxed planner problem with pollution then admits the recursive formulation of Section 4.1 with the emission window $\mathbf{S}_- = (Q_{t-s}, \dots, Q_{t-1})$ as the only new state, the accumulation constraint as the only new restriction, and the first-order system of the recursive problem as that of the sequence problem. In that system, (i) given the emission price ν_t , every date- t history solves the no-externality problem with period utility $\tilde{U}_t$ and the resource cost of the dirty good raised from 1 to $1 + \nu_t$; (ii) the pollution-stock multiplier μ_t is common to all date- t histories and equals the damage integral over the date- t cross-section; and (iii) the emission price collects the discounted pollution-stock multipliers, $\nu_t = \sum_{j=0}^s R^{-j} \mu_{t+j} \Phi_j$.

Proof. Step 1: the aggregates are deterministic and internalizing them is a finite-dimensional addition. Skills are the only shocks and they are idiosyncratic, allocations are measurable functions of skill histories, and agents form a continuum, so the cross-sectional distribution of histories at each date coincides with the ex-ante law of a single agent's history. Aggregate emissions Q_t are therefore numbers known at date 0 and so are the pollution stocks S_t . The relaxed planner problem is then equivalent to a joint choice of the allocation and two vectors $(Q_0, \dots, Q_T)$ and $(S_0, \dots, S_T)$, subject to the consistency constraint $Q_t = \int_0^\infty \mathbb{E}_0[q_t(\theta^t) | \theta] dF_0(\theta)$ and the accumulation constraint $S_t = S(Q_{t-s}, \dots, Q_t)$ at each date. Substituting the two

constraints back into the objective recovers the original problem, so the two problems select the same allocations.

Step 2: Lagrangian. Attach the multiplier ν_t in present value ν_t/R^t to the date- t consistency constraint and μ_t in present value μ_t/R^t to the date- t accumulation constraint, with the sign convention of the Hamiltonian in Appendix A.1. The Lagrangian of the relaxed problem is

$$\begin{aligned} \mathcal{L} = \mathcal{L}^{alloc}(\{c_t, q_t, y_t\}; \{S_t\}) &+ \sum_{t=0}^T \frac{\nu_t}{R^t} \left(\int_0^\infty \mathbb{E}_0[q_t(\theta^t)|\theta] dF_0(\theta) - Q_t \right) \\ &- \sum_{t=0}^T \frac{\mu_t}{R^t} [S_t - S(Q_{t-s}, \dots, Q_t)] \end{aligned}$$

where $\mathcal{L}^{alloc}$ collects every term of the Section 4.1 Lagrangian: the resource cost, the local incentive constraints, and the promise- and threat-keeping constraints, all with period utility $\tilde{U}_t$, as Lemma A.1 requires. The pollution stocks enter $\mathcal{L}^{alloc}$ only as the dates of $\tilde{U}_t$. The aggregates enter only the two constraint blocks.

Step 3: given the aggregates, the allocation problem is the Section 4.1 problem, re-dated and re-priced. Fix $(Q_t, S_t, \nu_t)_{t=0}^T$. Merging the consistency block with the resource cost turns the date- t flow cost of a history into

$$c_t(\theta^t) + (1 + \nu_t) q_t(\theta^t) - y_t(\theta^t)$$

plus the term $-\nu_t Q_t$, a constant with respect to the allocation. The allocation problem is therefore the no-externality problem with two changes: period utility $\tilde{U}_t$ and dirty-good resource cost $1 + \nu_t$. Both stay inside the class the Section 4.1 reduction covers. Dating the utility adds nothing beyond the date dependence already present in f_t and $\bar{\alpha}_t$, and over the finite horizon the recursion is a backward induction from $V_{T+1} = 0$, so no stationarity is used. Re-pricing keeps the period cost linear, hence continuous and convex, and increasing in q for $\nu_t > -1$. Since ν_t will turn out to be the social cost of carbon, this asks only that the carbon price not subsidize the dirty good by more than its unit resource cost. The factorization of the incentive-constraint set into current controls and continuation promises never involves the objective, so the re-pricing leaves it untouched. Given the aggregates, the allocation problem is therefore recursive in the Section 4.1 states $(\hat{w}, \hat{w}_2, \theta_-)$ and its value functions solve the Bellman equation of that section with $\tilde{U}_t$ and the re-priced dirty good. This is (i).

Step 4: first-order conditions for the aggregates. The pollution stock S_t appears in $\mathcal{L}$ in two places: the accumulation block, with derivative $-\mu_t/R^t$, and the date- t utility terms of $\mathcal{L}^{alloc}$,

at every date- t history. Within a history, utility enters two constraints: the local incentive constraint with costate ψ and the definition of u with multiplier ϕ , exactly as in the Hamiltonian of Appendix A.1. Differentiating each with respect to S ,

$$\frac{\partial}{\partial S} \psi \left[-U_l \left(c, q, \frac{y}{\theta}, S \right) \frac{l}{\theta} + \beta w_2 \right] = -\psi U_{Sl} \frac{l}{\theta} \quad \frac{\partial}{\partial S} \phi \left[u - U \left(c, q, \frac{y}{\theta}, S \right) - \beta w \right] = -\phi U_S$$

The pollution stock is common to all date- t histories, so stationarity of $\mathcal{L}$ in S_t sums these contributions across the date- t cross-section,

$$\mu_t = \int_0^\infty \left(-\psi U_{Sl} \frac{l}{\theta} - \phi U_S \right) d\theta$$

where the integral runs over the date- t population of skill histories. This is the first-order condition for S in Appendix A.1 whose type integral is to be read over that cross-section: at $t = 0$ the cross-section is F_0 and the reading is literal and at later dates the social-cost-of-carbon formula applies it by writing the date- $(t + j)$ damage integrals as population averages, the $\mathbb{E}_t \int(\cdot) f_{t+j} d\theta$ terms of Proposition 4.2. Because there is no aggregate risk, these averages are deterministic: $\mathbb{E}_t$ denotes a cross-section, the same for every date- t history, not a forecast specific to the current history. In particular, μ_t is one number per date, which is (ii).

Current aggregate emissions Q_t appear in the consistency block with derivative $-\nu_t/R^t$ and, in the accumulation blocks of dates t , through $t + s$ since $S_\tau = S(Q_{\tau-s}, \dots, Q_\tau)$ carries Q_t as an argument exactly when $t \leq \tau \leq t + s$. Stationarity of $\mathcal{L}$ in Q_t gives

$$-\frac{\nu_t}{R^t} + \sum_{\tau=t}^{t+s} \frac{\mu_\tau}{R^\tau} \frac{\partial S_\tau}{\partial Q_t} = 0 \quad \Longleftrightarrow \quad \nu_t = \sum_{j=0}^s \frac{1}{R^j} \mu_{t+j} \Phi_j$$

which is (iii).

Step 5: recursive problem of Section 4.1 encodes the same system. Two checks remain: the state and the first-order conditions. From date t on, the aggregates matter only through the dated utilities $\tilde{U}_\tau$ for $\tau \geq t$, hence through $S_\tau = S(Q_{\tau-s}, \dots, Q_\tau)$ for $\tau \geq t$, and each of these depends on past aggregates only through the window $\mathbf{S}_- = (Q_{t-s}, \dots, Q_{t-1})$, the rest being current and future choices. Appending $\mathbf{S}_-$ to the Section 4.1 state is therefore sufficient and the window updates recursively, appending Q_t and dropping Q_{t-s} . The conditions: differentiate the Hamiltonian of Appendix A.1. The condition for q carries the extra term $\left[\mu_t S_q + \frac{1}{R} \frac{\partial V_{t+1}}{\partial Q} \right] f$,

so the dirty good's effective resource cost is $1 + \nu_t$ with $\nu_t = \mu_t S_q + \frac{1}{R} \frac{\partial V_{t+1}}{\partial Q_t}$, and the envelope chain of that appendix resolves $\frac{\partial V_{t+1}}{\partial Q_t} = \sum_{j=1}^s R^{-(j-1)} \mu_{t+j} \Phi_j$, so that

$$\nu_t = \mu_t \Phi_0 + \frac{1}{R} \sum_{j=1}^s \frac{1}{R^{j-1}} \mu_{t+j} \Phi_j = \sum_{j=0}^s \frac{1}{R^j} \mu_{t+j} \Phi_j$$

The condition for S is the population damage integral of Step 4. The conditions for (c, l, u, w, w_2) contain no aggregate term and are those of the Section 4.1 problem with $\tilde{U}_t$, matching Step 3. The two first-order systems coincide condition by condition. $\square$

The argument concerns the relaxed problem, as everywhere in the paper: the envelope condition stands in for global incentive compatibility, and the sufficiency caveat of Section 3 is unchanged since Lemma A.1 maps the externality economy into the class for which that caveat was stated. Section 6 verifies global incentive compatibility numerically at the computed optima.

A.3 Labor wedge

Dividing the first-order condition for c of Appendix A.1 by U_c and using $\gamma_c \equiv U_{cl}/U_c$, with the multipliers $\phi, \psi, \lambda_1, \lambda_2$ and the costate boundary conditions $\psi(0) = \psi(\infty) = 0$ as introduced there,

$$\phi = \frac{f}{U_c} - \frac{\psi}{\theta} \gamma_c. \quad (14)$$

From the first-order condition for l and since $U_{ll}l + U_l = U_l(1 + \epsilon)$ with $\epsilon \equiv U_{ll}l/U_l$,

$$-\theta f - \phi U_l = \frac{\psi}{\theta} U_l(1 + \epsilon) \implies \phi = -\frac{\theta f}{U_l} - \frac{\psi}{\theta}(1 + \epsilon)$$

Equating the two expressions for ϕ and collecting the terms in ψ ,

$$\frac{f}{U_c} + \frac{\theta f}{U_l} = -\frac{\psi}{\theta}(1 + \epsilon - \gamma_c)$$

The labor wedge is $1 - \tau^y = -U_l/(\theta U_c)$, so $\theta U_c/U_l = -1/(1 - \tau^y)$, making the left-hand side $\frac{f}{U_c} \left(1 - \frac{1}{1 - \tau^y}\right) = -\frac{f}{U_c} \frac{\tau^y}{1 - \tau^y}$. Substituting and rearranging,

$$\frac{\tau^y}{1 - \tau^y} = (1 + \epsilon - \gamma_c) \frac{\psi U_c}{\theta f} = A_t(\theta) \frac{\psi U_c}{\theta f}, \quad A_t(\theta) \equiv 1 + \epsilon_t(\theta) - \gamma_{c,t}(\theta) \quad (15)$$

Next, substitute ϕ from (14) into $\dot{\psi} = \lambda_1 \bar{\alpha}_t f - \lambda_2 f_2 - \phi$,

$$\dot{\psi} - \frac{\gamma_c}{\theta} \psi = \lambda_1 \bar{\alpha}_t f - \lambda_2 f_2 - \frac{f}{U_c}$$

This is a linear first-order ordinary differential equation. Multiplying by the integrating factor $\exp \left[- \int^\theta \gamma_c(\tilde{x}) d\tilde{x}/\tilde{x} \right]$ and integrating from θ to ∞ with the boundary condition $\psi(\infty) = 0$,

$$\psi(\theta) = \int_\theta^\infty \exp \left[- \int_\theta^x \gamma_c(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] \left(\frac{f(x)}{U_c(x)} - \lambda_1 \bar{\alpha}_t(x) f(x) + \lambda_2 f_2(x) \right) dx. \quad (16)$$

The remaining boundary condition $\psi(0) = 0$ pins down λ_1 ,

$$\lambda_{1,t} = \frac{\int_0^\infty \exp \left[- \int_0^x \gamma_{c,t}(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] \left(\frac{1}{U_{c,t}(x)} f_t(x) + \lambda_{2,t} f_{2,t}(x) \right) dx}{\int_0^\infty \exp \left[- \int_0^x \gamma_{c,t}(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] \bar{\alpha}_t(x) f_t(x) dx}. \quad (17)$$

Substituting (16) into (15),

$$\frac{\tau^y}{1 - \tau^y} = (1 + \epsilon - \gamma_c) \frac{U_c(\theta)}{\theta f(\theta)} \int_\theta^\infty \exp \left[- \int_\theta^x \gamma_c(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] \left(\frac{f(x)}{U_c(x)} - \lambda_1 \bar{\alpha}_t(x) f(x) + \lambda_2 f_2(x) \right) dx$$

I split the integral into the $[f/U_c - \lambda_1 \bar{\alpha}_t f]$ part, which yields the intratemporal term $A_t B_t C_t$, and the $\lambda_2 f_2$ part, which yields the intertemporal term. The three factors are collected in (19) below: B_t is the inverse hazard rate $(1 - F_t)/(\theta f_t)$ and C_t the average of the redistributive term over types above θ , so the $1 - F_t(\theta)$ inserted to make C_t an average cancels against B_t and the product is the integral itself. Splitting it this way is what puts the decomposition in the form Golosov et al. (2016) use. For the first part, carry $U_c(\theta)/U_c(x)$ into the integrand. Along the optimal allocation $U_{c,t}$ depends on the type through $c(\theta)$, $q(\theta)$, and $l(\theta) = y(\theta)/\theta$; the pollution stock is common to the date- t cross-section, so $\partial S_t / \partial \theta = 0$ and no term in U_{cS} appears. Hence,

$$\frac{d}{d\theta} \ln U_{c,t} = \frac{U_{cc} \dot{c} + U_{cq} \dot{q} + U_{cl} \dot{l}}{U_c} = -\sigma_t \frac{\dot{c}}{c} - \sigma_{cq,t} \frac{\dot{q}}{q} + \gamma_c \frac{\dot{l}}{l}$$

Since $l = y/\theta$ gives $\dot{l}/l = \dot{y}/y - 1/\theta$, integrating from θ to x and multiplying by the integrating factor gives the following:

$$\frac{U_{c,t}(\theta)}{U_{c,t}(x)} \exp \left[- \int_{\theta}^x \gamma_c(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] = \exp \left[\int_{\theta}^x \left(\sigma_t(\tilde{x}) \frac{\dot{c}_t(\tilde{x})}{c_t(\tilde{x})} + \sigma_{cq,t}(\tilde{x}) \frac{\dot{q}_t(\tilde{x})}{q_t(\tilde{x})} - \gamma_c(\tilde{x}) \frac{\dot{y}_t(\tilde{x})}{y_t(\tilde{x})} \right) d\tilde{x} \right], \quad (18)$$

Here, the $\gamma_c/\tilde{x}$ term from $\dot{l}/l$ cancels the integrating factor. The cross-curvature $\sigma_{cq,t}$ enters because, under nonseparability, extracting resources from a higher type shifts the marginal utility of the numéraire through the adjustment of both goods. The first part therefore equals $A_t B_t C_t$. For the second part, two preliminaries pin down $\lambda_{2,t}$. The first-order condition for w_2 rearranges to $\partial V_{t+1}/\partial w_2 = -\beta R \psi/f$, and the envelope theorem applied to the period- $(t+1)$ problem gives $\partial V_{t+1}/\partial \hat{w}_2 = -\lambda_{2,t+1}$, the multiplier with which threat keeping enters the Lagrangian. Equating the two expressions and shifting the date back one period,

$$\lambda_{2,t}(\theta^{t-1}) = \beta R \frac{\psi_{t-1}}{f_{t-1}}$$

Next, evaluating (15) at $t-1$ and solving for the costate,

$$\frac{\psi_{t-1}}{f_{t-1}} = \frac{\tau_{t-1}^y}{1 - \tau_{t-1}^y} \frac{\theta_{t-1}}{U_{c,t-1} A_{t-1}}$$

The $\lambda_2 f_2$ block of the wedge is $\lambda_{2,t}$, a constant in the current type, times the same outer structure as the first part,

$$A_t(\theta) \frac{U_{c,t}(\theta)}{\theta f_t(\theta)} \lambda_{2,t} \int_{\theta}^{\infty} \exp \left[- \int_{\theta}^x \gamma_{c,t}(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] f_{2,t}(x) dx$$

Substituting the two preliminaries and collecting the ratios,

$$\beta R \frac{\tau_{t-1}^y}{1 - \tau_{t-1}^y} \frac{A_t(\theta)}{A_{t-1}} \frac{U_{c,t}(\theta)}{U_{c,t-1}} \frac{\theta_{t-1} \int_{\theta}^{\infty} \exp \left[- \int_{\theta}^x \gamma_{c,t}(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] f_{2,t}(x) dx}{\theta f_t(\theta)} = \beta R \frac{\tau_{t-1}^y}{1 - \tau_{t-1}^y} D_t(\theta)$$

Collecting both parts,

$$\frac{\tau^y}{1 - \tau^y} = A_t(\theta) B_t(\theta) C_t(\theta) + \beta R \frac{\tau_{t-1}^y}{1 - \tau_{t-1}^y} D_t(\theta)$$

$$\begin{aligned}
A_t(\theta) &= 1 + \epsilon_t(\theta) - \gamma_{c,t}(\theta) \\
B_t(\theta) &= \frac{1 - F_t(\theta)}{\theta f_t(\theta)} \\
C_t(\theta) &= \int_{\theta}^{\infty} \exp \left[\int_{\theta}^x \left(\sigma_t(\tilde{x}) \frac{\dot{c}_t(\tilde{x})}{c_t(\tilde{x})} + \sigma_{cq,t}(\tilde{x}) \frac{\dot{q}_t(\tilde{x})}{q_t(\tilde{x})} - \gamma_{c,t}(\tilde{x}) \frac{\dot{y}_t(\tilde{x})}{y_t(\tilde{x})} \right) d\tilde{x} \right] (1 - \lambda_{1,t} \bar{\alpha}_t(x) U_{c,t}(x)) \frac{f_t(x) dx}{1 - F_t(\theta)} \\
D_t(\theta) &= \frac{A_t(\theta)}{A_{t-1}} \frac{U_{c,t}(\theta)}{U_{c,t-1}} \frac{\theta_{t-1} \int_{\theta}^{\infty} \exp \left[- \int_{\theta}^x \gamma_{c,t}(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] f_{2,t}(x) dx}{\theta f_t(\theta)}
\end{aligned} \tag{19}$$

This is Golosov et al. (2016)'s labor-wedge decomposition, with the cross-curvature $\sigma_{cq,t}$ in C_t .

A.4 Savings wedge

The savings wedge follows from the same first-order conditions of Appendix A.1. I define the incentive credit χ_t , the share of the cost of delivering utility through the numéraire that the planner recoups because consumption complements labor, relaxing the incentive constraint:

$$\chi_t(\theta^t) \equiv \frac{\gamma_{c,t}}{1 + \epsilon_t - \gamma_{c,t}} \frac{\tau_t^y}{1 - \tau_t^y}$$

Using $\frac{\tau_t^y}{1 - \tau_t^y} = \frac{\psi U_c}{\theta f} (1 + \epsilon_t - \gamma_{c,t})$, $\phi = f/U_c - \psi \gamma_c/\theta$ from (14) can be rewritten as follows:

$$\frac{\phi}{f} = \frac{1}{U_c} (1 - \gamma_c \psi U_c / (\theta f)) = \frac{1 - \chi_t}{U_{c,t}}. \tag{20}$$

$\chi_t > 0$ when the numéraire is complementary with labor and labor is taxed and zero when preferences are separable in the numéraire against labor or labor is undistorted. The first-order condition for w and the envelope condition $\partial V_t / \partial \hat{w} = \lambda_{1,t}$ one period ahead give

$$\lambda_{1,t+1}(\theta^t) = \beta R \frac{\phi(\theta)}{f(\theta)} = \beta R \frac{1 - \chi_t}{U_{c,t}}. \tag{21}$$

A second expression for $\lambda_{1,t}$ comes from integrating the costate equation over the whole type space. With $\psi(0) = \psi(\infty) = 0$,

$$0 = \int_0^{\infty} \dot{\psi} d\theta = \lambda_{1,t} \int_0^{\infty} \bar{\alpha}_t f d\theta - \lambda_{2,t} \int_0^{\infty} f_2 d\theta - \int_0^{\infty} \phi d\theta$$

The Pareto weights are normalized so that $\int \bar{\alpha}_t f d\theta = 1$, and $\int f_t(\theta|\theta_-) d\theta = 1$ for every θ_- implies $\int f_{2,t} d\theta = 0$. Hence $\lambda_{1,t} = \int_0^\infty \phi d\theta$, which by (20) is

$$\lambda_{1,t} = \mathbb{E}_{t-1} \left[\frac{1 - \chi_t}{U_{c,t}} \right], \quad (22)$$

where $\mathbb{E}_{t-1}$ integrates over θ_t conditional on θ_{t-1} . The multiplier on promise keeping is set before the current draw is known. Evaluating (22) at $t + 1$ and equating with (21),

$$\frac{1 - \chi_t}{U_{c,t}} = \frac{1}{\beta R} \mathbb{E}_t \left[\frac{1 - \chi_{t+1}}{U_{c,t+1}} \right]. \quad (23)$$

Defining the incentive-adjusted marginal utility $\widehat{U}_{c,t} \equiv U_{c,t}/(1 - \chi_t)$,

$$\frac{1}{\widehat{U}_{c,t}} = \frac{1}{\beta R} \mathbb{E}_t \left[\frac{1}{\widehat{U}_{c,t+1}} \right]$$

This is the inverse Euler equation in the adjusted marginal utility $\widehat{U}_c$. Delivering a util through the numéraire costs $1/U_{c,t}$ in resources and, when $U_{cl} > 0$, the unit lowers the information rent $-U_l l/\theta$, an incentive gain worth $\chi_t/U_{c,t}$.

A.5 Commodity wedge

As in the previous proofs, starting from the first-order conditions of Appendix A.1,

$$\phi = \frac{f}{U_c} - \frac{\psi U_{cl} \frac{l}{\theta}}{U_c}$$

Because c and q share the same ϕ ,

$$\phi = \frac{f}{U_q} - \frac{\psi U_{ql} \frac{l}{\theta}}{U_q} + \frac{\nu_t f}{U_q}$$

Hence,

$$\frac{f}{U_q} - \frac{\psi U_{ql} \frac{l}{\theta}}{U_q} + \frac{\nu_t f}{U_q} = \frac{f}{U_c} - \frac{\psi U_{cl} \frac{l}{\theta}}{U_c}$$

Multiplying by U_q and U_c ,

$$U_c \left(f - \psi U_{ql} \frac{l}{\theta} + \nu_t f \right) = U_q \left(f - \psi U_{cl} \frac{l}{\theta} \right)$$

Using $\gamma_c \equiv \frac{U_{cl} l}{U_c}$ and $\gamma_q \equiv \frac{U_{ql} l}{U_q}$, or, equivalently, $\gamma_c U_c = U_{cl} l$ and $\gamma_q U_q = U_{ql} l$,

$$U_c \left(f - \frac{\psi}{\theta} \gamma_q U_q + \nu_t f \right) = U_q \left(f - \frac{\psi}{\theta} \gamma_c U_c \right)$$

Then, putting the γ terms (as well as the externality term) on one side,

$$U_c f - U_q f = \frac{\psi}{\theta} U_c U_q (\gamma_q - \gamma_c) - \nu_t U_c f$$

Finally, dividing by $U_q f$,

$$\frac{U_c}{U_q} - 1 = \frac{\psi U_c}{\theta f} (\gamma_q - \gamma_c) - \nu_t \frac{U_c}{U_q}$$

Note that, by definition, $1 + \tau^q = U_q/U_c$, so $\frac{U_c}{U_q} - 1 = -\frac{\tau^q}{1 + \tau^q}$. Multiplying through by -1 ,

$$\frac{\tau^q}{1 + \tau^q} = \frac{\psi U_c}{\theta f} (\gamma_c - \gamma_q) + \nu_t \frac{U_c}{U_q}$$

Since $\frac{U_c}{U_q} = \frac{1}{1 + \tau^q}$,

$$\frac{\tau^q}{1 + \tau^q} = \frac{\psi U_c}{\theta f} (\gamma_c - \gamma_q) + \frac{\nu_t}{1 + \tau^q}$$

Rearranging,

$$\frac{\tau^q - \nu_t}{1 + \tau^q} = \frac{\psi U_c}{\theta f} (\gamma_c - \gamma_q)$$

The prefactor $\psi U_c/(\theta f)$ is the same object that drives the labor wedge. The first-order conditions for c and l and the costate equation for ψ are untouched by the externality, so the

integrating-factor steps of Appendix A.3 carry over verbatim. Solving (15) for the prefactor and substituting the $ABCD$ decomposition,

$$\frac{\psi U_c}{\theta f} = \frac{1}{A_t} \frac{\tau^y}{1 - \tau^y} = B_t C_t + \beta R \frac{\tau_{t-1}^y}{1 - \tau_{t-1}^y} \frac{D_t}{A_t}$$

Multiplying by the complementarity gap $\gamma_{c,t} - \gamma_{q,t}$,

$$\frac{\tau_t^q - \nu_t}{1 + \tau_t^q} = (\gamma_{c,t} - \gamma_{q,t}) \left[B_t(\theta) C_t(\theta) + \beta R \frac{\tau_{t-1}^y}{1 - \tau_{t-1}^y} \frac{D_t(\theta)}{A_t(\theta)} \right]$$

with A_t, B_t, C_t, D_t as in Appendix A.3. Alternatively, multiplying $\frac{\psi U_c}{\theta f} = \frac{1}{A_t} \frac{\tau^y}{1 - \tau^y}$ by the gap,

$$\frac{\tau_t^q - \nu_t}{1 + \tau_t^q} = \frac{\gamma_{c,t} - \gamma_{q,t}}{1 + \epsilon_t - \gamma_{c,t}} \frac{\tau_t^y}{1 - \tau_t^y}$$

The commodity wedge inherits every dynamic feature of the labor wedge. Its shape is that of the labor wedge, scaled by the elasticity term. The externality leaves this block untouched and enters only by replacing τ^q with the deviation $\tau^q - \nu_t$ on the left.

A.6 The social cost of carbon

Now consider the shadow price of current emissions ν_t of (13). Its first piece is the date- t pollution-stock multiplier μ_t , the marginal damage of a unit of the current pollution stock. Substituting ϕ from the first-order condition for c of Appendix A.1 into the first-order condition for S , then dividing and multiplying by f ,

$$\mu_t = \int_0^\infty \left(-\psi U_{Sl} \frac{l}{\theta} - \left[\frac{f}{U_c} - \frac{\psi U_{cl} l}{U_c} \right] U_S \right) d\theta = \int_0^\infty \left(-\frac{\psi}{\theta f} U_{Sl} l - \left[1 - \frac{\psi}{\theta f} U_{cl} l \right] \frac{U_S}{U_c} \right) f d\theta$$

Rewrite $\psi/\theta f$ by noting that

$$\frac{\psi}{\theta f} = \left(\frac{\tau^y}{1 - \tau^y} \right) \frac{1}{U_c} \frac{1}{1 + \epsilon - \gamma_c}$$

Hence, also defining $\xi \equiv \frac{U_{Sl} l}{U_S}$,

$$\begin{aligned}
\mu_t &= \int \left(- \left(\frac{\tau^y}{1 - \tau^y} \right) \frac{1}{1 + \epsilon - \gamma_c} \frac{U_{Sl} l}{U_c} - \left[1 - \left(\frac{\tau^y}{1 - \tau^y} \right) \frac{1}{1 + \epsilon - \gamma_c} \frac{U_{cl} l}{U_c} \right] \frac{U_S}{U_c} \right) f d\theta \\
&= \int \left(- \left(\frac{\tau^y}{1 - \tau^y} \right) \frac{1}{1 + \epsilon - \gamma_c} \frac{U_{Sl} l}{U_c} \frac{U_S}{U_S} - \left[1 - \left(\frac{\tau^y}{1 - \tau^y} \right) \frac{\gamma_c}{1 + \epsilon - \gamma_c} \right] \frac{U_S}{U_c} \right) f d\theta \\
&= \int \left(- \left(\frac{\tau^y}{1 - \tau^y} \right) \frac{\xi}{1 + \epsilon - \gamma_c} \frac{U_S}{U_c} - \left[1 - \left(\frac{\tau^y}{1 - \tau^y} \right) \frac{\gamma_c}{1 + \epsilon - \gamma_c} \right] \frac{U_S}{U_c} \right) f d\theta \\
&= - \int \left[1 + \left(\frac{\tau^y}{1 - \tau^y} \right) \frac{\xi - \gamma_c}{1 + \epsilon - \gamma_c} \right] \frac{U_S}{U_c} f d\theta
\end{aligned}$$

The second piece of ν_t prices the echo of current emissions in future pollution stocks. By the envelope theorem applied to the period- $(t + 1)$ problem, Q_t enters continuation costs only through the accumulation constraints of the next s periods: $S_{t+j} = S(Q_{t+j-s}, \dots, Q_{t+j})$ carries Q_t as an argument for $j \leq s$ and drops it afterward. Each envelope step peels off one date,

$$\frac{\partial V_{t+1}}{\partial Q_t} = \mu_{t+1} \Phi_1 + \frac{1}{R} \frac{\partial V_{t+2}}{\partial Q_t} \quad \Phi_j \equiv \frac{\partial S_{t+j}}{\partial Q_t}$$

and iterating until Q_t drops out ($\Phi_j = 0$ for $j > s$ by construction),

$$\frac{\partial V_{t+1}}{\partial Q_t} = \sum_{j=1}^s \frac{1}{R^{j-1}} \mu_{t+j} \Phi_j$$

Substituting into (13), with $\Phi_0 = S_q$,

$$\nu_t = \mu_t \Phi_0 + \frac{1}{R} \sum_{j=1}^s \frac{1}{R^{j-1}} \mu_{t+j} \Phi_j = \sum_{j=0}^s \frac{1}{R^j} \mu_{t+j} \Phi_j$$

Substituting the damage block above for each μ_{t+j} and writing the date- $(t + j)$ population integrals as expectations over future skill histories,

$$\text{SCC}_t \equiv \nu_t = - \sum_{j=0}^s \frac{1}{R^j} \mathbb{E}_t \int \left[1 + \left(\frac{\tau_{t+j}^y}{1 - \tau_{t+j}^y} \right) \frac{\xi_{t+j} - \gamma_{c,t+j}}{1 + \epsilon_{t+j} - \gamma_{c,t+j}} \right] \frac{U_{S,t+j}}{U_{c,t+j}} \Phi_j f_{t+j} d\theta$$

Here, s is the number of periods emissions affect the pollution stock.

A.7 Retirement

This subsection establishes the retirement structure the rest of the paper draws on. Retirement dates are $t \geq T_R$. Output is zero and the period utility is $U(c, q, 0, S_t)$, so the current type enters neither preferences nor technology.

Lemma A.2 (Retirement kills the costate).

At every retirement date an optimal allocation is measurable with respect to the retirement-entry history θ^{T_R-1} , the marginal continuation values vanish, $w_{2,t} = 0$ for $t \geq T_R - 1$, the retirement component problem is deterministic history by history, and the costate on the local incentive constraint is $\psi_t = 0$ for $t \geq T_R$. At the junction, the envelope condition of the last working date loses its future-rent term, $\dot{u}_{T_R-1} = U_\theta$.

Proof. Step 1: payoff-irrelevant reports cannot be separated. At a retirement date the report moves the assignment but neither preferences nor an output requirement, so the utility an agent obtains from reporting $\hat{\theta}$,

$$U(c(\hat{\theta}), q(\hat{\theta}), 0, S_t) + \beta w(\hat{\theta})$$

does not depend on the true type. Incentive compatibility then forces it to be constant in $\hat{\theta}$ on the support. Were it higher at some report, every type would prefer that report and truth-telling would fail for a positive mass of types. Hence, u_t is constant across the date- t draws given θ^{t-1} at every retirement date. The assignment can then be taken flat as well: every draw faces the same delivery problem, minimizing $c + q + \frac{1}{R}V_{t+1}$ over the bundles and continuation promises that deliver the common utility, so an optimal assignment common to all draws exists, and without loss the allocation from T_R on is measurable with respect to θ^{T_R-1} .

Step 2: the marginal continuation value dies at the junction. For $t \geq T_R - 1$, the marginal continuation value integrates a constant against the derivative of a density. By Step 1, $u_{t+1}(\theta^t, \theta')$ is constant in θ' , equal to some $\bar{u}_{t+1}(\theta^t)$, so

$$w_{2,t}(\theta^t) = \int_0^\infty u_{t+1}(\theta^t, \theta') \partial_{\theta_t} f_{t+1}(\theta' | \theta_t) d\theta' = \bar{u}_{t+1}(\theta^t) \frac{\partial}{\partial \theta_t} \int_0^\infty f_{t+1}(\theta' | \theta_t) d\theta' = 0$$

since the density integrates to one at every θ_t . At the junction, $w_2 = 0$ enters the date- $(T_R - 1)$ problem as a feasibility restriction rather than a first-order condition: the retirement block can deliver no other value, so the choice of w_2 disappears, the last working date's costate is left free, and its envelope condition reads $\dot{u} = U_\theta$.

Step 3: the retirement block is deterministic and carries no costate. With the allocation measurable in θ^{T_R-1} and $w_2 = 0$, the envelope condition at retirement dates reads $\dot{u} =$

$U_\theta + \beta w_2 = 0$, which the flat profile satisfies identically. The type integral drops from the component problem, which becomes deterministic history by history,

$$V_t(\hat{w}, \mathbf{S}_-) = \min_{c,q,w,S} c + q + \frac{1}{R} V_{t+1}(w, \mathbf{S}) \quad \text{s.t.} \quad \hat{w} = U(c, q, 0, S) + \beta w$$

with the emission state and its aggregate constraint carried as in Section 4.1. There is no envelope constraint and no threat-keeping constraint, hence no costate and no threat multiplier: $\psi_t = 0$ and $\lambda_{2,t} = 0$ for $t \geq T_R$. The working-age first-order system confirms the consistency of this reading: at $\lambda_2 = 0$, $\bar{\alpha}_t = 1$, and a flat allocation, the condition for the numéraire gives $\phi = f/U_c$, promise keeping gives $\lambda_{1,t} = \int \phi d\theta = 1/U_{c,t}$, and the costate equation integrates to

$$\dot{\psi} = \lambda_1 f - \phi = \frac{f}{U_{c,t}} - \frac{f}{U_{c,t}} = 0$$

which, with $\psi(0) = 0$, gives $\psi(\theta) = 0$ at every θ .

Step 4: the relaxed problem alone would not deliver this. With persistent skills, retirement draws remain informative about working-life types and the relaxed problem, which keeps only the envelope condition, would tilt retirement utility against those draws to relax the last working date's incentive constraint through w_2 . Step 1 rules the tilt out: nothing disciplines a payoff-irrelevant report, so no such tilt is globally incentive compatible. The measurability restriction of Step 1 is therefore a restriction to the implementable class, not an approximation, and it is how the life-cycle block of Section 6, following Farhi and Werning (2013), treats retirement.

Step 5: the formulas do not need the costate. In retirement, the formulas of interest are the commodity wedge, the savings wedge, and the social cost of carbon. In each of them, ψ enters multiplied by one of γ_c , γ_q , or ξ . Each of these statistics is proportional to l through its defining cross-partial, so each vanishes at $l = 0$. The retirement conclusions, a zero screening tax, the standard inverse Euler equation and Pigouvian carbon pricing therefore hold whatever the costate. $\square$

B Derivations: best carbon-only reform

B.1 Validity of the recursive formulation

Incentive compatibility and the first-order approach: Both constraints restrict the functions (c_t, q_t, y_t) that the planner may choose. No reporting strategy enters either constraint, and neither appears on either side of the incentive inequalities, so incentive compatibility is decided as usual. The one-shot-deviation reduction of Fernandes and Phelan (2000) and the envelope theorem of Kapička (2013) therefore apply.

Neither constraint involves the promised value u , the continuation promise w , or the marginal continuation value w_2 . The labor-wedge constraint restricts the current bundle and the current pollution stock; the savings-wedge constraint also reaches the date- $(t + 1)$ bundles through $\mathbb{E}_t[U_{c,t+1}]$, which part (ii) converts into a state carried alongside them. The conditions for those three are therefore those of Appendix A.1, such that the costate equation is unchanged in form, its boundary conditions $\psi(0) = \psi(\infty) = 0$ still hold, and the envelope identities for $\hat{w}$ and $\hat{w}_2$ carry over. What changes is ϕ , which the condition for c defines and which both constraints reach.

Part (i): The labor-wedge constraint involves only the current bundle $(c(\theta), q(\theta), y(\theta))$ and the current pollution stock at every date- t history, with no promised value, no continuation object, and no other date's controls. It enters the date- t component problem the way the resource cost does, pointwise with multiplier $\eta(\theta)f_t$, and adds no state.

Part (ii): Fix a history θ^t with $t < T$. The savings-wedge constraint reads

$$U_{c,t}(\theta^t) = (1 - \bar{\tau}_t^s(\theta_t)) \beta R \int_0^\infty U_{c,t+1}(\theta^t, \theta') f_{t+1}(\theta' | \theta_t) d\theta'$$

The left side is pinned by the date- t allocation at θ^t while the right side integrates over possible future allocations conditional on the current skill. Isolating the integral, define

$$M(\theta^t) \equiv \frac{U_{c,t}(\theta^t)}{(1 - \bar{\tau}_t^s(\theta_t)) \beta R}$$

This is a function of date- t controls alone, so constraint (6) can be rewritten as

$$M(\theta^t) = \int_0^\infty U_{c,t+1}(\theta^t, \theta') f_{t+1}(\theta' | \theta_t) d\theta'$$

That is, the constraint requires marginal utility at date $t+1$ to average to $M(\theta^t)$ over the draws that follow θ^t . The draws that follow θ^t are succeeded by the date- $(t+1)$ problem at the state Section 4.1 writes $(w(\theta^t), w_2(\theta^t), \theta_t, \mathbf{S})$. Two histories reaching that same state can compute different requirements since M is read off the date- t allocation and not off the promises. The requirement can therefore be appended to that state as $\hat{M}$. Two terms enter the Hamiltonian. The first delivers the inherited requirement.

$$\vartheta_t \left[\int_0^\infty U_c \left(c(\theta), q(\theta), \frac{y(\theta)}{\theta}, S \right) f_t(\theta|\theta_-) d\theta - \hat{M} \right]$$

The second is the continuation value's dependence on the requirement passed forward, which Appendix B.2 differentiates by the chain rule through M . Only the first involves $\hat{M}$, so $\partial V_t / \partial \hat{M} = -\vartheta_t$ by the envelope theorem. The date-0 problem is the continuation of no history, so it carries no such term and $\vartheta_0 = 0$. The date- T problem has no successor, so it passes no requirement, $\vartheta_{T+1} = 0$, and (7) reads $\eta_T^s = \vartheta_T U_{c,T}$.

An allocation satisfying (6) at every history hands every component problem a requirement it meets, and component problems that each meet their inherited requirement assemble into an allocation satisfying (6) at every history. Minimizing the same cost over the same constraint set, the two problems have the same first-order conditions, which are the ones in Appendix B.2.

B.2 First-order conditions

The Hamiltonian is that of Appendix A.1 with two added terms and a continuation value that now carries the marginal-utility requirement M as an argument:

$$\begin{aligned} \mathcal{H} = & \left[c + q - \theta l + \frac{1}{R} V_{t+1}(w, w_2, M(c, q, l, S), \theta, \mathbf{S}) \right] f \\ & + \psi \left[-U_l(c, q, y/\theta, S) \frac{l}{\theta} + \beta w_2 \right] \\ & - \lambda_1 u(\theta) \bar{\alpha}_t f_t \\ & + \lambda_2 u(\theta) f_2 \\ & + \phi [u - U(c, q, y/\theta, S) - \beta w] \\ & - \mu [S - S(\dots, Q)] \\ & + \eta \left[-\frac{U_l(c, q, y/\theta, S)}{\theta U_c(c, q, y/\theta, S)} - (1 - \bar{\tau}_t^y(\theta)) \right] f \\ & + \vartheta_t \left[U_c(c, q, y/\theta, S) - \hat{M} \right] f \end{aligned}$$

Differentiating the continuation term with respect to a control $z \in \{c, q, l, S\}$, by the chain rule through M and the envelope condition at $t + 1$,

$$\frac{1}{R} \frac{\partial V_{t+1}}{\partial \hat{M}} \frac{\partial M}{\partial z} f = -\frac{\vartheta_{t+1}(\theta^t)}{(1 - \bar{\tau}_t^s) \beta R^2} U_{cz} f$$

Adding the $\vartheta_t U_{cz} f$ term, each first-order condition gains $(\eta_t^s / U_{c,t}) U_{cz} f$ with η_t^s as in (7), alongside the labor-constraint term $\eta f \Gamma_z$. The first-order conditions are

$$\begin{aligned} \frac{\partial \mathcal{H}}{\partial c} &: f - \psi U_{cl} \frac{l}{\theta} + \eta f \Gamma_c + \frac{\eta_t^s}{U_c} U_{cc} f = \phi U_c \\ \frac{\partial \mathcal{H}}{\partial q} &: f - \psi U_{ql} \frac{l}{\theta} + \nu_t f + \eta f \Gamma_q + \frac{\eta_t^s}{U_c} U_{cq} f = \phi U_q \\ \frac{\partial \mathcal{H}}{\partial l} &: -\theta f - \phi U_l + \eta f \Gamma_l + \frac{\eta_t^s}{U_c} U_{cl} f = \frac{1}{\theta} \psi [U_{ll} + U_l] \\ \frac{\partial \mathcal{H}}{\partial u} &: \dot{\psi} = \lambda_1 \bar{\alpha}_t f - \lambda_2 f_2 - \phi \\ \frac{\partial \mathcal{H}}{\partial w} &: \frac{1}{R} \frac{\partial V_{t+1}}{\partial w} f = \phi \beta \\ \frac{\partial \mathcal{H}}{\partial w_2} &: \frac{1}{R} \frac{\partial V_{t+1}}{\partial w_2} f = -\psi \beta \\ \frac{\partial \mathcal{H}}{\partial S} &: \int_0^\infty \left(-\psi U_{Sl} \frac{l}{\theta} - \phi U_S + \eta f \Gamma_S + \frac{\eta_t^s}{U_c} U_{cS} f \right) d\theta = \mu \end{aligned}$$

with $\nu_t \equiv \mu_t S_q + \frac{1}{R} \partial V_{t+1} / \partial Q$ the shadow price of current emissions. The derivatives of the labor-wedge constraint are

$$\begin{aligned} \Gamma_c &= (1 - \bar{\tau}_t^y) \frac{\sigma}{c} - \frac{\gamma_c}{y} \\ \Gamma_q &= (1 - \bar{\tau}_t^y) \frac{\sigma_{cq}}{q} - \frac{\gamma_q (1 + \tau^q)}{y} \\ \Gamma_l &= (1 - \bar{\tau}_t^y) \frac{\epsilon - \gamma_c}{l} \\ \Gamma_S &= -\frac{U_S}{U_c} \left[\frac{\xi}{y} + (1 - \bar{\tau}_t^y) \frac{\xi_c}{c} \right] \end{aligned} \tag{24}$$

All four are $\partial \Gamma / \partial z$ evaluated with $U_l = -\theta U_c (1 - \bar{\tau}_t^y)$, which collapses the quotient rule to $\partial \Gamma / \partial z = -U_{lz} / (\theta U_c) - (1 - \bar{\tau}_t^y) U_{cz} / U_c$. The factor $1 + \tau^q$ in Γ_q comes from the first piece:

γ_q is defined against U_q , so $U_{ql} = \gamma_q U_q / l = \gamma_q (1 + \tau^q) U_c / l$. It is what cancels the prefactor $1/(1 + \tau^q)$ in Appendix B.9 and lets $\gamma_{q,t}/y_t$ pair with $\gamma_{c,t}/y_t$ there. Each savings term is η_t^s times the log-derivative of U_c in the corresponding direction since $U_{cz}/U_c = \partial \ln U_c / \partial z$. Written with elasticities, these are $-\sigma/c$ for the numéraire, $-\sigma_{cq}/q$ for the dirty good, γ_c/l for hours, and $\xi_c (U_S/U_c)/c$ for the pollution stock, with $\xi_c \equiv U_{Sc} c / U_S$ the complementarity of the pollutant with the clean good.

B.3 Statistics

That $\Gamma_c > 0$ signals normality of leisure follows from the standard problem's labor-wedge constraint. Give an agent with $-U_l/(\theta U_c) = 1 - \bar{\tau}_t^y$ a unit of the numéraire at fixed hours. The margin she requires moves by $\partial [-U_l/(\theta U_c)] / \partial c = -U_{lc}/(\theta U_c) + U_l U_{cc}/(\theta U_c^2)$. Using $U_l = -\theta U_c (1 - \bar{\tau}_t^y)$, this is $\Gamma_{c,t} = (1 - \bar{\tau}_t^y) \sigma_t / c_t - \gamma_{c,t} / y_t$. The first term means that diminishing marginal utility of consumption raises the margin she requires and the second term means that complementarity between the numéraire and labor lowers it. Normality of leisure is a statement about unearned income at a fixed wage. Her budget has slope $\theta (1 - \bar{\tau}_t^y)$ at the assigned point, whether the schedule is indexed on the type or on earnings, so a perturbation dm of unearned income gives $dc = \theta (1 - \bar{\tau}_t^y) dl + dm$. Differentiating, her first-order condition for hours becomes

$$\Gamma_{c,t} [\theta (1 - \bar{\tau}_t^y) dl + dm] + \Gamma_{l,t} dl = 0 \quad \implies \quad \frac{\partial l}{\partial m} = -\frac{\Gamma_{c,t}}{\theta (1 - \bar{\tau}_t^y) \Delta_t}$$

This uses $\theta (1 - \bar{\tau}_t^y) \Gamma_{c,t} + \Gamma_{l,t} = \theta (1 - \bar{\tau}_t^y) \Delta_t$, which is (24) collected at $y = \theta l$ against (8), with $\Gamma_{l,t}$ and Δ_t . Under an earnings-indexed schedule, both carry the progressivity parameter p_t of Appendix B.13 and the identity is unchanged. The denominator is positive under $\Delta_t > 0$, so hours fall in unearned income when $\Gamma_{c,t} > 0$. The argument is unaffected by the savings-wedge constraint, which does not enter the agent's hours choice at the frozen wedge.

Δ_t has a reading of its own, and it is the one Section 5.1 assumes. Fix the dirty-good assignment and perturb (c, l) along an indifference curve of an agent holding the margin at $-U_l/(\theta U_c) = 1 - \bar{\tau}_t^y$: utility compensation imposes $dc = \theta (1 - \bar{\tau}_t^y) dl$, and differentiating the hours condition along the perturbation,

$$0 = U_c d(1 - \bar{\tau}_t^y) + \left[\theta (1 - \bar{\tau}_t^y)^2 U_{cc} + 2 (1 - \bar{\tau}_t^y) U_{cl} + \frac{U_{ll}}{\theta} \right] dl$$

In the statistics of (1) and (2) the bracket equals $-U_c (1 - \bar{\tau}_t^y) \theta \Delta_t$, so

$$s_{yy,t} \equiv \frac{\partial y}{\partial (1 - \bar{\tau}_t^y)} \Big|_u = \theta \frac{dl}{d(1 - \bar{\tau}_t^y)} \Big|_u = \frac{1}{(1 - \bar{\tau}_t^y) \Delta_t} > 0 \quad (25)$$

the compensated response of earnings to the net-of-wedge rate, holding the commodity split fixed: the hours margin's own compensated response, the object the labor-wedge constraint prices, not the full two-good demand response. The same bracket, up to a factor of θ , is the agent's second-order condition. Let her face the linear budget $c = \theta(1 - \bar{\tau}_t^y)l + m$ at the frozen net wage with the dirty-good assignment held fixed, and choose l to maximize utility. Then $dU/dl = \theta(1 - \bar{\tau}_t^y)U_c + U_l$, whose vanishing is (5), and

$$\frac{d^2U}{dl^2} = \theta^2(1 - \bar{\tau}_t^y)^2 U_{cc} + 2\theta(1 - \bar{\tau}_t^y)U_{cl} + U_{ll} = -\theta^2(1 - \bar{\tau}_t^y)U_c \Delta_t$$

which is negative exactly when $\Delta_t > 0$. So the assumption says the agent's hours choice at the frozen wedge is an interior maximum rather than a minimum: a restriction on the agent's problem, not on the planner's. Under an earnings-indexed schedule both displays carry $\Delta_t + p_t/y_t$: the schedule's own dependence on earnings enters through $d(1 - \bar{\tau}_t^y)/dl = -p_t(1 - \bar{\tau}_t^y)\theta/y_t$ and shifts each bracket by exactly the term that replaces Δ_t with $\Delta_t + p_t/y_t$, which is the $\Gamma_{l,t}$ of Appendix B.13.

B.4 The labor multiplier

Divide the first-order condition for c by f and use $U_{cl} = \gamma_c U_c$ and $U_{cc}/U_c = -\sigma/c$.

$$\frac{U_c \phi}{f} = 1 - \gamma_c \frac{\psi U_c}{\theta f} + \eta \Gamma_c - \eta_t^s \frac{\sigma}{c} \equiv 1 - \chi_t \quad (26)$$

Here, χ is the incentive credit of Appendix A.4 generalized to the case with frozen wedges and later used in Proposition 5.2. Solving for ϕ twice, once from the condition for c and once from the condition for l using $U_{ll} + U_l = U_l(1 + \epsilon)$,

$$\begin{aligned} \phi &= \frac{f}{U_c} - \frac{\psi}{\theta} \gamma_c + \eta \frac{\Gamma_c}{U_c} f + \eta_t^s \frac{U_{cc}}{U_c} \frac{f}{U_c} \\ \phi &= -\frac{\theta f}{U_l} - \frac{\psi}{\theta} (1 + \epsilon) + \eta \frac{\Gamma_l}{U_l} f + \eta_t^s \frac{U_{cl}}{U_c} \frac{f}{U_l} \end{aligned}$$

Equating the two and multiplying through by U_l ,

$$\left(\frac{U_l}{\theta U_c} + 1 \right) \theta f = -\frac{\psi U_l}{\theta} (1 + \epsilon - \gamma_c) + \eta U_l \left[\frac{\Gamma_l}{U_l} - \frac{\Gamma_c}{U_c} \right] f + \eta_t^s \left[\frac{U_{cl}}{U_c} - \frac{U_{cc} U_l}{U_c U_c} \right] f$$

Because $\bar{\tau}_t^y = 1 + U_l/(\theta U_c)$, divide by $\theta f(1 - \bar{\tau}_t^y)$ and substitute $U_l = -\theta U_c(1 - \bar{\tau}_t^y)$ term by term. On the left, $U_l/(\theta U_c) = -(1 - \bar{\tau}_t^y)$ leaves $[1 - (1 - \bar{\tau}_t^y)]/(1 - \bar{\tau}_t^y) = \bar{\tau}_t^y/(1 - \bar{\tau}_t^y)$. On the right, the first term becomes $A_t \psi U_c/(\theta f)$ with $A_t = 1 + \epsilon - \gamma_c$, and the labor-wedge-multiplier term becomes $-\eta [\Gamma_l U_c/U_l - \Gamma_c]$ which by (24) equals

$$-\eta \left[-\frac{\epsilon - \gamma_c}{y} - (1 - \bar{\tau}_t^y) \frac{\sigma}{c} + \frac{\gamma_c}{y} \right] = \eta \left[(1 - \bar{\tau}_t^y) \frac{\sigma}{c} + \frac{\epsilon - 2\gamma_c}{y} \right] = \eta \Delta_t$$

Using $U_d/U_c = \gamma_c/l$, $U_{cc}/U_c = -\sigma/c$, and $U_l/U_c = -\theta(1 - \bar{\tau}_t^y)$, the savings-wedge-multiplier term is

$$\frac{\eta_t^s}{\theta(1 - \bar{\tau}_t^y)} \left[\frac{\gamma_c}{l} - \frac{\sigma}{c} \theta(1 - \bar{\tau}_t^y) \right] = -\eta_t^s \left[\frac{\sigma}{c} - \frac{\gamma_c}{(1 - \bar{\tau}_t^y)y} \right] = -\eta_t^s \Lambda_t \quad \text{with } \Lambda_t \text{ as in (8)}$$

Putting everything together,

$$\frac{\bar{\tau}_t^y}{1 - \bar{\tau}_t^y} = A_t \frac{\psi U_{c,t}}{\theta f_t} + \eta_t \Delta_t - \eta_t^s \Lambda_t \quad (27)$$

Next, define the desired labor wedge $\tau_t^{y,*}$ and define $\tau_t^{y,*}/(1 - \tau_t^{y,*}) \equiv A_t \psi U_{c,t}/(\theta f_t)$. This is the same function with the same statistics as the standard labor wedge, evaluated at the constrained allocation. It is what the planner's own labor condition returns there, not what a fully free margin would deliver, since lifting the constraint would move ψ and the allocation with it. Moving it to the left of (27) and adding $\eta_t^s \Lambda_t$ to both sides,

$$E_t \equiv \frac{\bar{\tau}_t^y}{1 - \bar{\tau}_t^y} - \frac{\tau_t^{y,*}}{1 - \tau_t^{y,*}} = \eta_t \Delta_t - \eta_t^s \Lambda_t$$

$$\eta_t \Delta_t = E_t + \eta_t^s \Lambda_t$$

This is the expression in Lemma 5.2, with E_t being the labor wedge gap. To infer its sign, divide by Δ_t , which is positive by assumption,

$$\eta_t = \frac{E_t + \eta_t^s \Lambda_t}{\Delta_t} \quad (28)$$

η_t therefore carries the sign of $E_t + \eta_t^s \Lambda_t$, which is the reading of Lemma 5.2.

B.5 The intertemporal condition

The chain of Appendix A.4 is unchanged in form because neither constraint enters the conditions for u , w and w_2 or the boundary conditions. What the constraints reach is the value of ϕ/f , which (26) evaluates at the generalized incentive credit. The first-order condition for w and the envelope condition give

$$\lambda_{1,t+1} = \beta R \frac{\phi}{f} = \beta R \frac{1 - \chi_t}{U_{c,t}}$$

Integrating the costate equation over $(0, \infty)$ with $\psi(0) = \psi(\infty) = 0$,

$$\lambda_{1,t} = \int_0^\infty \phi d\theta = \mathbb{E}_{t-1} \left[\frac{1 - \chi_t}{U_{c,t}} \right]$$

Evaluating the second at $t + 1$ and equating with the first,

$$\frac{1 - \chi_t}{U_{c,t}} = \frac{1}{\beta R} \mathbb{E}_t \left[\frac{1 - \chi_{t+1}}{U_{c,t+1}} \right] \quad (29)$$

This is the generalized inverse Euler equation of Proposition 5.2. Multiplying (29) by $U_{c,t}$ and substituting the frozen Euler equation,

$$1 - \chi_t = (1 - \bar{\tau}_t^s) \mathbb{E}_t[U_{c,t+1}] \mathbb{E}_t \left[\frac{1 - \chi_{t+1}}{U_{c,t+1}} \right]$$

Because $\mathbb{E}[X] \mathbb{E}[Y] = \mathbb{E}[XY] - \text{Cov}(X, Y)$ with $X = U_{c,t+1}$ and $Y = (1 - \chi_{t+1})/U_{c,t+1}$,

$$1 - \chi_t = (1 - \bar{\tau}_t^s) \left\{ \mathbb{E}_t[1 - \chi_{t+1}] - \text{Cov}_t \left(U_{c,t+1}, \frac{1 - \chi_{t+1}}{U_{c,t+1}} \right) \right\} \quad (30)$$

This is the general form that the covariance display below specializes.

Both (29) and (30) are statements in χ_t , which by (26) carries the two multipliers together. The law of motion is to be an equation in the savings multiplier alone, so split the delivery cost into the part free of η_t^s and the part that carries it:

$$\mathcal{G}_t \equiv 1 - \gamma_{c,t} \frac{\psi U_{c,t}}{\theta f_t} + \eta_t \Gamma_{c,t} \quad \text{so that} \quad 1 - \chi_t = \mathcal{G}_t - \eta_t^s \frac{\sigma_t}{c_t} \quad (31)$$

$\mathcal{G}_t$ is what a util delivered through the numéraire costs once the incentive credit and the frozen labor schedule are counted when the savings-wedge constraint is inactive. Substituting (31) collects η^s on one side and everything else on the other. I maintain $\mathcal{G}_t > 0$ at every date. By (31), it equals one when the numéraire is separable from labor and the labor wedge is not frozen, but nothing signs it in general, and both the definition below and the factoring further down need it. Define the desired savings wedge as

$$1 - \tau_t^{s,*} \equiv \frac{\mathcal{G}_t}{\mathbb{E}_t[\mathcal{G}_{t+1}]}$$

This is the intertemporal margin the mechanism delivers when only the labor wedge schedule is frozen. When the date- $(t+1)$ draw is degenerate, the covariance vanishes and the conditional expectation drops. Writing (31) on both sides,

$$\mathcal{G}_t - \eta_t^s \frac{\sigma_t}{c_t} = (1 - \bar{\tau}_t^s) \left[\mathcal{G}_{t+1} - \eta_{t+1}^s \frac{\sigma_{t+1}}{c_{t+1}} \right]$$

Collecting the η^s terms on the left and factoring the positive $\mathcal{G}_{t+1}$ out of the right,

$$\eta_t^s \frac{\sigma_t}{c_t} - (1 - \bar{\tau}_t^s) \eta_{t+1}^s \frac{\sigma_{t+1}}{c_{t+1}} = \mathcal{G}_{t+1} \left[\frac{\mathcal{G}_t}{\mathcal{G}_{t+1}} - (1 - \bar{\tau}_t^s) \right] = \mathcal{G}_{t+1} (\bar{\tau}_t^s - \tau_t^{s,*}) \quad (32)$$

This is the law of motion of Proposition 5.2 and, with $\mathcal{G}_{t+1} > 0$, its right-hand side carries the sign of the wedge gap $\bar{\tau}_t^s - \tau_t^{s,*}$. Away from a degenerate draw, the same steps applied to (30) carry the date- $(t+1)$ terms as conditional expectations and leave the covariance standing.

$$\eta_t^s \frac{\sigma_t}{c_t} - (1 - \bar{\tau}_t^s) \mathbb{E}_t \left[\eta_{t+1}^s \frac{\sigma_{t+1}}{c_{t+1}} \right] = \mathbb{E}_t[\mathcal{G}_{t+1}] (\bar{\tau}_t^s - \tau_t^{s,*}) + (1 - \bar{\tau}_t^s) \text{Cov}_t \left(U_{c,t+1}, \frac{1 - \chi_{t+1}}{U_{c,t+1}} \right) \quad (33)$$

At a constant $\chi_{t+1} \equiv \bar{\chi}$, the covariance is $(1 - \bar{\chi}) \text{Cov}_t(U_{c,t+1}, 1/U_{c,t+1})$. Its second factor is negative whenever $U_{c,t+1}$ is non-degenerate, since $\mathbb{E}_t[U_{c,t+1} \cdot 1/U_{c,t+1}] = 1$ while $\mathbb{E}_t[U_{c,t+1}] \mathbb{E}_t[1/U_{c,t+1}] > 1$ by Jensen's inequality and $U_{c,t+1} > 0$ by Assumption 3.1. The product is negative because $1 - \bar{\chi} > 0$: by (26) it equals $U_{c,t+1} \phi_{t+1}/f_{t+1}$, and $\phi > 0$ because delivering utility costs resources. It stays negative unless χ_{t+1} varies enough with $U_{c,t+1}$ to reverse it and it is negative outright when $1 - \chi_{t+1}$ is independent of $U_{c,t+1}$ with positive mean. Risk therefore adds a negative drift to the wedge gap, offsetting a positive error and deepening a negative one.

When the labor schedule is left free, $\eta_t = 0$ and $\mathcal{G}_t = 1 - \gamma_{c,t}\psi U_{c,t}/(\theta f_t)$. The law of motion keeps its form, with the labor-constraint term $+\eta_t\Gamma_{c,t}$ absent from the savings-free delivery cost and from the desired savings wedge. Setting $\bar{\tau}_t^s = \tau_t^{s,*}$ at every date solves (32) with $\eta^s = 0$.

From T_R on, the same term is absent for a different reason. The labor-wedge constraint is not imposed there, so $\eta_t = 0$ and $\psi = 0$ by Corollary 5.3 and $\eta_t\Gamma_{c,t}$ is absent from χ_t and $\mathcal{G}_t$ rather than evaluated at $y_t = 0$, where the $1/y_t$ that $\Gamma_{c,t}$ carries is undefined. The retiree's delivery cost is $\mathcal{G}_t = 1$ and the desired savings wedge is zero, so the law of motion reads $\eta_t^s\sigma_t/c_t - (1 - \bar{\tau}_t^s)\eta_{t+1}^s\sigma_{t+1}/c_{t+1} = \bar{\tau}_t^s$. That is, any nonzero frozen wedge keeps injecting error through retirement.

B.6 The sign of the savings multiplier

The date-0 problem is the continuation of no history, so $\vartheta_0 = 0$. Likewise, the date- T problem has no successor and passes no requirement forward, so there is no ϑ_{T+1} term. Equation (7) carries the two multipliers in a single bracket, $\vartheta_t - \vartheta_{t+1}(\theta^t)/\kappa_t$ with $\kappa_t \equiv (1 - \bar{\tau}_t^s)\beta R^2 > 0$ because $\bar{\tau}_t^s < 1$ and $\beta, R > 0$. At $t = 0$, the leading term vanishes, leaving $\eta_0^s = -\vartheta_1(\theta^0)U_{c,0}/\kappa_0$. At $t = T$, the trailing term vanishes, leaving $\eta_T^s = \vartheta_T U_{c,T}$. Solving (7) for ϑ_{t+1} gives:

$$\vartheta_{t+1}(\theta^t) = \kappa_t \left[\vartheta_t - \frac{\eta_t^s}{U_{c,t}} \right] \quad (34)$$

Fix a history. Write $P_t \equiv \prod_{j=0}^{t-1} \kappa_j$, with $P_0 = 1$ and $P_t > 0$ for all $t > 0$, and define $x_t \equiv \vartheta_t/P_t$. Dividing (34) by $P_{t+1} = P_t\kappa_t$ leaves the lagged term with no coefficient.

$$x_{t+1} = x_t - \frac{\eta_t^s}{P_t U_{c,t}} \quad (35)$$

η_t^s is then the increment the date- t constraint adds to the normalized multiplier. Summing (35) from $t = 0$ to $T - 1$ telescopes the left side to $x_T - x_0$. $\vartheta_0 = 0$ gives $x_0 = 0$. The terminal condition $\eta_T^s = \vartheta_T U_{c,T}$ gives $x_T = \eta_T^s/(P_T U_{c,T})$. Collecting both endpoints on one side,

$$\sum_{t=0}^T \frac{\eta_t^s}{P_t U_{c,t}} = 0 \quad (36)$$

The weights $1/(P_t U_{c,t})$ are strictly positive so, if $\eta_t^s \geq 0$ at every date along the history, every term of (36) is non-negative and a sum of non-negative terms equals zero only if each term equals zero. So $\eta_t^s = 0$ at every date along the history, and (34) run forward from $\vartheta_0 = 0$ then gives $\vartheta_t = 0$ at every date along it. The case $\eta_t^s \leq 0$ at every date is the same argument with

the inequalities reversed. Nothing compares two histories at the same date and the history was arbitrary, so a schedule leaving η^s one-signed at every history leaves it zero at every history. By contraposition, if η^s is nonzero at some date along the history, then (36) means terms of both signs must appear.

This disciplines the sign of η_t^s and not the size of ϑ_t . The chain (34) compounds ϑ_t at $(1 - \bar{\tau}_t^s)\beta R^2$ per period, so it grows with the accumulated bending of (10) and can end the life cycle orders of magnitude above the per-period gap $\bar{\tau}_t^s - \tau_t^{s,*}$, even when the schedule sits close to its desired counterpart at every date. The formulas of Section 5.1 price η_t^s , a difference of adjacent multipliers that nets out the accumulated level, and by Proposition 5.2 its drift is of the order of that gap. A large ϑ_t and a small carbon-price deviation therefore coexist: ϑ_t measures how far the constraint has bent the path of marginal utility, η_t^s how much bending the current date adds.

B.7 The direction of the reversal

At a degenerate draw, the law of motion pins where the multiplier can change sign and what sign it ends with. Let $z_t \equiv \eta_t^s \sigma_t / c_t$. I assume $\sigma_t > 0$ throughout, which makes z_t carry the sign of η_t^s ¹³, and read (32) as a backward recursion.

$$z_t = (1 - \bar{\tau}_t^s) z_{t+1} + \mathcal{G}_{t+1} (\bar{\tau}_t^s - \tau_t^{s,*}) \quad (37)$$

Write $a_t \equiv 1 - \bar{\tau}_t^s$, positive because $\bar{\tau}_t^s < 1$, and $g_t \equiv \mathcal{G}_{t+1} (\bar{\tau}_t^s - \tau_t^{s,*})$, which carries the sign of the wedge gap where $\mathcal{G}_{t+1} > 0$. Iterating (37) down from the last date,

$$z_t = a_{t,T} z_T + \sum_{s=t}^{T-1} a_{t,s} g_s, \quad a_{t,s} \equiv \prod_{j=t}^{s-1} a_j > 0 \quad (38)$$

Every $a_{t,s}$ is a product of positive terms, so each z_t is a positive multiple of z_T plus a positively weighted sum of the gaps from t onward. Equation (36) then fixes the terminal value. In these variables, it reads $\sum_t \omega_t z_t = 0$ with $\omega_t \equiv c_t / (\sigma_t P_t U_{c,t}) > 0$. Substituting (38) and collecting,

$$z_T = - \frac{\sum_{s < T} W_s g_s}{\sum_t \omega_t a_{t,T}}, \quad W_s \equiv \sum_{t \leq s} \omega_t a_{t,s} > 0 \quad (39)$$

Every weight is strictly positive, so the multiplier ends the life cycle negative when the wedge

¹³ Assumption 3.1 delivers only $\sigma_t \geq 0$. The strict version, which the calibration of Section 6 satisfies, is what the argument here needs.

gaps are positive in weighted average. This holds whatever the pattern of underdistortion or overdistortion over the life cycle. Whenever that weighted sum is nonzero, $z_T \neq 0$, so η_t^s is nonzero at some date and Lemma 5.3 gives it both signs along the history such that a crossing occurs.

Where a crossing can sit is settled date by date by the sign of the local gap. Where $g_t > 0$, $z_t \leq 0$ forces $z_{t+1} < 0$, so the multiplier cannot rise through zero and only a downward crossing is available. Where $g_t < 0$, $z_t \geq 0$ forces $z_{t+1} > 0$, so only an upward crossing is available. Where the gap is zero, $z_t = (1 - \bar{\tau}_t^s) z_{t+1}$ and the two dates share a sign, so no crossing occurs. Each case rules a direction out rather than forcing one in. Whether the multiplier crosses at a given date depends on its level there. Retirement is where the gap's sign is known in advance since $\mathcal{G}_{t+1} = 1$ and $\tau_t^{s,*} = 0$ leave $g_t = \bar{\tau}_t^s$, which is positive at any positive capital income tax, meaning retirement can host only a downward crossing.

Away from the degenerate draw case, the reversal result still survives because it never reads the law of motion away from retirement. There, the draw is degenerate whatever the skill process and $\mathcal{G}_{t+1} = 1$ with $\tau_t^{s,*} = 0$, so $\eta_t^s = 0$ at every date would reduce (32) to $0 = \bar{\tau}_t^s$, which any nonzero capital income tax rules out. Lemma 5.3, free of both the draw and the first-order system, then gives both signs along every history. However, the results on the terminal sign of (39), the date-by-date restriction on where a crossing can sit, as well as the count and direction that follow from them do not apply.

All three read (32), whereas risk replaces this with (33), which relates the realized z_t to the conditional mean $\mathbb{E}_t[z_{t+1}]$ rather than to z_{t+1} itself. The date-by-date restriction then weakens: the most $z_t \leq 0$ can deliver is $\mathbb{E}_t[z_{t+1}] < 0$, a statement about the average successor and not about any particular one. The terminal sign fares worse, since (38) unrolls the recursion from one date to the next and there is no such chain to unroll here.

B.8 The net shadow price

Substituting (28) into the net shadow price $\zeta_t \equiv \eta_t (1 - \bar{\tau}_t^y) - \eta_t^s$ and collecting terms in η_t^s ,

$$\zeta_t = (1 - \bar{\tau}_t^y) \frac{E_t + \eta_t^s \Lambda_t}{\Delta_t} - \eta_t^s = \frac{(1 - \bar{\tau}_t^y) E_t}{\Delta_t} + \eta_t^s \left[\frac{(1 - \bar{\tau}_t^y) \Lambda_t}{\Delta_t} - 1 \right]$$

By (8), $(1 - \bar{\tau}_t^y) \Lambda_t = \Gamma_{c,t}$, so the bracket is $(\Gamma_{c,t} - \Delta_t) / \Delta_t$. The two statistics carry the term $(1 - \bar{\tau}_t^y) \sigma_t / c_t$ identically, so it cancels in their difference and

$$\Gamma_{c,t} - \Delta_t = -\frac{\gamma_{c,t}}{y_t} - \frac{\epsilon_t - 2\gamma_{c,t}}{y_t} = \frac{\gamma_{c,t} - \epsilon_t}{y_t} = -\frac{A_t - 1}{y_t}$$

The last equality uses $A_t \equiv 1 + \epsilon_t - \gamma_{c,t}$. Substituting the bracket gives (9). The coefficient on E_t is positive because $\Delta_t > 0$ and $1 - \bar{\tau}_t^y = -U_l/(\theta U_c) > 0$ by (5) and Assumption 3.1. The coefficient on η_t^s , taken with its sign, is $-(A_t - 1)/(y_t \Delta_t)$, which is negative wherever $A_t > 1$. This proves the identity and the coefficient signs. Appendix B.13 records what an earnings-indexed schedule does to them.

Part (i) of Corollary 5.2. Fix a history and suppose $\eta^s \neq 0$ along it. Lemma 5.3, whose walk runs along a fixed history, then delivers dates with $\eta_t^s > 0$ and dates with $\eta_t^s < 0$ along that one. Nothing in that argument restricts where they fall, though at a degenerate draw (37) puts the positive dates first when the savings schedule overdistorts at every date. Over the working life, the coefficient is negative under $A_t > 1$. From the retirement year onward, the decomposition has no content, but Corollary 5.3 gives $\zeta_t = -\eta_t^s$, a coefficient of -1 . The coefficient is therefore negative at every date of the life cycle, so the contribution carries the opposite sign to the multiplier throughout and a multiplier taking both signs delivers a contribution of both signs.

The two regimes are not patched together at T_R . Clearing y_t out of the working-life coefficient,

$$-\frac{A_t - 1}{y_t \Delta_t} = -\frac{\epsilon_t - \gamma_{c,t}}{y_t (1 - \bar{\tau}_t^y) \sigma_t / c_t + \epsilon_t - 2\gamma_{c,t}} \quad (40)$$

$\gamma_{c,t} = U_{cl_t}/U_{c,t}$ vanishes with labor supply, as does y_t itself. That is, wherever ϵ_t stays bounded away from zero the coefficient tends to -1 . Retirement's -1 is the limit of the working-life coefficient rather than a second case, which is why $A_t > 1$ and no other threshold is what carries one sign across the seam.

Part (ii) of Corollary 5.2. E_t is not a within-date object since $\tau_t^{y,*}$ carries the costate, which the persistence term of Proposition 4.1 ties to date $t - 1$. What the labor side lacks is not a link across dates, but the pair of endpoint conditions that gives Lemma 5.3 its force, and the identity carries no cross-date term in E_t . With the schedule free to move date by date, any sign profile for the labor contribution is therefore consistent with the identity.

Part (iii) of Corollary 5.2. Read (9) at a schedule that overdistorts labor at every date. The labor term $(1 - \bar{\tau}_t^y) E_t / \Delta_t$ is then positive at every date whatever the sign of η_t itself, which (28) leaves ambiguous wherever $\eta_t^s \Delta_t < 0$. With normal leisure, this is wherever $\eta_t^s < 0$. Write $L_t > 0$ for the labor term of (9), so that ζ_t is L_t plus the savings term, which by Appendix B.7 rises through zero exactly once when the savings schedule overdistorts throughout and the date- $(t + 1)$ draw is degenerate. Under weak separability, the carbon-price deviation is $-\zeta_t \mathcal{I}_t$ by Corollary 5.1, carrying the sign opposite to ζ_t where the dirty good is normal, so a larger labor term sits against a lower deviation at every age. This reads the identity at one allocation and is not a comparative static in $\bar{\tau}_t^y$, which would move Δ_t , η_t^s and $\tau_t^{y,*}$ as well.

What that does to the crossing takes more care than adding a constant to a rising profile.

Raising L_t raises ζ_t pointwise, so the set of ages carrying a positive deviation, $\{t : \zeta_t < 0\}$, can only shrink. Where that set is an initial segment of the life cycle, shrinking it is the crossing coming forward. A large enough labor level empties it and leaves the deviation negative throughout. That is what the identity supports and it is weaker than a crossing date falling smoothly in the labor level, which would need the savings term increasing in t . Appendix B.7 counts the savings term's crossings and rules out downward ones. It does not make the profile monotone between them.

This is a reading of the identity at fixed multipliers rather than a comparative static in the schedule. Freezing labor moves η_t into $\mathcal{G}_t$, hence into $\tau_t^{s,*}$ and the law of motion, so the savings term responds too. What the identity delivers is that the two channels enter additively at the solved allocation, not that one can be moved with the other held still.

B.9 Commodity wedge

Divide the first-order condition for q by f , using $U_{ql}l = \gamma_q U_q$ and $U_{cq}/U_c = -\sigma_{cq}/q$.

$$1 - \gamma_q \frac{\psi U_q}{\theta f} + \nu_t + \eta \Gamma_q - \eta_t^s \frac{\sigma_{cq}}{q} = \frac{\phi U_q}{f}$$

Dividing by U_q , substituting $\phi/f = (1 - \chi_t)/U_c$ from (26), and multiplying by U_c ,

$$\frac{U_c}{U_q} \left(1 + \nu_t + \eta \Gamma_q - \eta_t^s \frac{\sigma_{cq}}{q} \right) - \gamma_q \frac{\psi U_c}{\theta f} = 1 - \gamma_c \frac{\psi U_c}{\theta f} + \eta \Gamma_c - \eta_t^s \frac{\sigma}{c}$$

With $1 + \tau^q = U_q/U_c$, multiply through by $1 + \tau^q$ and rearrange terms.

$$\frac{\tau_t^q - \nu_t}{1 + \tau_t^q} = (\gamma_c - \gamma_q) \frac{\psi U_c}{\theta f} - \eta \Gamma_c + \eta_t^s \frac{\sigma}{c} + \frac{1}{1 + \tau^q} \left[\eta \Gamma_q - \eta_t^s \frac{\sigma_{cq}}{q} \right]$$

Substituting Γ_c and Γ_q from (24),

$$\begin{aligned} \frac{\tau_t^q - \nu_t}{1 + \tau_t^q} &= (\gamma_{c,t} - \gamma_{q,t}) \frac{\psi U_{c,t}}{\theta f_t} - \eta_t (1 - \bar{\tau}_t^y) \frac{\sigma_t}{c_t} + \eta_t \frac{\gamma_{c,t}}{y_t} + \eta_t^s \frac{\sigma_t}{c_t} \\ &\quad + \eta_t (1 - \bar{\tau}_t^y) \frac{\sigma_{cq,t}}{q_t (1 + \tau_t^q)} - \eta_t \frac{\gamma_{q,t}}{y_t} - \eta_t^s \frac{\sigma_{cq,t}}{q_t (1 + \tau_t^q)} \\ &= (\gamma_{c,t} - \gamma_{q,t}) \left(\frac{\psi U_{c,t}}{\theta f_t} + \frac{\eta_t}{y_t} \right) - [\eta_t (1 - \bar{\tau}_t^y) - \eta_t^s] \left[\frac{\sigma_t}{c_t} - \frac{\sigma_{cq,t}}{q_t (1 + \tau_t^q)} \right] \end{aligned}$$

The first bracket is the net shadow price ζ_t and the second bracket is the income-effect gap. It is immediate that $\mathcal{I}_t > 0$ if and only if the dirty good is normal. Let q^d be the dirty-good demand from the household's split of expenditure x at the prevailing consumer price, and let $\mathcal{D} \equiv U_{qq} - 2(1 + \tau^q)U_{cq} + (1 + \tau^q)^2 U_{cc}$ be the curvature of utility along that budget line, negative because the split sets $U_q = (1 + \tau^q)U_c$ and Assumption 3.1 signs $\mathcal{D} = -(2U_c U_q U_{cq} - U_c^2 U_{qq} - U_q^2 U_{cc})/U_c^2$. This is the $\mathcal{D}$ of Section 5.2 at $\bar{p} = 1 + \tau^q$, the frozen price having no counterpart in this problem. Differentiating the split's tangency along the budget line,

$$\frac{\partial q^d}{\partial x} = \frac{(1 + \tau^q)U_{cc} - U_{cq}}{\mathcal{D}} = \frac{U_c}{\mathcal{D}} \left[-(1 + \tau^q) \frac{\sigma}{c} + \frac{\sigma_{cq}}{q} \right] = \frac{U_c (1 + \tau^q)}{|\mathcal{D}|} \mathcal{I}_t$$

From T_R on, the conditions for c and q are redone rather than continued. The labor-wedge constraint is not imposed there, so $\eta_t = 0$, and $\psi = 0$ by Appendix A.7. The conditions for c and q lose their costate and labor-constraint terms, leaving $\phi U_c/f = 1 - \eta_t^s \sigma/c$ and $1 + \nu_t - \eta_t^s \sigma_{cq}/q = \phi U_q/f$. Dividing the second by U_q , multiplying by U_c , and equating the two,

$$\frac{1 + \nu_t}{1 + \tau_t^q} = 1 - \eta_t^s \left[\frac{\sigma_t}{c_t} - \frac{\sigma_{cq,t}}{q_t (1 + \tau_t^q)} \right] = 1 - \eta_t^s \mathcal{I}_t \quad \implies \quad \frac{\tau_t^q - \nu_t}{1 + \tau_t^q} = \eta_t^s \mathcal{I}_t$$

This is Corollary 5.3. No separability assumption enters and nothing is read at $y_t = 0$. Both $\gamma_{c,t}$ and $\gamma_{q,t}$ carry the factor l_t , so they disappear at $l_t = 0$ and the Corlett-Hague term is absent for that reason as well as through ψ .

Appendix B.10 identifies $\nu_t = \text{SCC}_t$ in both, completing Proposition 5.1 and Corollary 5.3.

B.10 The constrained social cost of carbon

Take the first-order condition for S , divide the integrand by f , and substitute $\phi/f = (1 - \chi_t)/U_c$ together with $U_S l = \xi U_S$, Γ_S from (24) and $U_{cS}/U_c = \xi_c (U_S/U_c)/c$. The integrand becomes

$$-\frac{U_S}{U_c} \left\{ \xi \frac{\psi U_c}{\theta f} + (1 - \chi_t) + \eta \left[\frac{\xi}{y} + (1 - \bar{\tau}_t^y) \frac{\xi_c}{c} \right] - \eta_t^s \frac{\xi_c}{c} \right\}$$

Expanding $1 - \chi_t = 1 - \gamma_c \frac{\psi U_c}{\theta f} + \eta \left[(1 - \bar{\tau}_t^y) \frac{\sigma}{c} - \frac{\gamma_c}{y} \right] - \eta_t^s \frac{\sigma}{c}$ and collecting,

$$\begin{aligned} \{\cdot\} &= 1 + (\xi - \gamma_c) \frac{\psi U_c}{\theta f} + \eta \frac{\xi - \gamma_c}{y} + [\eta (1 - \bar{\tau}_t^y) - \eta_t^s] \frac{\sigma + \xi_c}{c} \\ &= 1 + (\xi - \gamma_c) \left(\frac{\psi U_c}{\theta f} + \frac{\eta}{y} \right) + \zeta_t \frac{\sigma + \xi_c}{c} \end{aligned}$$

The date- t pollution-stock multiplier is

$$\mu_t = - \int_0^\infty \left[1 + (\xi - \gamma_c) \left(\frac{\psi U_c}{\theta f} + \frac{\eta}{y} \right) + \zeta_t \frac{\sigma + \xi_c}{c} \right] \frac{U_S}{U_c} f d\theta \quad (41)$$

Iterating the emission-price recursion $\nu_t = \mu_t S_q + \frac{1}{R} \partial V_{t+1} / \partial Q$ forward as in Appendix A.6 with $\Phi_j \equiv \partial S_{t+j} / \partial Q_t$ and $\Phi_j = 0$ for $j > s$,

$$\nu_t = \text{SCC}_t = - \sum_{j=0}^s \frac{1}{R^j} \mathbb{E}_t \int \left[1 + (\xi_{t+j} - \gamma_{c,t+j}) \left(\frac{\psi_{t+j} U_{c,t+j}}{\theta f_{t+j}} + \frac{\eta_{t+j}}{y_{t+j}} \right) + \zeta_{t+j} \frac{\sigma_{t+j} + \xi_{c,t+j}}{c_{t+j}} \right] \frac{U_{S,t+j}}{U_{c,t+j}} \Phi_j f_{t+j} d\theta$$

The sum runs over dates so, read at a working-age history, it reaches histories with $t + j \geq T_R$, at which the labor-wedge constraint is not imposed. With $\psi = 0$ and no Γ_S term, the braces of (41) collapse to $1 - \eta_{t+j}^s (\sigma_{t+j} + \xi_{c,t+j}) / c_{t+j} = 1 + \zeta_{t+j} (\sigma_{t+j} + \xi_{c,t+j}) / c_{t+j}$, which is the displayed bracket with the screening term deleted. Only the savings channel prices damages from T_R on, which is Corollary 5.3 read inside the damage stream rather than at the commodity margin.

Setting $\eta = 0$ gives the weight under a frozen savings schedule alone, setting $\eta^s = 0$ gives the weight under a frozen labor schedule alone, and setting both to zero leaves the formula in Appendix A.6. Under $U = V(h(c, q, S), l)$, which gives $\gamma_c = \gamma_q$ and $\xi = \gamma_c$ together, the bracket collapses to $1 + \zeta_{t+j} (\sigma_{t+j} + \xi_{c,t+j}) / c_{t+j}$, so weak separability is not enough to restore a Pigouvian social cost of carbon once a margin is frozen.

B.11 Carbon-only reforms and the frozen wedges

This subsection proves Lemma 5.5 and Corollary 5.4, then records the construction that supports the constrained optimum inside the class and its standing.

The household's problem. Given the tax functions T_t^y , the capital income tax rate τ^k , a carbon-only reform $\{\mathcal{T}_t\}$, initial assets $a_0 = 0$, and an aggregate pollution stock which no single household moves, the household chooses plans $\{c_t, q_t, l_t, a_{t+1}\}$ measurable in its skill history to maximize lifetime expected utility subject to the budget constraint.

$$\mathbb{E}_0 \left[\sum_{t=0}^T \beta^t U(c_t, q_t, l_t, S_t) \right] \\ \text{s.t.}$$

$$c_t(\theta^t) + q_t(\theta^t) + \mathcal{T}_t(q^t(\theta^t)) + a_{t+1}(\theta^t) = \theta_t l_t(\theta^t) - T_t^y(\theta_t l_t(\theta^t)) + \tilde{R} a_t(\theta^{t-1})$$

with $\tilde{R} = 1 + (R - 1)(1 - \tau^k)$, $a_{T+1} = 0$, $l_t = 0$ from T_R on, and labor supply and saving interior at the dates where they are choices.

Proof of Lemma 5.5. The household's Lagrangian is

$$\mathcal{L} = \mathbb{E}_0 \left[\sum_{t=0}^T \beta^t \left\{ U(c_t, q_t, l_t, S_t) + \mu_t \left[\theta_t l_t - T_t^y(\theta_t l_t) + \tilde{R} a_t - c_t - q_t - \mathcal{T}_t(q^t) - a_{t+1} \right] \right\} \right]$$

Every tariff is measurable in (q^τ, τ) , so $\partial \mathcal{T}_\tau / \partial l_t = \partial \mathcal{T}_\tau / \partial a_{t+1} = 0$ at every τ , while history dependence carries q_t into every later tariff. The first-order conditions are

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial c_t} &: U_{c,t} = \mu_t \\ \frac{\partial \mathcal{L}}{\partial l_t} &: -U_{l,t} = \mu_t \theta_t (1 - T_t^{y'}(y_t)) \\ \frac{\partial \mathcal{L}}{\partial a_{t+1}} &: \mu_t = \beta \tilde{R} \mathbb{E}_t[\mu_{t+1}] \\ \frac{\partial \mathcal{L}}{\partial q_t} &: U_{q,t} = \mu_t \left(1 + \frac{\partial \mathcal{T}_t}{\partial q_t} \right) + \sum_{j=1}^{T-t} \beta^j \mathbb{E}_t \left[\mu_{t+j} \frac{\partial \mathcal{T}_{t+j}}{\partial q_t} \right] \end{aligned} \tag{42}$$

The labor supply conditions hold at $t < T_R$. Substituting $\mu_t = U_{c,t}$ from the condition for the numéraire into the other three, dividing the labor supply condition by $\theta_t U_{c,t}$ and the condition for the dirty good by $U_{c,t}$, and using $\tilde{R} = (1 - \bar{\tau}^s) R$, the definition of the frozen savings wedge, in the condition for saving,

$$\begin{aligned} -\frac{U_{l,t}}{\theta_t U_{c,t}} &= 1 - T_t^{y'}(y_t) \\ U_{c,t} &= (1 - \bar{\tau}^s) \beta R \mathbb{E}_t[U_{c,t+1}] \\ 1 + \tau_t^q &\equiv \frac{U_{q,t}}{U_{c,t}} = 1 + \frac{\partial \mathcal{T}_t}{\partial q_t} + \sum_{j=1}^{T-t} \beta^j \frac{\mathbb{E}_t[U_{c,t+j} \partial \mathcal{T}_{t+j} / \partial q_t]}{U_{c,t}} \end{aligned}$$

The first two are the conditions of the lemma, which Section 5.1 imposes on the allocation as the frozen-wedge constraints (5) and (6), the labor condition at the earnings-indexed schedule $\bar{\tau}_t^y(y) = T_t^{y'}(y)$ of Appendix B.13 and the saving condition at the constant wedge. The third

collects the current marginal tariff and the exposure of future tariffs to the current purchase, discounted and weighted by marginal utility. Every term on its right is an instrument choice, and no tariff derivative appears anywhere else in (42). This proves the lemma.

Proof of Corollary 5.4. Suppose a carbon-only reform implements the optimum of Section 4. By Lemma 5.5 the optimum then satisfies the frozen Euler equation at the constant $\bar{\tau}^s = (1 - \tilde{R}/R)$ at every history. Two steps rule that out on the savings margin at every τ^k , and a third shows the labor margin fails on its own.

Step 1: a nonzero rate fails at retirement dates. At a retirement date the optimum is deterministic along the history and the costate vanishes (Lemma A.2), while $\gamma_{c,t} = 0$ at $l = 0$, so the incentive credit is zero, $\chi_t = \chi_{t+1} = 0$. The generalized inverse Euler equation (23) at a degenerate draw then reads $1/U_{c,t} = (1/\beta R)(1/U_{c,t+1})$, that is,

$$U_{c,t} = \beta R U_{c,t+1}$$

The frozen Euler equation requires $U_{c,t} = (1 - \bar{\tau}^s) \beta R U_{c,t+1}$ at the same history. With $U_c > 0$, both hold only at $\bar{\tau}^s = 0$, hence at $\tilde{R} = R$, which is $\tau^k = 0$ since the tax falls on the interest component $R - 1 > 0$.

Step 2: the zero rate fails at working-life histories. At $\tau^k = 0$ the frozen Euler equation reads $U_{c,t} = \beta R \mathbb{E}_t[U_{c,t+1}]$. By hypothesis the optimum carries a strictly positive savings wedge at some working-life history: $1 - \tau_t^s = U_{c,t}/(\beta R \mathbb{E}_t[U_{c,t+1}]) < 1$ there, so $U_{c,t} < \beta R \mathbb{E}_t[U_{c,t+1}]$, and the frozen Euler equation fails. The hypothesis is not restrictive. Under weak separability it follows from the inverse Euler equation: $1/U_{c,t} = (1/\beta R) \mathbb{E}_t[1/U_{c,t+1}]$ gives

$$1 - \tau_t^s = \frac{U_{c,t}}{\beta R \mathbb{E}_t[U_{c,t+1}]} = \frac{1}{\mathbb{E}_t[U_{c,t+1}] \mathbb{E}_t[1/U_{c,t+1}]} \leq 1$$

where the inequality is the Cauchy–Schwarz bound $\mathbb{E}[X] \mathbb{E}[1/X] \geq \left(\mathbb{E} \left[\sqrt{X} \cdot 1/\sqrt{X} \right] \right)^2 = 1$, strict unless $U_{c,t+1}$ is degenerate; this is the positive intertemporal wedge of Golosov et al. (2003). Under the nonseparable preferences of Section 6 the hypothesis is read off the solved allocation, whose savings wedge runs from 0.56% at entry to 0.02% at 63, the last age before the retirement junction.

No constant matches a path that takes both values, so Steps 1 and 2 together cover every τ^k .

Step 3: the labor margin fails independently. By the lemma, every class equilibrium carries labor wedge $T_t^{y'}(y_t)$, a function of earnings and the date. The optimum's labor wedge (15) is $A_t \psi U_{c,t}/(\theta f_t)$, which depends on the history through the costate and the allocation. Under

persistent shocks two date- t histories assigned equal earnings generically differ in it, so the optimal wedge is not measurable in the pair (y_t, t) and no tax function of earnings reproduces it. Even at a configuration in which the optimal wedge happened to be measurable in earnings, implementation would further require that function to coincide with the statutory $T_t^{y'}$, which it does not. This proves the corollary.

The supporting tariff. Conversely, a solution of the constrained problem of Section 5.1 is supported inside the class by the following construction. Set the asset path to zero and define the on-path tariff at the solved purchase histories by budget balance,

$$\mathcal{T}_t(q^t(\theta^t)) = \theta_t l_t(\theta^t) - T_t^y(\theta_t l_t(\theta^t)) - c_t(\theta^t) - q_t(\theta^t)$$

and extend it off path so that purchase plans away from the solved menu forfeit the intercepts. Wherever the solved purchase histories separate skill histories, the menu of tariff positions is in bijection with the reporting menu of the mechanism, and choosing a position is the report; separation is the generic case in the calibration, where the dirty good is normal and strictly increasing in the within-date state, and the construction requires it, since a tariff conditioned on purchases alone cannot deliver two intercepts at one pooled purchase history.

At a position, the household's remaining choices are hours, the clean good, and saving at $\tilde{R}$, and its problem in those variables is concave: U is concave, the budget is linear in c and a , and after-tax earnings $y - T_t^y(y)$ are concave in y under the earnings-indexed schedule with $p_t \in (0, 1)$. At the truthful position the assigned bundle satisfies the household's first-order conditions in all three variables, which is exactly what the frozen-wedge constraints assert, so it solves the position problem globally. A deviation is therefore a position change combined with re-optimization at the new position, and at the truthful allocation no such deviation carries a first-order gain: the envelope condition prices the position change, and the household's first-order conditions price the re-optimization. Global gains are the sufficiency caveat the first-order approach carries everywhere in the paper; the one-shot audit of Appendix D.4 evaluates deviations across menu positions at the assigned bundles, and re-optimization at a misreported position sits with the same caveat. The construction also locates the levels freedom of the text: the intercepts are transfers conditioned on purchases rather than on earnings.

Two boundaries of the construction are worth recording. The income-side instruments offer no level freedom of their own: λ_t^T enters the marginal rate $\lambda_t^T(1 - p_t)y^{-p_t}$, so it cannot move the level of the schedule without moving the frozen rates, and every intercept in the class therefore sits on the tariff side. And the construction certifies nonemptiness for the sequence problem, which is what makes the constrained optimum well posed; it does not certify that every $(\hat{w}, \hat{w}_2, \hat{M})$ in the nominal state space is deliverable, a coverage question for the solver rather than for the theory.

B.12 The optimal flat pair

Let the two schedules be flat at levels $\bar{\tau}_t^y$ and $\bar{\tau}_t^s$ and treat each as a choice variable. Write $\mathcal{L}$ for the date-0 Lagrangian of the problem of Section 5.1: the resource cost with every constraint appended, whose period block is the Hamiltonian of Appendix B.2. The rate $\bar{\tau}_t^y$ enters it only through the date- t labor-wedge constraints, one per history, so its derivative collects those terms and nothing else:

$$\frac{\partial \mathcal{L}}{\partial \bar{\tau}_t^y} = \int_0^\infty \eta_t(\theta) f_t(\theta) d\theta = \mathbb{E}_0[\eta_t] = 0$$

The rate $\bar{\tau}_t^s$ enters only through the requirement $M(\theta)$ passed to the successor, so

$$\frac{\partial \mathcal{L}}{\partial \bar{\tau}_t^s} = \int_0^\infty \frac{1}{R} \frac{\partial V_{t+1}}{\partial \hat{M}} \frac{\partial M(\theta)}{\partial \bar{\tau}_t^s} f_t(\theta) d\theta = - \int_0^\infty \frac{\vartheta_{t+1}(\theta) U_{c,t}(\theta)}{(1 - \bar{\tau}_t^s)^2 \beta R^2} f_t(\theta) d\theta = 0$$

which, with $(1 - \bar{\tau}_t^s)^2 \beta R^2 > 0$ constant across the date- t cross-section, gives $\mathbb{E}_0[\vartheta_{t+1} U_{c,t}] = 0$. Taking the cross-sectional expectation of (7) then gives $\mathbb{E}_0[\eta_t^s] = \mathbb{E}_0[\vartheta_t U_{c,t}]$, which equals zero at $t = 0$ because $\vartheta_0 = 0$.

For the closed form, solve (27) for η_t and impose $\mathbb{E}_0[\eta_t] = 0$:

$$\mathbb{E}_0 \left[\frac{1}{\Delta_t} \left(\frac{\bar{\tau}_t^y}{1 - \bar{\tau}_t^y} - A_t \frac{\psi U_{c,t}}{\theta f_t} + \eta_t^s \Lambda_t \right) \right] = 0$$

The weight is $1/\Delta_t$, which Appendix B.3 reads as the compensated earnings response $s_{yy,t} = 1/((1 - \bar{\tau}_t^y) \Delta_t)$ of (25). With $\bar{\tau}_t^y$ constant across the cross-section, the zero-average condition rearranges to

$$\frac{\bar{\tau}_t^y}{1 - \bar{\tau}_t^y} = \frac{\mathbb{E}_0 \left[s_{yy,t} \left(A_t \frac{\psi U_{c,t}}{\theta f_t} - \eta_t^s \Lambda_t \right) \right]}{\mathbb{E}_0[s_{yy,t}]}$$

which is Proposition 5.5. At $\eta_t^s = 0$ the savings term drops and the rate averages the desired wedges alone.

B.13 An earnings-indexed schedule

Section 5.1 leaves the frozen labor schedule arbitrary. The one this paper freezes is the two-parameter tax and transfer function of Bénabou (2002) and Heathcote et al. (2017),

$$T_t(y) = y - \lambda_t^T y^{1-p_t} \implies 1 - \bar{\tau}_t^y(y) = \lambda_t^T (1 - p_t) y^{-p_t} \quad (43)$$

where p_t indexes progressivity and λ_t^T the level.

Indexing the schedule on earnings rather than on type changes one statistic and nothing else, which is what lets Section 5.1 be read at (43) throughout. Three things have to be shown: that $\Gamma_{c,t}$, $\Gamma_{q,t}$ and $\Gamma_{S,t}$ survive intact, since $\bar{\tau}_t^y$ now depends on the type through assigned earnings $y_t(\theta^t)$; that $\Gamma_{l,t}$ and hence Δ_t each gain a term in p_t ; and that the resulting substitution $\epsilon_t \mapsto \epsilon_t + p_t$ stays inside Δ_t everywhere downstream of (8) but one.

With the frozen labor wedge following (43), $1 - \bar{\tau}_t^y(y) = \lambda_t^T (1 - p_t) y^{-p_t}$ with $y = \theta l$. Differentiating,

$$\frac{\partial \bar{\tau}_t^y}{\partial y} = \lambda_t^T (1 - p_t) p_t y^{-p_t-1} = \frac{p_t (1 - \bar{\tau}_t^y)}{y}$$

The constraint is $\Gamma = -U_l/(\theta U_c) - (1 - \bar{\tau}_t^y(y))$. Its derivatives with respect to c , q , and S are unchanged, because $\bar{\tau}_t^y$ depends on none of them. Its derivative with respect to l gains one term through $y = \theta l$:

$$\Gamma_l = (1 - \bar{\tau}_t^y) \frac{\epsilon - \gamma_c}{l} + \frac{\partial \bar{\tau}_t^y}{\partial y} \theta = (1 - \bar{\tau}_t^y) \frac{\epsilon - \gamma_c}{l} + \frac{p_t (1 - \bar{\tau}_t^y)}{l} = (1 - \bar{\tau}_t^y) \frac{\epsilon - \gamma_c + p_t}{l}$$

using $\theta/y = 1/l$. Only Δ_t changes, since Γ_l enters the derivation of Appendix B.4 solely through $-\eta [\Gamma_l U_c/U_l - \Gamma_c]$:

$$-\eta \left[-\frac{\epsilon - \gamma_c + p_t}{y} - (1 - \bar{\tau}_t^y) \frac{\sigma}{c} + \frac{\gamma_c}{y} \right] = \eta \left[(1 - \bar{\tau}_t^y) \frac{\sigma}{c} + \frac{\epsilon + p_t - 2\gamma_c}{y} \right]$$

so $\Delta_t \mapsto \Delta_t + p_t/y_t$, that is

$$\Gamma_{l,t} = (1 - \bar{\tau}_t^y) \frac{\epsilon_t - \gamma_{c,t} + p_t}{l_t} \quad \text{and} \quad \Delta_t = (1 - \bar{\tau}_t^y) \frac{\sigma_t}{c_t} + \frac{\epsilon_t + p_t - 2\gamma_{c,t}}{y_t}$$

The same p_t/y_t reaches the compensated-perturbation bracket of Appendix B.3, so $s_{yy,t}$ carries the shifted Δ_t as well.

Every other object downstream of (8) is untouched but one. The exception is the cancellation of Appendix B.8, where $\Gamma_{c,t}$ is unchanged while Δ_t is not, so the substitution survives the difference:

$$\Gamma_{c,t} - \Delta_t = -\frac{A_t - 1}{y_t} - \frac{p_t}{y_t} = -\frac{A_t - 1 + p_t}{y_t}$$

Equation (9) therefore holds with $A_t - 1 + p_t$ in the numerator of the coefficient on η_t^s and $\Delta_t + p_t/y_t$ in both denominators, so that the coefficient reads $-(A_t - 1 + p_t)/(y_t\Delta_t + p_t)$: p_t appears outside Δ_t as well as inside it, not instead of it, and the condition signing that coefficient weakens from $A_t > 1$ to $A_t - 1 + p_t > 0$. Because $p_t > 0$ for a progressive schedule, Δ_t rises: the schedule's convexity in earnings makes the agent's objective more concave in hours, so an interior labor choice is easier to sustain than under a schedule flat in earnings, and the interior-hours requirement $\Delta_t > 0$ is weaker than under a type-indexed schedule.

Freezing a tax function rather than a type-indexed wedge schedule is the more faithful exercise. A tax code conditions on earnings, so freezing (43) denies the planner the within-date, type-by-type level adjustment Kaplow (2012) relies on: on the range of assigned earnings the schedule is pinned up to a single constant. It does not deny the planner a transfer that varies with the history, which the mechanism retains, and the deviations of Corollary 5.1 survive that freedom.

C Derivations: a given carbon price

This appendix derives the results of Section 5.2 in one pass. The commodity wedge is held to a schedule $\bar{p}_t(\theta) \equiv 1 + \bar{\tau}_t^q(\theta)$: the given-schedule results read every condition at an arbitrary exogenous schedule, and the optimal-anonymous-price results at a flat schedule, $\bar{p}_t(\theta) = 1 + \bar{\tau}_t$, whose level is chosen optimally. The shadow emission price ν_t of (13) is carried throughout; the no-externality benchmark sets $\nu_t = 0$. Multiplier conventions follow Appendix A.3, and $\eta^q(\theta)f_t(\theta|\theta_-)$ is the multiplier on the wedge constraint. Two shorthands recur: the ad-valorem deviation and its desired counterpart,

$$\varpi_t(\theta) \equiv \frac{\bar{\tau}_t^q(\theta) - \nu_t}{1 + \bar{\tau}_t^q(\theta)} \quad \varpi_t^*(\theta^t) \equiv (\gamma_{c,t} - \gamma_{q,t}) \frac{\psi U_{c,t}}{\theta f_t}$$

which at a flat schedule are the ϖ_t and ϖ_t^* of Section 5.2.

C.1 The problem and first-order conditions

The Hamiltonian is that of Appendix A.1 plus the constraint term $\eta^q [U_q/U_c - \bar{p}] f$. Write $\Gamma^q \equiv U_q/U_c - \bar{p}$ and Γ_z^q for its partial derivatives; each follows from the quotient rule,

$$\Gamma_z^q = \frac{U_{qz}}{U_c} - \frac{U_q}{U_c} \frac{U_{cz}}{U_c} = \frac{U_{qz}}{U_c} - \bar{p} \frac{U_{cz}}{U_c}$$

evaluated at the constrained allocation, where $U_q/U_c = \bar{p}$. For the numéraire, with $U_{qc}/U_c = -\sigma_{cq}/q$ and $U_{cc}/U_c = -\sigma/c$,

$$\Gamma_c^q = -\frac{\sigma_{cq}}{q} + \bar{p} \frac{\sigma}{c} = \bar{p} \left[\frac{\sigma}{c} - \frac{\sigma_{cq}}{\bar{p}q} \right] = \bar{p} \mathcal{I}_t, \quad (44)$$

the income-effect gap (3) evaluated at $1 + \tau^q = \bar{p}$. For the dirty good, define $\sigma_{q,t} \equiv -U_{qq} q_t / U_q$, the own-curvature of the dirty good's marginal utility, so that $U_{qq}/U_c = (U_{qq}q/U_q)(U_q/U_c)/q = -\bar{p} \sigma_q/q$ and

$$\Gamma_q^q = -\bar{p} \frac{\sigma_q}{q} + \bar{p} \frac{\sigma_{cq}}{q} = -\bar{p} \frac{\sigma_q - \sigma_{cq}}{q} \equiv -\bar{p} \mathcal{J}_t, \quad (45)$$

with $\mathcal{J}_t \equiv (\sigma_{q,t} - \sigma_{cq,t})/q_t$ the mirror image of $\mathcal{I}_t$. For labor, with $U_{ql}/U_c = (U_{ql}l/U_q)(U_q/U_c)/l = \bar{p} \gamma_q/l$ and $U_{cl}/U_c = \gamma_c/l$,

$$\Gamma_l^q = \bar{p} \frac{\gamma_q}{l} - \bar{p} \frac{\gamma_c}{l} = -\bar{p} \frac{\gamma_c - \gamma_q}{l}. \quad (46)$$

For the pollution stock, with $\xi_c \equiv U_{Sc}c/U_S$ and $\xi_q \equiv U_{Sq}q/U_S$ the complementarities of the pollutant with each good, $U_{qS}/U_c = \xi_q(U_S/U_c)/q$ and $U_{cS}/U_c = \xi_c(U_S/U_c)/c$, so

$$\Gamma_S^q = \frac{U_S}{U_c} \left[\frac{\xi_q}{q} - \bar{p} \frac{\xi_c}{c} \right]. \quad (47)$$

The first-order conditions are those of Appendix A.1, each augmented by $\eta^q f \Gamma_z^q$:

$$\begin{aligned} \frac{\partial \mathcal{H}}{\partial c} &: f - \psi U_{cl} \frac{l}{\theta} + \eta^q f \Gamma_c^q = \phi U_c \\ \frac{\partial \mathcal{H}}{\partial q} &: f - \psi U_{ql} \frac{l}{\theta} + \nu_t f + \eta^q f \Gamma_q^q = \phi U_q \\ \frac{\partial \mathcal{H}}{\partial l} &: -\theta f - \phi U_l + \eta^q f \Gamma_l^q = \frac{1}{\theta} \psi [U_{ll}l + U_l] \\ \frac{\partial \mathcal{H}}{\partial S} &: \int_0^\infty \left(-\psi U_{Sl} \frac{l}{\theta} - \phi U_S + \eta^q f \Gamma_S^q \right) d\theta = \mu \end{aligned}$$

with the conditions for u , w , and w_2 unchanged, so the costate equation, the boundary conditions $\psi(0) = \psi(\infty) = 0$, and the promise-keeping chain carry over. In the no-externality economy $\nu_t = 0$ and the pollution-stock condition is absent. When the level of a flat schedule is itself chosen, one more derivative is needed; it is taken in Appendix C.9.

C.2 Implementation and first-order approach

Fix a common price $\bar{p} = 1 + \bar{\tau}$ and consider the agent's split of expenditure x ,

$$\tilde{U}(x, l; \bar{\tau}) = \max_{c, q \geq 0} U(c, q, l) \quad \text{s.t.} \quad c + \bar{p}q = x$$

The first-order condition of the split is $U_q = \bar{p} U_c$. If we substitute c using the budget constraint, $U(x - \bar{p}q, q, l)$ and therefore $\tilde{U}_q = -U_c \bar{p} + U_q$ and $\tilde{U}_{qq} = U_{cc} \bar{p}^2 - 2U_{cq} \bar{p} + U_{qq}$. Assumption 3.1 implies that $\mathcal{D} < 0$ and that an interior split exists and is unique. A bundle (c, q) therefore satisfies the wedge constraint $U_q/U_c = \bar{p}$ if and only if it is the split of $x = c + \bar{p}q$.

For the envelope identities, differentiate $\tilde{U} = U(c^d, q^d, l)$, writing $c^d(x, l; \bar{p})$ and $q^d(x, l; \bar{p})$ for the maximizers of the split problem and $\mathcal{D} \equiv \tilde{U}_{qq} < 0$ for the curvature of utility along the

budget line, the statistic of Section 5.2 that carries the identities below. Differentiating the budget over x gives $\frac{\partial c^d}{\partial x} + \bar{p} \frac{\partial q^d}{\partial x} = 1$. By the first-order condition,

$$\tilde{U}_x = U_c \frac{\partial c^d}{\partial x} + U_q \frac{\partial q^d}{\partial x} = U_c \left(\frac{\partial c^d}{\partial x} + \bar{p} \frac{\partial q^d}{\partial x} \right) = U_c$$

Differentiating the budget in l at fixed x gives $\frac{\partial c^d}{\partial l} + \bar{p} \frac{\partial q^d}{\partial l} = 0$.

$$\tilde{U}_l = U_l + U_c \left(\frac{\partial c^d}{\partial l} + \bar{p} \frac{\partial q^d}{\partial l} \right) = U_l$$

A deviator who reports $\hat{\theta}$ receives the bundle assigned to $\hat{\theta}$, meaning the pair $(x(\hat{\theta}), y(\hat{\theta}))$. They split $x(\hat{\theta})$ at the same common price, so the deviation space is that of a one-good economy in (x, y) with period utility $\tilde{U}(x, y/\theta; \bar{\tau})$. The type enters $\tilde{U}$ only through $l = y/\theta$, so the envelope condition is $\dot{u} = -\tilde{U}_l l/\theta + \beta w_2 = -U_l l/\theta + \beta w_2$ and the first-order approach applies as in the one-good economy of Appendix A.3.

C.3 Demand identities

Differentiating the tangency $U_q = \bar{p}U_c$ together with the budget gives the demand derivatives. In x at fixed $l, \bar{p}$, the budget gives $\partial_x c^d = 1 - \bar{p} \partial_x q^d$. Substituting into $U_{qc} \partial_x c^d + U_{qq} \partial_x q^d = \bar{p}(U_{cc} \partial_x c^d + U_{cq} \partial_x q^d)$ and collecting,

$$\mathcal{D} \partial_x q^d = -(U_{cq} - \bar{p}U_{cc}) \implies \partial_x q^d = \frac{U_{cq} - \bar{p}U_{cc}}{|\mathcal{D}|} = \frac{\bar{p}U_c \mathcal{I}_t}{|\mathcal{D}|}, \quad (48)$$

where the last step uses $U_{cq} - \bar{p}U_{cc} = U_c(\sigma \bar{p}/c - \sigma_{cq}/q) = \bar{p}U_c \mathcal{I}_t$ from the definitions of the statistics. Since $\partial_x q^d$ is the income expansion of dirty-good demand, $\mathcal{I}_t > 0$ exactly when the dirty good is normal. The budget then gives $\partial_x c^d = 1 - m_q$ with $m_q \equiv \bar{p} \partial_x q^d$, which satisfies $U_q \mathcal{J}_t = \bar{p}U_{cq} - U_{qq}$ and hence $\partial_x c^d = U_q \mathcal{J}_t / |\mathcal{D}|$ is positive when c is normal. In l at fixed $x, \bar{p}$, the budget gives $\partial_l c^d = -\bar{p} \partial_l q^d$ and the differentiated tangency yields $\mathcal{D} \partial_l q^d = \bar{p}U_{cl} - U_{ql}$, with $U_{cl} = \gamma_c U_c / l$ and $U_{ql} = \gamma_q \bar{p} U_c / l$,

$$\partial_l q^d = \frac{\bar{p}U_c(\gamma_c - \gamma_q)}{l \mathcal{D}} = \frac{\bar{p}U_c(\gamma_q - \gamma_c)}{l |\mathcal{D}|}. \quad (49)$$

Compensated own-price response: along an indifference curve $dc = -\bar{p}dq$ and differentiating the tangency in $\bar{p}$ at constant utility gives $\mathcal{D} d\bar{p} = U_c d\bar{p}$. Hence,

$$s_{qq} \equiv \frac{\partial q^d}{\partial \bar{p}} \Big|_u = \frac{U_c}{\mathcal{D}} = -\frac{U_c}{|\mathcal{D}|} < 0 \quad \frac{\partial q^d}{\partial \bar{p}} = s_{qq} - q^d \partial_x q^d \quad (\text{Slutsky})$$

Three identities tie the constraint statistics to the demand objects. Combining (49) with s_{qq} ,

$$l \partial_l q^d = -\bar{p} |s_{qq}| (\gamma_c - \gamma_q). \quad (50)$$

Adding $\bar{p} \mathcal{I}_t = (U_{cq} - \bar{p} U_{cc})/U_c$ and $\mathcal{J}_t = (\bar{p} U_{cq} - U_{qq})/U_q = (\bar{p} U_{cq} - U_{qq})/(\bar{p} U_c)$,

$$\Delta_t^q \equiv \bar{p} \mathcal{I}_t + \mathcal{J}_t = \frac{\bar{p} U_{cq} - \bar{p}^2 U_{cc} + \bar{p} U_{cq} - U_{qq}}{\bar{p} U_c} = \frac{-\mathcal{D}}{\bar{p} U_c} = \frac{1}{\bar{p} |s_{qq}|} > 0. \quad (51)$$

Combining (48) and (51), the marginal budget share is the numéraire-side curvature share,

$$m_q = \bar{p} \partial_x q^d = \frac{\bar{p}^2 U_c \mathcal{I}_t}{|\mathcal{D}|} = \frac{\bar{p} \mathcal{I}_t}{\Delta_t^q}. \quad (52)$$

C.4 The multiplier

Dividing the first-order condition for c by $U_c f$, using $U_c l = \gamma_c U_c$ and (44),

$$\frac{U_c \phi}{f} = 1 - \gamma_c \frac{\psi U_c}{\theta f} + \eta^q \bar{p} \mathcal{I}_t. \quad (53)$$

Dividing the first-order condition for q by $U_q f$, multiplying by U_c , and substituting $U_c/U_q = 1/\bar{p}$ and (45),

$$\frac{U_c \phi}{f} = \frac{1 + \nu_t}{\bar{p}} - \gamma_q \frac{\psi U_c}{\theta f} + \eta^q \frac{\Gamma_q^q}{\bar{p}} = \frac{1 + \nu_t}{\bar{p}} - \gamma_q \frac{\psi U_c}{\theta f} - \eta^q \mathcal{J}_t. \quad (54)$$

Equating (53) and (54) and collecting,

$$1 - \frac{1 + \nu_t}{\bar{p}} = (\gamma_c - \gamma_q) \frac{\psi U_c}{\theta f} - \eta^q (\bar{p} \mathcal{I}_t + \mathcal{J}_t)$$

Since $1 - (1 + \nu_t)/\bar{p} = (\bar{\tau}_t^q - \nu_t)/(1 + \bar{\tau}_t^q) = \varpi_t$ with Δ_t^q from (51),

$$\eta_t^q \Delta_t^q = \varpi_t^* - \varpi_t \quad \Longleftrightarrow \quad \eta_t^q = \bar{p}_t |s_{qq,t}| (\varpi_t^* - \varpi_t). \quad (55)$$

This is Lemma 5.7 with $\text{SCC}_t \equiv \nu_t$; a type-varying schedule changes nothing in the display. At $\eta_t^q = 0$, (55) is the primitive corrective-wedge condition of Appendix A.5, so the multiplier vanishes where the frozen deviation equals the desired one.

C.5 Labor wedge

From the first-order condition for l , using $U_{ll}l + U_l = U_l(1 + \epsilon)$,

$$\phi = -\frac{\theta f}{U_l} - \frac{\psi}{\theta}(1 + \epsilon) + \frac{\eta_t^q f \Gamma_l^q}{U_l}$$

With $U_l = -(1 - \tau_t^y)\theta U_c$, where τ_t^y is the endogenous labor wedge, and (46),

$$\frac{\Gamma_l^q}{U_l} = \frac{-\bar{p}(\gamma_c - \gamma_q)/l}{-(1 - \tau_t^y)\theta U_c} = \frac{\bar{p}(\gamma_c - \gamma_q)}{(1 - \tau_t^y)y U_c}$$

Multiplying by U_c/f ,

$$\frac{U_c \phi}{f} = \frac{1}{1 - \tau_t^y} - (1 + \epsilon) \frac{\psi U_c}{\theta f} + \eta_t^q \frac{\bar{p}(\gamma_c - \gamma_q)}{(1 - \tau_t^y)y} \quad (56)$$

Equating (53) and (56) and collecting, with $1/(1 - \tau_t^y) - 1 = \tau_t^y/(1 - \tau_t^y)$ and $A_t = 1 + \epsilon - \gamma_c$,

$$\frac{\tau_t^y}{1 - \tau_t^y} = A_t \frac{\psi U_c}{\theta f} + \eta_t^q \bar{p}_t \left[\mathcal{I}_t - \frac{\gamma_{c,t} - \gamma_{q,t}}{(1 - \tau_t^y)y_t} \right] \quad (57)$$

This is the primitive labor wedge of Proposition 5.3. The bracket is the response of the implied commodity wedge $\ln(U_q/U_c)$ to a utility-compensated unit of earnings. Along $(dc, dl) = (1 - \tau_t^y, 1/\theta)$, we have $d\ln(U_q/U_c) = (1 - \tau_t^y)\mathcal{I}_t + (\gamma_q - \gamma_c)/y$. Dividing by $1 - \tau_t^y$ gives the bracket. Under weak separability the bracket is $\mathcal{I}_t$. Using (55) with $\varpi^* = 0$ and (52),

$$\eta_t^q \bar{p}_t \mathcal{I}_t = -\frac{\bar{p}_t \mathcal{I}_t}{\Delta_t^q} \varpi_t = -m_{q,t} \varpi_t = m_{q,t} \frac{\nu_t - \bar{\tau}_t^q}{1 + \bar{\tau}_t^q}$$

which is Corollary 5.5.

C.6 Savings wedge

Define $\chi_t^q \equiv \gamma_{c,t} \frac{\psi U_{c,t}}{\theta f_t} - \eta_t^q \bar{p}_t \mathcal{I}_t$, so that (53) reads

$$\frac{\phi}{f} = \frac{1 - \chi_t^q}{U_{c,t}}.$$

The promise-keeping chain of Appendix A.4 is untouched, since the wedge constraint enters none of its steps. The first-order condition for w and the envelope condition $\partial V_{t+1}/\partial \hat{w} = \lambda_{1,t+1}$ give

$$\lambda_{1,t+1}(\theta^t) = \beta R \frac{\phi(\theta)}{f(\theta)} = \beta R \frac{1 - \chi_t^q}{U_{c,t}}.$$

Integrating the costate equation over the type space with $\psi(0) = \psi(\infty) = 0$, $\int \bar{\alpha}_t f d\theta = 1$, and $\int f_2 d\theta = 0$ gives the second expression

$$\lambda_{1,t} = \int_0^\infty \phi d\theta = \mathbb{E}_{t-1} \left[\frac{1 - \chi_t^q}{U_{c,t}} \right].$$

Evaluating this at $t + 1$ and equating with the previous display one period apart,

$$\frac{1 - \chi_t^q}{U_{c,t}} = \frac{1}{\beta R} \mathbb{E}_t \left[\frac{1 - \chi_{t+1}^q}{U_{c,t+1}} \right]$$

Substituting the multiplier (55) and the curvature-share identity (52) assembles the adjustment into composite form,

$$\chi_t^q = \gamma_c \frac{\psi U_c}{\theta f} - m_q (\varpi_t^* - \varpi_t) = [\gamma_c - m_q(\gamma_c - \gamma_q)] \frac{\psi U_c}{\theta f} + m_q \varpi_t = \tilde{\gamma}_t \frac{\psi U_{c,t}}{\theta f_t} + m_{q,t} \varpi_t$$

with $\tilde{\gamma} \equiv (1 - m_q)\gamma_c + m_q\gamma_q$ as in Proposition C.1, and $m_q\varpi = (\bar{\tau}^q - \nu) \partial_x q^d$ is the commodity revenue net of marginal damages returned by a marginal dollar of expenditure. It remains to pass from the inverse Euler equation to the savings wedge. The savings wedge τ_t^s is defined by the agent's intertemporal condition $U_{c,t} = \beta R (1 - \tau_t^s) \mathbb{E}_t[U_{c,t+1}]$, so

$$1 - \tau_t^s = \frac{U_{c,t}}{\beta R \mathbb{E}_t[U_{c,t+1}]}.$$

The generalized inverse Euler equation gives $\beta R = U_{c,t} \mathbb{E}_t[(1 - \chi_{t+1}^q)/U_{c,t+1}] / (1 - \chi_t^q)$. Substituting,

$$1 - \tau_t^s = \frac{1 - \chi_t^q}{\mathbb{E}_t[(1 - \chi_{t+1}^q)/U_{c,t+1}] \mathbb{E}_t[U_{c,t+1}]}, \quad (58)$$

the exact wedge. To approximate it, write $g_{t+1} \equiv \ln U_{c,t+1} = \bar{g} + \hat{g}$ with $\mathbb{E}_t \hat{g} = 0$, and treat the fluctuation $\hat{g}$ and the adjustment χ_{t+1}^q as small. Expanding $e^{\pm g_{t+1}}$ to second order in $\hat{g}$ and keeping first order in χ^q ,

$$\begin{aligned}\mathbb{E}_t[U_{c,t+1}] &= e^{\bar{g}} \left(1 + \frac{1}{2} \text{Var}_t \hat{g} \right), \\ \mathbb{E}_t \left[\frac{1 - \chi_{t+1}^q}{U_{c,t+1}} \right] &= e^{-\bar{g}} \left(1 - \mathbb{E}_t \chi_{t+1}^q + \text{Cov}_t(\chi_{t+1}^q, \hat{g}) + \frac{1}{2} \text{Var}_t \hat{g} \right),\end{aligned}$$

where $\mathbb{E}_t \hat{g} = 0$ turns the linear term of the second expansion into the covariance. Their product is $1 - \mathbb{E}_t \chi_{t+1}^q + \text{Cov}_t(\chi_{t+1}^q, \hat{g}) + \text{Var}_t \hat{g}$ to this order, so inverting (58),

$$1 - \tau_t^s = (1 - \chi_t^q) \left[1 + \mathbb{E}_t \chi_{t+1}^q - \text{Cov}_t(\chi_{t+1}^q, \hat{g}) - \text{Var}_t \hat{g} \right] \approx 1 - \chi_t^q + \mathbb{E}_t \chi_{t+1}^q - \text{Cov}_t(\chi_{t+1}^q, \hat{g}) - \text{Var}_t \hat{g}.$$

Since $\text{Cov}_t(\chi_{t+1}^q, \hat{g}) = \text{Cov}_t(\chi_{t+1}^q, \ln U_{c,t+1})$,

$$\tau_t^s \approx \text{Var}_t(\ln U_{c,t+1}) + (\chi_t^q - \mathbb{E}_t \chi_{t+1}^q) + \text{Cov}_t(\chi_{t+1}^q, \ln U_{c,t+1}),$$

the second-order form of Proposition 5.3. The baseline expansion of Appendix A.4 is the special case $\chi^q \mapsto \chi$.

C.7 Composite statistics and the comprehensive wedge

Proposition C.1 (Composite statistics).

Define

$$\tilde{\gamma} \equiv (1 - m_q) \gamma_c + m_q \gamma_q \quad \tilde{\epsilon} \equiv \epsilon - \frac{\bar{p}^2 U_c^2 (\gamma_c - \gamma_q)^2}{l |U_l| |\mathcal{D}|} \leq \epsilon \quad \tilde{A} \equiv 1 + \tilde{\epsilon} - \tilde{\gamma}$$

Then $\tilde{\gamma} = \tilde{U}_{xl} l / \tilde{U}_x$ and $\tilde{\epsilon} = \tilde{U}_{ll} l / \tilde{U}_l$ for the indirect utility of Lemma 5.6. Under weak separability $\gamma_c = \gamma_q$, so $\tilde{\gamma} = \gamma_c$, $\tilde{\epsilon} = \epsilon$, and the restricted problem is that of Section 4.

Substituting the multiplier (55) into the labor condition (57) yields the comprehensive form of the wedge. Using $\bar{p}/\Delta^q = \bar{p}^2 |s_{qq}|$ from (51), the correction term expands into four pieces,

$$\eta^q \bar{p} \left[\mathcal{I} - \frac{\gamma_c - \gamma_q}{(1 - \tau^y) y} \right] = \left[m_q - \frac{\bar{p}^2 |s_{qq}| (\gamma_c - \gamma_q)}{(1 - \tau^y) y} \right] (\varpi^* - \varpi)$$

and, expanding with $\varpi^* = (\gamma_c - \gamma_q) \psi U_c / (\theta f)$,

$$\begin{aligned}
m_q \varpi^* &= m_q(\gamma_c - \gamma_q) \frac{\psi U_c}{\theta f} = (\gamma_c - \tilde{\gamma}) \frac{\psi U_c}{\theta f} \\
-\frac{\bar{p}^2 |s_{qq}|(\gamma_c - \gamma_q)}{(1 - \tau^y)y} \varpi^* &= -\frac{\bar{p}^2 |s_{qq}|(\gamma_c - \gamma_q)^2}{(1 - \tau^y)y} \frac{\psi U_c}{\theta f} = -(\epsilon - \tilde{\epsilon}) \frac{\psi U_c}{\theta f} \\
-m_q \varpi &= -(\bar{\tau}^q - \nu) \partial_x q^d \\
+\frac{\bar{p}^2 |s_{qq}|(\gamma_c - \gamma_q)}{(1 - \tau^y)y} \varpi &= -\frac{\bar{\tau}^q - \nu}{1 - \tau^y} \cdot \frac{\partial_l q^d}{\theta}
\end{aligned}$$

where the second line uses $l|U_l| = (1 - \tau^y)\theta l U_c = (1 - \tau^y)y U_c$ to match Proposition C.1's Le Chatelier correction, $\epsilon - \tilde{\epsilon} = \bar{p}^2 U_c^2 (\gamma_c - \gamma_q)^2 / (l|U_l||\mathcal{D}|) = \bar{p}^2 |s_{qq}|(\gamma_c - \gamma_q)^2 / ((1 - \tau^y)y)$, and the fourth line uses $\partial_l q^d / \theta = -\bar{p} |s_{qq}|(\gamma_c - \gamma_q) / y$ from (50) and $\varpi \bar{p} = \bar{\tau}^q - \nu$. Moving the two fiscal pieces to the left-hand side of (57) and collecting the $\psi U_c / (\theta f)$ terms on the right,

$$\frac{\tau_t^y}{1 - \tau_t^y} + \frac{\bar{\tau}_t^q - \nu_t}{1 - \tau_t^y} \left[(1 - \tau_t^y) \partial_x q^d + \frac{\partial_l q^d}{\theta} \right] = [A_t + (\gamma_c - \tilde{\gamma}) - (\epsilon - \tilde{\epsilon})] \frac{\psi U_c}{\theta f} = \tilde{A}_t \frac{\psi U_c}{\theta f}$$

which is the comprehensive wedge Ω_t of (12), with $\bar{\tau}^q - \nu$ the screening tax and the bracket the compensated demand response $dq^d/dy|_u$. Along a perturbation dy with $dx = (1 - \tau^y)dy$, utility is unchanged and $dq^d = [(1 - \tau^y)\partial_x q^d + \partial_l q^d/\theta]dy$.

The composite statistics are also the curvatures of the indirect utility of Lemma 5.6, which proves the second claim of Proposition C.1. Since $\tilde{U}_x = U_c(c^d, q^d, l)$ and $\partial_l c^d = -\bar{p} \partial_l q^d$,

$$\tilde{U}_{xl} = U_{cl} + (U_{cq} - \bar{p} U_{cc}) \partial_l q^d$$

Dividing by $\tilde{U}_x = U_c$, multiplying by l , and substituting (48) and (49),

$$\frac{\tilde{U}_{xl} l}{\tilde{U}_x} = \gamma_c + \frac{l(U_{cq} - \bar{p} U_{cc})}{U_c} \cdot \frac{\bar{p} U_c (\gamma_q - \gamma_c)}{l|\mathcal{D}|} = \gamma_c + m_q (\gamma_q - \gamma_c) = \tilde{\gamma}$$

Similarly $\tilde{U}_l = U_l(c^d, q^d, l)$ gives, using $U_{lq} - \bar{p} U_{lc} = (\bar{p} U_c / l)(\gamma_q - \gamma_c)$,

$$\tilde{U}_{ll} = U_{ll} + (U_{lq} - \bar{p} U_{lc}) \partial_l q^d = U_{ll} + \frac{\bar{p}^2 U_c^2 (\gamma_c - \gamma_q)^2}{l^2 |\mathcal{D}|}$$

and dividing by $\tilde{U}_l = U_l < 0$ flips the sign of the positive correction,

$$\frac{\tilde{U}_u l}{\tilde{U}_l} = \epsilon - \frac{\bar{p}^2 U_c^2 (\gamma_c - \gamma_q)^2}{l |U_l| |\mathcal{D}|} = \tilde{\epsilon} \leq \epsilon$$

Under weak separability, the corrections vanish and the problem is that of Appendix A.3.

C.8 The costate and the comprehensive-wedge decomposition

Proposition C.2 (Comprehensive-wedge decomposition).

At every history,

$$\Omega_t = \tilde{A}_t(\theta) \tilde{B}_t(\theta) \tilde{C}_t(\theta) + \beta R \Omega_{t-1} \tilde{D}_t(\theta)$$

where $\tilde{B}_t = B_t$ is the hazard ratio of Proposition 4.1 and $\tilde{C}_t, \tilde{D}_t$ are its C_t and D_t with the integrating kernel built on the composite $\tilde{\gamma}$ and, in $\tilde{C}_t$, the redistributive weight carrying the revenue term $1 - (\bar{\tau}_t^q - \nu_t) \partial_x q^d - \lambda_{1,t} \bar{\alpha}_t U_{c,t}$, the commodity credit net of marginal damages.

Substituting $\phi = f/U_c - \psi \gamma_c / \theta + \eta^q \bar{p} \mathcal{I}_t f / U_c$ from (53) into the costate equation and eliminating the multiplier with $\eta^q \bar{p} \mathcal{I}_t = m_q(\varpi^* - \varpi)$ and $m_q \varpi^* f / U_c = m_q(\gamma_c - \gamma_q) \psi / \theta$,

$$\dot{\psi} - \frac{\tilde{\gamma}}{\theta} \psi = \lambda_1 \bar{\alpha}_t f - \lambda_2 f_2 - [1 - m_q \varpi] \frac{f}{U_c}$$

Solving this ODE with the integrating factor $\exp[-\int \tilde{\gamma} d\tilde{x}/\tilde{x}]$ and $\psi(\infty) = 0$,

$$\psi(\theta) = \int_{\theta}^{\infty} \exp \left[- \int_{\theta}^x \tilde{\gamma}(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] \left([1 - m_q(x) \varpi(x)] \frac{f(x)}{U_c(x)} - \lambda_1 \bar{\alpha}_t(x) f(x) + \lambda_2 f_2(x) \right) dx$$

Here, $\psi(0) = 0$ pins $\lambda_{1,t}$ as in (17). To carry $U_c(\theta)$ into the integrand, multiply the identity (18) by $\exp[\int_{\theta}^x (\gamma_c - \tilde{\gamma}) d\tilde{x}/\tilde{x}]$, with $\gamma_c - \tilde{\gamma} = m_q(\gamma_c - \gamma_q)$:

$$\frac{U_c(\theta)}{U_c(x)} \exp \left[- \int_{\theta}^x \tilde{\gamma}(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] = \exp \left[\int_{\theta}^x \left(\sigma \frac{\dot{c}}{c} + \sigma_{cq} \frac{\dot{q}}{q} - \gamma_c \frac{\dot{y}}{y} + m_q(\gamma_c - \gamma_q) \frac{1}{\tilde{x}} \right) d\tilde{x} \right] \equiv \tilde{K}(\theta, x)$$

By Appendix C.7, $\Omega_t = \tilde{A}_t \psi U_c / (\theta f)$, so multiplying the costate solution by $\tilde{A}_t U_c / (\theta f)$ and splitting the integral as in Appendix A.3,

$$\Omega_t = \tilde{A}_t(\theta) \tilde{B}_t(\theta) \tilde{C}_t(\theta) + \beta R \Omega_{t-1} \tilde{D}_t(\theta)$$

$$\begin{aligned}
\tilde{B}_t(\theta) &= \frac{1 - F_t(\theta)}{\theta f_t(\theta)} \\
\tilde{C}_t(\theta) &= \int_{\theta}^{\infty} \tilde{K}(\theta, x) (1 - m_q(x) \varpi(x) - \lambda_{1,t} \bar{\alpha}_t(x) U_{c,t}(x)) \frac{f_t(x) dx}{1 - F_t(\theta)} \\
\tilde{D}_t(\theta) &= \frac{\tilde{A}_t(\theta) U_{c,t}(\theta) \theta_{t-1} \int_{\theta}^{\infty} \exp \left[- \int_{\theta}^x \tilde{\gamma}_t(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] f_{2,t}(x) dx}{\tilde{A}_{t-1} U_{c,t-1} \theta f_t(\theta)}
\end{aligned}$$

The persistence block uses $\lambda_{2,t} = \beta R \psi_{t-1} / f_{t-1}$ and, from $\Omega_{t-1} = \tilde{A}_{t-1} \psi_{t-1} U_{c,t-1} / (\theta_{t-1} f_{t-1})$,

$$\frac{\psi_{t-1}}{f_{t-1}} = \frac{\theta_{t-1} \Omega_{t-1}}{\tilde{A}_{t-1} U_{c,t-1}}$$

Hence, the recursing object is the comprehensive wedge, which is Proposition C.2. The revenue term $-m_q \varpi$ inside the $\tilde{C}_t$ weight is the only trace of the second good in the redistributive valuation.

C.9 The optimal uniform price

Let the schedule be flat, $\bar{\tau}_t^q(\theta) = \bar{\tau}_t$, with the level chosen by the planner. The level enters the date-0 Lagrangian only through the date- t wedge constraints, each contributing $\eta^q f \partial[U_q/U_c - (1 + \bar{\tau}_t)] / \partial \bar{\tau}_t = -\eta^q f$. The R^{-t} discounting and the transition densities convert the sum over date- t histories into the unconditional population expectation, as in Step 4 of Proposition A.1. By the envelope theorem,

$$\frac{dW_0}{d\bar{\tau}_t} = -\frac{1}{R^t} \mathbb{E}_0[\eta_t^q], \quad W_0 \equiv -\mathcal{L}_0$$

Hence, the optimal level sets $\mathbb{E}_0[\eta_t^q] = 0$. In the reduction bookkeeping, the same condition emerges from Roy's identity offsetting the incentive value of compensation. The constraint formulation nets the two automatically and only the substitution distortion against the screening benefit survives. Substituting (55) with $\bar{p}_t$ and ϖ_t common across the date- t cross-section,

$$\mathbb{E}_0[\eta_t^q] = \bar{p}_t \{ \mathbb{E}_0[|s_{qq,t}| \varpi_t^*] - \varpi_t \mathbb{E}_0[|s_{qq,t}|] \} = 0 \quad \implies \quad \varpi_t = \frac{\mathbb{E}_0 \left[|s_{qq,t}| (\gamma_{c,t} - \gamma_{q,t}) \frac{\psi U_{c,t}}{\theta f_t} \right]}{\mathbb{E}_0[|s_{qq,t}|]}$$

Multiplying this by $\bar{p}_t \mathbb{E}_0[|s_{qq}|]$ and replacing $\bar{p} |s_{qq}| (\gamma_c - \gamma_q) \psi U_c / (\theta f) = -(\psi U_c / (\theta f)) l \partial_l q^d$ and $\bar{p} \varpi_t = \bar{\tau}_t - \nu_t$,

$$(\bar{\tau}_t - \nu_t) \mathbb{E}_0[|s_{qq,t}|] = -\mathbb{E}_0\left[\frac{\psi U_{c,t}}{\theta f_t} l_t \partial_l q_t^d\right]$$

At history-specific prices the same condition without the integral gives $\varpi(\theta^t) = \varpi^*(\theta^t)$, the unconstrained wedge of Appendix A.5. The uniform tax is its $|s_{qq}|$ -weighted average. Under weak separability, every history-specific wedge is zero, so $\bar{\tau}_t = \nu_t$; in retirement, $\psi = 0$ by Lemma A.2, so $\psi U_c/(\theta f) = 0$ and again $\bar{\tau}_t = \nu_t$ (both cases of Corollary 5.6).

C.10 The cost of uniformity

By the envelope theorem applied history by history, the date- t resource value of moving a single history's price is $-\eta^q(\theta)f(\theta)$ per unit of $\bar{\tau}^q(\theta)$. By (55),

$$-\eta^q f = \bar{p} |s_{qq}| (\varpi - \varpi^*) f$$

This value zeroes out at $\varpi = \varpi^*$, reproducing the history-specific optimum. Per unit of the screening component ϖ the cost is $\bar{p}^2/(1+\nu)$ times this, since $\varpi = 1 - (1+\nu)/\bar{p}$ gives $d\bar{\tau}^q = \bar{p}^2 d\varpi/(1+\nu)$. The loss from setting ϖ rather than ϖ^* at a history is then, to second order

$$\int_{\varpi^*}^{\varpi} \frac{\bar{p}^3}{1+\nu} |s_{qq}| (v - \varpi^*) dv = \frac{1}{2} \frac{\bar{p}^3}{1+\nu} |s_{qq}| (\varpi - \varpi^*)^2$$

Aggregating over the date- t cross-section, with $\bar{p}$ common because the price is uniform and ν common because the emission price is an aggregate, and minimizing the resulting quadratic over the common ϖ yields the weighted mean $\varpi = \mathbb{E}_t^{|s|}[\varpi_t^*]$ of Appendix C.9 and, at that minimizer,

$$\text{Loss}_t \approx \frac{1}{2} \frac{(1 + \bar{\tau}_t)^3}{1 + \nu_t} \mathbb{E}_0[|s_{qq,t}|] \text{Var}_t^{|s|}(\varpi_t^*)$$

where $\mathbb{E}_t^{|s|}$ and $\text{Var}_t^{|s|}$ are the $|s_{qq,t}|$ -weighted population mean and variance over date- t histories, which is Proposition 5.4.

C.11 The externality and the social cost of carbon

The conditions above already carry the shadow emission price: every deviation is measured net of ν_t , so all of Section 5.2's results hold with $\bar{\tau}_t^q - \nu_t$ the screening tax and every statistic

evaluated at the externality allocation, and the optimal flat level of Appendix C.9 delivers the additive split of Proposition 5.4. What remains is the pollution-stock multiplier. Substitute the three pieces of the first-order condition for S ,

$$-\psi U_{sl} \frac{l}{\theta} = -\frac{\psi U_c}{\theta f} \xi \frac{U_S}{U_c} f \quad -\phi U_S = -(1 - \chi_t^q) \frac{U_S}{U_c} f \quad \eta^q f \Gamma_S^q = \eta^q \frac{U_S}{U_c} \left[\frac{\xi_q}{q} - \bar{p} \frac{\xi_c}{c} \right] f$$

the second from (53) and the third from (47), to get

$$\mu_t = - \int_0^\infty \left[\frac{\psi U_c}{\theta f} \xi + 1 - \chi_t^q - \eta^q \left(\frac{\xi_q}{q} - \bar{p} \frac{\xi_c}{c} \right) \right] \frac{U_S}{U_c} f d\theta$$

Expanding $1 - \chi_t^q = 1 - \gamma_c \frac{\psi U_c}{\theta f} + \eta^q \bar{p} \mathcal{I}_t$ with $\bar{p} \mathcal{I}_t = \bar{p} \sigma / c - \sigma_{cq} / q$ and collecting,

$$\mu_t = - \int_0^\infty \left[1 + (\xi - \gamma_c) \frac{\psi U_c}{\theta f} + \eta^q \left(\bar{p} \frac{\sigma + \xi_c}{c} - \frac{\sigma_{cq} + \xi_q}{q} \right) \right] \frac{U_S}{U_c} f d\theta$$

The shadow emission price aggregates these date- $(t+j)$ damages exactly as in Appendix A.6. Current emissions enter the continuation value only through the accumulation constraints of the next s periods, so each envelope step peels off one date,

$$\frac{\partial V_{t+1}}{\partial Q_t} = \sum_{j=1}^s \frac{1}{R^{j-1}} \mu_{t+j} \Phi_j, \quad \Phi_j \equiv \frac{\partial S_{t+j}}{\partial Q_t},$$

and, with $\Phi_0 = S_q$, (13) gives $\nu_t = \sum_{j=0}^s R^{-j} \mu_{t+j} \Phi_j$. Substituting this expression for each μ_{t+j} and writing the date- $(t+j)$ population integrals as expectations over future skill histories,

$$\text{SCC}_t \equiv \nu_t = - \sum_{j=0}^s \frac{1}{R^j} \mathbb{E}_t \int \left[1 + (\xi_{t+j} - \gamma_{c,t+j}) \frac{\psi_{t+j} U_{c,t+j}}{\theta f_{t+j}} + \eta_{t+j}^q \left(\bar{p}_{t+j} \frac{\sigma_{t+j} + \xi_{c,t+j}}{c_{t+j}} - \frac{\sigma_{cq,t+j} + \xi_{q,t+j}}{q_{t+j}} \right) \right] \frac{U_{S,t+j}}{U_{c,t+j}} \Phi_j f_{t+j} d\theta$$

whose bracket is the damage weight of Proposition 5.3 at each future history. The new piece is the response of the damage valuation to a budget-neutral clean-for-dirty swap: along $(dc, dq) = (\bar{p}, -1)$, which holds expenditure at consumer prices fixed, $d \ln(U_S/U_c) = \bar{p}(\sigma + \xi_c)/c - (\sigma_{cq} + \xi_q)/q$, using $\partial_c \ln(U_S/U_c) = (\sigma + \xi_c)/c$ and $\partial_q \ln(U_S/U_c) = (\sigma_{cq} + \xi_q)/q$. At $\eta^q = 0$ the weight collapses to the fiscal factor of Appendix A.6; at the optimal flat level the multiplier averages to zero but is not zero history by history, so the weight carries the term even when the level is chosen optimally, which is how the corrective part of Proposition 5.4's additive split is repriced relative to Proposition 4.2. The corrective component of the price is nonetheless uniform across types and histories, because damages depend only on aggregate emissions, so the emissions margin carries no information about types and cannot screen; the uniformity restriction binds only the Corlett-Hague screening component and never the corrective one.

D Quantitative exercise

D.1 Calibrating the given schedules to the US tax system

The schedules the exercise needs are not hypothetical. For the labor margin, take the earnings-indexed schedule (43) of Appendix B.13. Heathcote et al. (2017) estimate $p = 0.181$ for US households aged 25 to 60 by regressing log post-government on log pre-government earnings in the PSID, with an R^2 of 0.91. Heathcote and Tsujiyama (2021) pair it with $\lambda^T = 0.840$. Progressivity is flat in age. Re-estimating with both parameters conditioned on age, Heathcote et al. (2020) find no significant age dependence in p , so an age-invariant progressivity is a feature of the US system rather than an approximation error. Their null covers the slope alone; the level λ_t^T is left free to vary with age.

For the savings margin there is no counterpart to (43) with the same standing, since the fitted tax functions of Heathcote et al. (2017) and Guner et al. (2014) are functions of earnings and imply nothing about the intertemporal margin. The practical route is a linear capital income tax. With private saving at gross return R taxed at rate τ_t^k on the interest component, the household's Euler equation and the definition of the savings wedge give

$$1 - \bar{\tau}_t^s = \frac{1 + (R - 1)(1 - \tau_t^k)}{R} \quad (59)$$

so a single number fixes the whole schedule. With $\beta R = 1$ as in Section 6, (59) collapses to $\bar{\tau}_t^s = (1 - \beta)\tau_t^k$: at $\beta = 0.95$ the frozen wedge is one twentieth of the capital income tax rate, because the tax falls on the interest component alone. A wedge of under one percent of the gross return is therefore a tax of tens of percent on the margin it actually distorts, and I report the schedule in tax units throughout to keep that visible.

The rate (59) asks for is a marginal one on the household's own intertemporal margin, since it comes from the Euler equation on a marginal unit of saving. The natural US anchor is the Congressional Budget Office (2014) effective marginal tax rate on capital income, which it puts at 18 percent, and I take that as the benchmark. Two qualifications are worth stating, because the headline figure is not quite the object the mapping wants. It is a capital-stock-weighted composite of a 29 percent rate on business capital and a rate near zero on owner-occupied housing, which is close to half the stock; and it measures the total wedge between the return an investment must earn and the return the saver receives, so it includes a corporate layer that a household facing an exogenous gross return R never sees. The personal layer alone is what the mapping freezes, and the same study prices it: an average marginal rate of 27 percent on interest income, applied to the share of marginal saving that is not assigned to tax-deferred or Roth accounts, where the marginal rate is zero.¹⁴ The benchmark is also computed under 2014

¹⁴The composite rate is CBO's Table 2 and the sectoral weights its Table A-1; the marginal rate on interest

law, before the 2017 reform cut rates on business capital, so as a description of the current code it errs high rather than low.

The aggregate effective rates computed by the method of Mendoza et al. (1994), which put τ^k near 0.36 for the US (Trabandt and Uhlig 2011), are averages: capital tax revenue divided by capital income in the national accounts. They coincide with the marginal rate in the linear-tax representative-agent settings they were built for, which is precisely the identification a Mirrleesian model drops. They do reach state, local and property taxes that the federal marginal measure leaves out, so I carry 0.36 as an upper bound in the sensitivity analysis of Section 6.4 and use 0.18 everywhere else.

At $\tau^k = 0.18$ the frozen wedge is $\bar{\tau}^s = 0.009$, nine tenths of a percent of the gross return per year. Section 6 computes the full-optimum savings wedge as falling from 0.0056 to 0.0002 over the working life; in the same tax units that is a capital income tax of about 11% at labor-market entry declining to nearly zero by retirement. The US system therefore overdistorts the intertemporal margin at every age, and by more as the cohort ages, at any τ^k above 0.11, which every measure of the US rate clears.

The exercise inherits one sensitivity worth stating, and the deviations survive it. How far the given schedules sit from their desired counterparts, and therefore how large the net shadow price ζ_t of Section 5.1 is, depends on the Pareto weights $\alpha(\theta)$ of Section 3. Under a utilitarian objective Heathcote and Tsujiyama (2021) find a welfare gain of about 2% of consumption from replacing the estimated US schedule with the fully optimal one, so the schedule is far from optimal; under the weights that rationalize observed US progressivity within their parametric class the gain falls to about 0.05%, and one might expect the deviations of Corollary 5.1 to vanish with it. Re-solving the joint exercise at $\tau^k = 0.18$ under the family $\alpha(\theta) \propto \theta^{-\varrho}$ on the initial skill shows that they do not.¹⁵ Across the whole family the working-life mean of the deviation stays in $[-0.278, -0.249]$, against a utilitarian value of -0.264 ; the retirement mean stays in $[-1.038, -0.999]$, against -1.019 ; and the age at which η_t^s changes sign falls between 39 and 40 at every ϱ . Lemma 5.3 forces that reversal along every history without dating it, so the date is a property of the solve rather than of the lemma. A six-fold tilt in the weights therefore moves the deviation by six percent, and it saturates as the weights tilt toward the unlucky. The labor multiplier η_t moves the way its tightness reading predicts, its working-life mean shrinking by a quarter as ϱ falls, but it saturates well short of zero. No power weighting

income and the account shares are its Tables A-4 and A-3. The personal-layer figure is my own arithmetic from those two tables, not a number CBO publishes: at their fully-taxable shares it lands in the high teens. That it agrees with the 18 percent headline is a coincidence of construction rather than a corroboration, since the headline nets a corporate layer against a housing subsidy and this does neither.

¹⁵Primitive weights enter the planner problem only through the birth-period objective, so each ϱ re-solves the birth block against the same continuation surfaces; $\varrho = 0$ reproduces the production run, and $\varrho < 0$ weights the fortunate draws. The sweep runs $\varrho \in [-4, 1]$; at $\varrho = -4$ the weight ratio across the 1–99 percent range of the initial draw is six.

of the initial skill, however steep, rationalizes the frozen pair.

The weights have so little room because almost none of this economy’s inequality is theirs to reweight. Agents here are *ex ante* identical, so the primitive weights attach only to the information in the first draw; every later difference across agents is risk, and a welfarist planner prices risk with the agents’ own expected utility whatever α . A calibration with initial dispersion in the data’s range would hand the weights more of the labor gap, the margin Section 6.7 shows changing sign outright under the Golosov et al. (2016) wedge path. It would hand them none of the savings margin: the retiree’s desired savings wedge is zero under every admissible weighting while the frozen wedge stays $(1 - \beta)\tau^k > 0$, so the savings deviation is not the weights’ to remove. I take the utilitarian case throughout, matching the planner of Section 3; the sweep bounds what a reader with different distributional tastes should carry, and what moves is the working-life level, not the sign change of η_t^s . The carbon price’s own crossing is not invariant: it moves later as ϱ rises, from age 26 at $\varrho = -0.5$ to age 28 at the utilitarian weights and age 30 at $\varrho = 1$, and it leaves the life cycle at the young end once the weights tilt far enough toward the fortunate, with no crossing at any $\varrho \leq -1$. This is Lemma 5.4 again, since that crossing is where the savings term offsets the labor term rather than where either vanishes: tilting toward the fortunate lowers the early-life deviation until it is already negative at labor-market entry.

D.2 Solvers and numerical implementation

The three solvers share the architecture of the Farhi and Werning (2013) replication they are built on. Each period’s problem is solved by backward induction on a grid over the two multipliers dual to promise keeping and threat keeping. At every grid point the within-period problem is a two-point boundary value problem in the type dimension, integrated by an adaptive Dormand-Prince 5(4) scheme at relative tolerance 10^{-3} and absolute tolerance 10^{-6} and closed by shooting on the value promised at the bottom of the type support, with Brent’s method over expanding brackets. Values at the storage nodes are read from the dense-output interpolant of the same trajectory the shooting residual integrates, so the stored allocation and the residual that certifies it are the same object; the transversality condition is met to a median of 8×10^{-15} .

The frozen-world solvers add the marginal-utility requirement $\hat{M}$ of Lemma 5.1 as a third state and replace the shooting pass with direct collocation. The node conditions, the trapezoid collocation rows, the boundary rows and the row that delivers the inherited $\hat{M}$ are stacked and solved by Newton with an analytic sparse Jacobian, warm-started from the neighbouring date.

Grids differ across the stack because the state space does. The full optimum and the linear commodity tax run on 100×45 dual grids with 20 type nodes. The frozen world runs on 16×8 dual grids with 40 type nodes and an adaptive promise grid of 11 to 27 points: the third state has to be carried at every dual node, and the dual grid pays for it. This is the coarsening Section 6.1 refers to when it reads the incentive-compatibility audit against a control at the same grid. Shooting residuals are driven to 10^{-11} , pointwise roots to between 10^{-10} and 10^{-12} ,

and the Newton stacks to residuals of order 10^{-11} .

Welfare is measured in consumption-equivalent units, $\exp(\Delta u/\mathcal{A}) - 1$, with Δu the difference in mean simulated date-0 lifetime utility between the two allocations compared and $\mathcal{A} = (1 - \beta^{60})/(1 - \beta)$ the present value of a unit consumption stream, and it is computed from simulated lifetime utilities rather than from the grid value functions: at each simulated history the realized bundle is evaluated in the direct utility function, retirement included. Simulation draws one random-number stream per agent, keyed by the run seed, so the shock panel is identical across regimes and across thread counts and every comparison is a common-random-numbers comparison. What bounds the reported differences is not sampling error. For the full-optimum ladder at 300,000 histories the binding figure is the spread in the present-value resource residual across rungs, about 0.003 percentage points; the seed-to-seed sampling error on gains is an order of magnitude tighter.

One consequence for the quantitative exercise. The multiplier η_t is now determined at the fixed point between the assigned earnings $y_t(\theta^t)$ and the marginal rate the schedule attaches to them, since $\bar{\tau}_t^y$ appears on both sides of (27) through y . Under (43) the left-hand side of (27) is strictly increasing in y whenever $p_t \in (0, 1)$, since $\bar{\tau}_t^y/(1 - \bar{\tau}_t^y) = y^{p_t}/(\lambda_t^T(1 - p_t)) - 1$. The right-hand side moves with y too, through A_t , Δ_t , Λ_t and the costate, so any fixed point at which it is non-increasing in y is locally unique. Global uniqueness is not established, and the solver iterates to a fixed point rather than reading the schedule off the type.

D.3 Validation

Validation runs in three layers, and everything reported here was re-run on the production code.

The first layer checks the preference block. Every analytic derivative the solvers use is checked against central finite differences at a battery of parameter points, together with the structural identities the two-good extension has to satisfy: consistency of the indirect utility function, equality of the household's marginal rate of substitution with the consumer price, the budget and resource identities, the spending identity to under 10^{-9} , uniqueness of the root of the share-consistency residual, and the zero-twist limit in which the commodity wedge collapses to the Pigouvian rate. These pass at between 10^{-5} and 10^{-9} .

The second layer checks the solvers against each other on a common 48×24 dual grid with 20 type nodes. With the twist, damages, subsistence and the commodity tax all switched off, the linear-commodity-tax solver reproduces the Farhi and Werning (2013) replication: policies agree to 4×10^{-3} and the initial resource multiplier to 2.6×10^{-4} , and both converge under refinement to the published value. Re-solving both at relative tolerance 10^{-5} collapses the remaining gaps by more than two orders of magnitude, which identifies them as integration noise rather than a difference in the problem being solved. With the twist off and no externality the full-optimum

and linear solvers agree to 4.4×10^{-7} on the same multiplier, agree to machine precision at the last working date where the retirement closed form removes all continuation interpolation, and return a commodity wedge that is identically zero. With the twist off and a positive damage the full optimum sets the commodity wedge equal to marginal damage at every grid point and every period to 2.8×10^{-16} . That is the dynamic Atkinson-Stiglitz agreement of Section 6.1, and it holds to solver precision rather than approximately.

The third layer checks the frozen-world stack against the solvers it nests. At a zero labor multiplier the constrained node system is zeroed by the stored full-optimum solution to 5×10^{-13} , and at a zero savings multiplier the joint node system equals the system one stage below it exactly. The retirement chain reproduces the closed-form annuity at a zero frozen savings wedge to 1.8×10^{-15} . The stacked residual vanishes at stored full-optimum solutions to order 10^{-14} , every row of the joint stacked system vanishes at stored slices of the benchmark store to 8×10^{-10} , and warm re-convergence from a refreshed seed returns the stored solution within one Newton iteration. At the driver level, the frozen-world stack at the desired schedules approaches the full optimum at the second-order rate the type discretization implies, the gap in the initial multiplier falling from 1.2×10^{-2} to 3.8×10^{-3} as the type grid doubles from 20 nodes to 40; freezing the schedules at their desired counterparts returns multipliers indistinguishable from zero, which is the second agreement of Section 6.1; and the US freeze reproduces a recorded cross-solver anchor, two structurally different solvers agreeing on the deviation profile to better than 10^{-2} .

One check is a diagnostic rather than a gate, and it is worth saying why. Evaluating Proposition 5.1 directly at simulated states, with every multiplier reconstructed by interpolation from the stored solution, reproduces the solved deviation only to about 0.2 in absolute value, against a mean deviation of 0.26. The failure is one of conditioning on the read side rather than of the solution. The right-hand side of the proposition is a small residual of two large offsetting terms: at labor-market entry the labor and savings channels are -0.25 and $+0.31$, each several times the deviation they sum to, so errors of a few percent in the interpolated multipliers produce absolute misses of the same order as the deviation itself while leaving the solved allocation untouched. Separating the two sources confirms the diagnosis. The proposition's own algebra, evaluated against the node system at the same states and given the same reconstructed inputs, agrees to a mean of 0.024, an order of magnitude below the input error, and the node conditions the proposition collects are enforced exactly at grid points by the gates above. No number reported in this paper is computed from the identity.

D.4 The incentive-compatibility audit

The first-order approach behind every solver is verified ex post by a one-shot-deviation audit. At simulated states of each solved store, a deviating agent picks the best point on the entire reported-type menu and the audit records the utility gain over truth telling. Gains at the scale

Table 9: The one-shot-deviation audit across the solved stores. Entries are utility gains in utils at simulated states; “p99 gain” is the 99th percentile over the tightest window, “junction p99” the same statistic at the retirement junction, and the last column counts non-monotone reported-earnings menus among sampled states. The horizontal rule separates the full-optimum grid from the frozen-world production grid; the control row is the unconstrained solve at the frozen-world grid, whose gains are pure menu-interpolation noise.

economy	grid	p99 gain	junction p99	non-monotone menus
second best, benchmark	100×45	6.4×10^{-3}	2.5×10^{-3}	0/140
second best, $\gamma_c = \gamma_q$ placebo	100×45	7.4×10^{-3}	2.5×10^{-3}	0/840
Stone-Geary, flat Pigouvian rung	100×45	6.6×10^{-3}	2.5×10^{-3}	0/840
<i>control:</i> unconstrained, third-best grid	16×8	7.4×10^{-2}	6.0×10^{-4}	0/840
labor frozen (stage 1)	16×8	7.0×10^{-2}	5.8×10^{-4}	0/840
joint freeze, $\tau^k = 0.18$	16×8	4.4×10^{-2}	6.8×10^{-4}	0/840
joint freeze, $\tau^k = 0.36$	16×8	6.2×10^{-2}	1.2×10^{-3}	2/840
joint freeze, flat τ^{q*}	16×8	6.7×10^{-2}	9.1×10^{-4}	0/840

of the menu-interpolation noise are the audit’s resolution, not violations, and that scale differs by an order of magnitude across grids: the full optimum solves on 100×45 dual nodes, the frozen-world production grid on 16×8 . The frozen-world stores are therefore read against an unconstrained solve at the identical 16×8 grid, whose deviation gains are pure interpolation noise by construction.

Table 9 reports, for each store, the 99th-percentile deviation gain in the tightest re-simulation window, the set of simulated states whose reported type lies within half a log innovation standard deviation of the truth, $\hat{\sigma}$ being the log-skill innovation dispersion of Section 6.1, the retirement-junction row, where the continuation value is closed form and the audit is sharpest, and the count of non-monotone reported-earnings menus. Two readings carry the section’s claim. The four frozen-wedge stores sit at or below the unconstrained control at the identical grid, so the freeze introduces no deviation gains the grid itself does not already produce. And the junction rows sit well over an order of magnitude below the local windows on the frozen-world grid, so where the audit is sharp it is clean.